GRADE 7 COMMON CORE MATH

Answer Key

Main • Backward Branches • Forward 1 (Grade 8) • Forward 2 (Grade 9)

61 question sets • 1525 questions

Page numbers match the Question Collection.

Every page shows its question ID (for example S40-F2-Q3).

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SET 1 • 7.RP.A.1 • RATIOS & PROPORTIONAL RELATIONSHIPS

Compute unit rates associated with ratios of fractions

MAIN	Grade 7 main question	**7.RP.A.1**
BACKWARD 1	Find a unit rate from whole-number quantities	**6.RP.A.2**
BACKWARD 2	Divide a fraction by a fraction	**6.NS.A.1**
FORWARD 1	Compare unit rates shown in different representations	**8.EE.B.5**
FORWARD 2	Convert rates from one unit to another	**HSN-Q.A.1**

SET 1 • 7.RP.A.1

MAIN

Grade 7 main question • Grade 7 • Ratios & Proportional Relationships • 7.RP.A.1

Ana walks $\frac{3}{4}$ mile in $\frac{1}{3}$ hour at a steady pace.

How many miles does she walk per hour?

Miles per hour: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-M-Q1

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.1 • Set 1 • MAIN • Q1/5

8

QUESTION ID

S1-M-Q1

TYPE

Short answer

CORRECT ANSWER

2 1/4 miles per hour

GRADING NOTE 9/4 or 2.25 miles per hour is also correct.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-M-Q1

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.1 • Set 1 • MAIN • Q1/5

A recipe uses $\frac{2}{3}$ cup of sugar for every $\frac{1}{2}$ cup of flour.

How many cups of sugar are used per cup of flour?

Cups of sugar: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-M-Q2

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.1 • Set 1 • MAIN • Q2/5

9

QUESTION ID

S1-M-Q2

TYPE

Short answer

CORRECT ANSWER

1 1/3 cups

GRADING NOTE 4/3 cups is also correct.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-M-Q2

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.1 • Set 1 • MAIN • Q2/5

Leo paints $\frac{5}{6}$ of a wall in $\frac{1}{2}$ hour.

How many walls does he paint per hour?

A. $\frac{3}{5}$ wall

B. $1\frac{2}{3}$ walls

C. $\frac{5}{12}$ wall

D. $2\frac{1}{2}$ walls

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-M-Q3

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.1 • Set 1 • MAIN • Q3/5

10

QUESTION ID

S1-M-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. 1 2/3 walls

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-M-Q3

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.1 • Set 1 • MAIN • Q3/5

True or false?

A snail crawls $\frac{1}{8}$ meter in $\frac{1}{4}$ minute. Its speed is $\frac{1}{2}$ meter per minute.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-M-Q4

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.1 • Set 1 • MAIN • Q4/5

11

QUESTION ID

S1-M-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-M-Q4

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.1 • Set 1 • MAIN • Q4/5

True or false?

$\frac{2}{5}$ pound of cheese costs \$3. The price is \$1.20 per pound.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-M-Q5

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.1 • Set 1 • MAIN • Q5/5

12

QUESTION ID

S1-M-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE The price is \$3 ÷ 2/5 = \$7.50 per pound.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-M-Q5

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.1 • Set 1 • MAIN • Q5/5

SET 1 • 7.RP.A.1

BACKWARD 1

Find a unit rate from whole-number quantities • Grade 6 • Ratios & Proportional Relationships • 6.RP.A.2

A car travels 180 miles in 3 hours at a steady speed.

What is its speed in miles per hour?

Miles per hour: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-B1-Q1

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.2 • Set 1 • BACKWARD 1 • Q1/5

14

QUESTION ID

S1-B1-Q1

TYPE

Short answer

CORRECT ANSWER

60

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-B1-Q1

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.2 • Set 1 • BACKWARD 1 • Q1/5

6 notebooks cost $9.

What is the cost per notebook?

Cost: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-B1-Q2

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.2 • Set 1 • BACKWARD 1 • Q2/5

15

QUESTION ID

S1-B1-Q2

TYPE

Short answer

CORRECT ANSWER

$1.50

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-B1-Q2

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.2 • Set 1 • BACKWARD 1 • Q2/5

A printer prints 120 pages in 8 minutes.

How many pages does it print per minute?

A. 112

B. 960

C. 15

D. 128

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-B1-Q3

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.2 • Set 1 • BACKWARD 1 • Q3/5

16

QUESTION ID

S1-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. 15

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-B1-Q3

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.2 • Set 1 • BACKWARD 1 • Q3/5

True or false?

Jo reads 90 pages in 5 days at the same rate each day. That is 18 pages per day.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-B1-Q4

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.2 • Set 1 • BACKWARD 1 • Q4/5

17

QUESTION ID

S1-B1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-B1-Q4

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.2 • Set 1 • BACKWARD 1 • Q4/5

True or false?

4 pounds of grapes cost $24. The price is $8 per pound.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-B1-Q5

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.2 • Set 1 • BACKWARD 1 • Q5/5

18

QUESTION ID

S1-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE The price is $6 per pound.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-B1-Q5

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.2 • Set 1 • BACKWARD 1 • Q5/5

SET 1 • 7.RP.A.1

BACKWARD 2

Divide a fraction by a fraction • Grade 6 • The Number System • 6.NS.A.1

Divide.

$\frac{3}{4} \div \frac{1}{2}$

Quotient: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-B2-Q1

Grade 6 • The Number System • 6.NS.A.1 • Set 1 • BACKWARD 2 • Q1/5

20

QUESTION ID

S1-B2-Q1

TYPE

Short answer

CORRECT ANSWER

3/2 = 1 1/2

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-B2-Q1

Grade 6 • The Number System • 6.NS.A.1 • Set 1 • BACKWARD 2 • Q1/5

Divide.

$\frac{5}{6} \div \frac{1}{3}$

Quotient: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-B2-Q2

Grade 6 • The Number System • 6.NS.A.1 • Set 1 • BACKWARD 2 • Q2/5

21

QUESTION ID

S1-B2-Q2

TYPE

Short answer

CORRECT ANSWER

5/2 = 2 1/2

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-B2-Q2

Grade 6 • The Number System • 6.NS.A.1 • Set 1 • BACKWARD 2 • Q2/5

Divide.

$\frac{5}{8} \div \frac{1}{4}$

A. $1\frac{1}{4}$

B. $\frac{5}{32}$

C. $\frac{2}{5}$

D. $2\frac{1}{2}$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-B2-Q3

Grade 6 • The Number System • 6.NS.A.1 • Set 1 • BACKWARD 2 • Q3/5

22

QUESTION ID

S1-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. 2 1/2

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-B2-Q3

Grade 6 • The Number System • 6.NS.A.1 • Set 1 • BACKWARD 2 • Q3/5

True or false?

$\frac{1}{2} \div \frac{1}{4} = 2$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-B2-Q4

Grade 6 • The Number System • 6.NS.A.1 • Set 1 • BACKWARD 2 • Q4/5

23

QUESTION ID

S1-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-B2-Q4

Grade 6 • The Number System • 6.NS.A.1 • Set 1 • BACKWARD 2 • Q4/5

True or false?

$\frac{2}{3} \div \frac{1}{3} = \frac{2}{9}$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-B2-Q5

Grade 6 • The Number System • 6.NS.A.1 • Set 1 • BACKWARD 2 • Q5/5

24

QUESTION ID

S1-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-B2-Q5

Grade 6 • The Number System • 6.NS.A.1 • Set 1 • BACKWARD 2 • Q5/5

SET 1 • 7.RP.A.1

FORWARD 1

Compare unit rates shown in different representations • Grade 8 • Expressions & Equations • 8.EE.B.5

Which relationship has the greatest unit rate?

A. y = 4.2x

B. A table with the pairs (2, 8) and (5, 20)

C. A line through (0, 0) and (3, 12.3)

D. y = 3.9x

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-F1-Q1

Grade 8 • Expressions & Equations • 8.EE.B.5 • Set 1 • FORWARD 1 • Q1/5

26

QUESTION ID

S1-F1-Q1

TYPE

Multiple choice

CORRECT ANSWER

A. y = 4.2x

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-F1-Q1

Grade 8 • Expressions & Equations • 8.EE.B.5 • Set 1 • FORWARD 1 • Q1/5

Runner A: a graph of meters (y) against seconds (x) is a line through (0, 0) and (10, 45).

Runner B: d = 4.8t, where d is meters and t is seconds.

Which runner is faster, and by how many meters per second?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-F1-Q2

Grade 8 • Expressions & Equations • 8.EE.B.5 • Set 1 • FORWARD 1 • Q2/5

27

QUESTION ID

S1-F1-Q2

TYPE

Short answer

CORRECT ANSWER

Runner B, by 0.3 meter per second

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-F1-Q2

Grade 8 • Expressions & Equations • 8.EE.B.5 • Set 1 • FORWARD 1 • Q2/5

Plan A charges $0.15 per text message. The cost for Plan B is shown by a line through (0, 0) and (40, 5), where x is texts and y is dollars.

Which plan costs less per text?

Plan: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-F1-Q3

Grade 8 • Expressions & Equations • 8.EE.B.5 • Set 1 • FORWARD 1 • Q3/5

28

QUESTION ID

S1-F1-Q3

TYPE

Short answer

CORRECT ANSWER

Plan B

GRADING NOTE Plan B costs $0.125 per text, which is less than $0.15.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-F1-Q3

Grade 8 • Expressions & Equations • 8.EE.B.5 • Set 1 • FORWARD 1 • Q3/5

True or false?

The graph of y = 2.5x is steeper than the graph of a proportional relationship that passes through (4, 9).

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-F1-Q4

Grade 8 • Expressions & Equations • 8.EE.B.5 • Set 1 • FORWARD 1 • Q4/5

29

QUESTION ID

S1-F1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-F1-Q4

Grade 8 • Expressions & Equations • 8.EE.B.5 • Set 1 • FORWARD 1 • Q4/5

True or false?

A proportional relationship that passes through (6, 15) has a greater unit rate than y = 2.6x.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-F1-Q5

Grade 8 • Expressions & Equations • 8.EE.B.5 • Set 1 • FORWARD 1 • Q5/5

30

QUESTION ID

S1-F1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-F1-Q5

Grade 8 • Expressions & Equations • 8.EE.B.5 • Set 1 • FORWARD 1 • Q5/5

SET 1 • 7.RP.A.1

FORWARD 2

Convert rates from one unit to another • Grade 9 (High School) • Number & Quantity: Quantities • HSN-Q.A.1

A cheetah runs 66 feet per second.

How many miles per hour is this? (1 mile = 5,280 feet)

Miles per hour: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-F2-Q1

Grade 9 (High School) • Number & Quantity: Quantities • HSN-Q.A.1 • Set 1 • FORWARD 2 • Q1/5

32

QUESTION ID

S1-F2-Q1

TYPE

Short answer

CORRECT ANSWER

45

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-F2-Q1

Grade 9 (High School) • Number & Quantity: Quantities • HSN-Q.A.1 • Set 1 • FORWARD 2 • Q1/5

True or false?

Walking $\frac{1}{2}$ mile in $\frac{1}{10}$ hour is a speed of 440 feet per minute. (1 mile = 5,280 feet)

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-F2-Q2

Grade 9 (High School) • Number & Quantity: Quantities • HSN-Q.A.1 • Set 1 • FORWARD 2 • Q2/5

33

QUESTION ID

S1-F2-Q2

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-F2-Q2

Grade 9 (High School) • Number & Quantity: Quantities • HSN-Q.A.1 • Set 1 • FORWARD 2 • Q2/5

A pump moves 3 gallons every $\frac{1}{4}$ minute.

How many gallons per hour is this?

Gallons per hour: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-F2-Q3

Grade 9 (High School) • Number & Quantity: Quantities • HSN-Q.A.1 • Set 1 • FORWARD 2 • Q3/5

34

QUESTION ID

S1-F2-Q3

TYPE

Short answer

CORRECT ANSWER

720

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-F2-Q3

Grade 9 (High School) • Number & Quantity: Quantities • HSN-Q.A.1 • Set 1 • FORWARD 2 • Q3/5

A runner covers 400 meters in 80 seconds at a steady pace.

What is the speed in kilometers per hour?

A. 18

B. 300

C. 0.3

D. 5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-F2-Q4

Grade 9 (High School) • Number & Quantity: Quantities • HSN-Q.A.1 • Set 1 • FORWARD 2 • Q4/5

35

QUESTION ID

S1-F2-Q4

TYPE

Multiple choice

CORRECT ANSWER

A. 18

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-F2-Q4

Grade 9 (High School) • Number & Quantity: Quantities • HSN-Q.A.1 • Set 1 • FORWARD 2 • Q4/5

True or false?

60 miles per hour is the same speed as 1 mile per second.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-F2-Q5

Grade 9 (High School) • Number & Quantity: Quantities • HSN-Q.A.1 • Set 1 • FORWARD 2 • Q5/5

36

QUESTION ID

S1-F2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE 60 miles per hour is 1 mile per minute.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-F2-Q5

Grade 9 (High School) • Number & Quantity: Quantities • HSN-Q.A.1 • Set 1 • FORWARD 2 • Q5/5

SET 2 • 7.RP.A.2.a • RATIOS & PROPORTIONAL RELATIONSHIPS

Decide whether two quantities in a table are in a proportional relationship

MAIN	Grade 7 main question	**7.RP.A.2.a**
BACKWARD 1	Complete a table of equivalent ratios	**6.RP.A.3.a**
BACKWARD 2	Divide multi-digit decimals	**6.NS.B.3**
FORWARD 1	Decide whether a table shows a linear function	**8.F.A.3**
FORWARD 2	Recognize a quantity that changes at a constant rate per unit inte	**HSF-LE.A.1.b**

SET 2 • 7.RP.A.2.a

MAIN

Grade 7 main question • Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.a

True or false?

The table shows a proportional relationship between x and y.

x	2	5	8
y	7	17.5	28

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-M-Q1

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.a • Set 2 • MAIN • Q1/5

39

QUESTION ID

S2-M-Q1

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-M-Q1

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.a • Set 2 • MAIN • Q1/5

Which table shows a proportional relationship?

A. x: 2, 4, 6 and y: 3, 5, 7

B. x: 1, 2, 3 and y: 5, 6, 7

C. x: 3, 6, 9 and y: 12, 24, 36

D. x: 1, 3, 5 and y: 4, 12, 15

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-M-Q2

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.a • Set 2 • MAIN • Q2/5

40

QUESTION ID

S2-M-Q2

TYPE

Multiple choice

CORRECT ANSWER

C. x: 3, 6, 9 and y: 12, 24, 36

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-M-Q2

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.a • Set 2 • MAIN • Q2/5

The table shows the cost of renting a bike.

Is the cost proportional to the number of hours? Explain.

Hours	1	2	3
Cost ($)	4	9	14

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-M-Q3

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.a • Set 2 • MAIN • Q3/5

41

QUESTION ID

S2-M-Q3

TYPE

Short answer

CORRECT ANSWER

No. The ratios 4/1, 9/2, and 14/3 are not equal.

GRADING NOTE Any explanation showing the cost-to-hours ratios are not all equal.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-M-Q3

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.a • Set 2 • MAIN • Q3/5

Gallons of water and minutes are in a proportional relationship.

Find the two missing values in the table.

Minutes	2	4	6	?
Gallons	5	10	?	25

Missing values: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-M-Q4

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.a • Set 2 • MAIN • Q4/5

42

QUESTION ID

S2-M-Q4

TYPE

Short answer

CORRECT ANSWER

15 gallons for 6 minutes; 10 minutes for 25 gallons

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-M-Q4

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.a • Set 2 • MAIN • Q4/5

True or false?

The table shows a proportional relationship between x and y.

x	1	2	4
y	3	6	10

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-M-Q5

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.a • Set 2 • MAIN • Q5/5

43

QUESTION ID

S2-M-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-M-Q5

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.a • Set 2 • MAIN • Q5/5

SET 2 • 7.RP.A.2.a

BACKWARD 1

Complete a table of equivalent ratios • Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.a

The table shows equivalent ratios.

What is the missing value?

Cups of flour	2	4	6
Cups of milk	3	?	9

Missing value: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-B1-Q1

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.a • Set 2 • BACKWARD 1 • Q1/5

45

QUESTION ID

S2-B1-Q1

TYPE

Short answer

CORRECT ANSWER

6

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-B1-Q1

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.a • Set 2 • BACKWARD 1 • Q1/5

The table shows equivalent ratios.

What is the missing value?

Boys	5	10	15
Girls	4	8	?

Missing value: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-B1-Q2

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.a • Set 2 • BACKWARD 1 • Q2/5

46

QUESTION ID

S2-B1-Q2

TYPE

Short answer

CORRECT ANSWER

12

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-B1-Q2

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.a • Set 2 • BACKWARD 1 • Q2/5

Which ratio is equivalent to 3 : 5?

A. 9 : 15

B. 6 : 8

C. 3 : 15

D. 5 : 3

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-B1-Q3

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.a • Set 2 • BACKWARD 1 • Q3/5

47

QUESTION ID

S2-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. 9 : 15

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-B1-Q3

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.a • Set 2 • BACKWARD 1 • Q3/5

True or false?

2 : 7 and 6 : 21 are equivalent ratios.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-B1-Q4

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.a • Set 2 • BACKWARD 1 • Q4/5

48

QUESTION ID

S2-B1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-B1-Q4

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.a • Set 2 • BACKWARD 1 • Q4/5

True or false?

4 : 6 and 6 : 8 are equivalent ratios.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-B1-Q5

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.a • Set 2 • BACKWARD 1 • Q5/5

49

QUESTION ID

S2-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-B1-Q5

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.a • Set 2 • BACKWARD 1 • Q5/5

SET 2 • 7.RP.A.2.a

BACKWARD 2

Divide multi-digit decimals • Grade 6 • The Number System • 6.NS.B.3

Divide.

17.5 ÷ 5

Quotient: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-B2-Q1

Grade 6 • The Number System • 6.NS.B.3 • Set 2 • BACKWARD 2 • Q1/5

51

QUESTION ID

S2-B2-Q1

TYPE

Short answer

CORRECT ANSWER

3.5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-B2-Q1

Grade 6 • The Number System • 6.NS.B.3 • Set 2 • BACKWARD 2 • Q1/5

Divide.

8.4 ÷ 2.4

Quotient: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-B2-Q2

Grade 6 • The Number System • 6.NS.B.3 • Set 2 • BACKWARD 2 • Q2/5

52

QUESTION ID

S2-B2-Q2

TYPE

Short answer

CORRECT ANSWER

3.5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-B2-Q2

Grade 6 • The Number System • 6.NS.B.3 • Set 2 • BACKWARD 2 • Q2/5

Divide.

9.6 ÷ 3

A. 3.2

B. 32

C. 0.32

D. 3.02

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-B2-Q3

Grade 6 • The Number System • 6.NS.B.3 • Set 2 • BACKWARD 2 • Q3/5

53

QUESTION ID

S2-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. 3.2

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-B2-Q3

Grade 6 • The Number System • 6.NS.B.3 • Set 2 • BACKWARD 2 • Q3/5

True or false?

$28 \div 3.5 = 8$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-B2-Q4

Grade 6 • The Number System • 6.NS.B.3 • Set 2 • BACKWARD 2 • Q4/5

54

QUESTION ID

S2-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-B2-Q4

Grade 6 • The Number System • 6.NS.B.3 • Set 2 • BACKWARD 2 • Q4/5

True or false?

$4.5 \div 1.5 = 0.3$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-B2-Q5

Grade 6 • The Number System • 6.NS.B.3 • Set 2 • BACKWARD 2 • Q5/5

55

QUESTION ID

S2-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-B2-Q5

Grade 6 • The Number System • 6.NS.B.3 • Set 2 • BACKWARD 2 • Q5/5

SET 2 • 7.RP.A.2.a

FORWARD 1

Decide whether a table shows a linear function • Grade 8 • Functions • 8.F.A.3

True or false?

The table shows a linear function.

x	0	1	2	3
y	4	9	14	19

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-F1-Q1

Grade 8 • Functions • 8.F.A.3 • Set 2 • FORWARD 1 • Q1/5

57

QUESTION ID

S2-F1-Q1

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-F1-Q1

Grade 8 • Functions • 8.F.A.3 • Set 2 • FORWARD 1 • Q1/5

Which table shows a linear function that is NOT proportional?

A. x: 0, 1, 2 and y: 1, 2, 4

B. x: 0, 1, 2 and y: 0, 4, 8

C. x: 0, 1, 2 and y: 3, 5, 7

D. x: 0, 1, 2 and y: 0, 1, 4

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-F1-Q2

Grade 8 • Functions • 8.F.A.3 • Set 2 • FORWARD 1 • Q2/5

58

QUESTION ID

S2-F1-Q2

TYPE

Multiple choice

CORRECT ANSWER

C. x: 0, 1, 2 and y: 3, 5, 7

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-F1-Q2

Grade 8 • Functions • 8.F.A.3 • Set 2 • FORWARD 1 • Q2/5

The table shows a linear function.

What is its rate of change?

x	1	2	3	4
y	2	5	8	11

Rate of change: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-F1-Q3

Grade 8 • Functions • 8.F.A.3 • Set 2 • FORWARD 1 • Q3/5

59

QUESTION ID

S2-F1-Q3

TYPE

Short answer

CORRECT ANSWER

3

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-F1-Q3

Grade 8 • Functions • 8.F.A.3 • Set 2 • FORWARD 1 • Q3/5

Is the function in the table linear? Explain.

x	1	2	3	4
y	1	4	9	16

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-F1-Q4

Grade 8 • Functions • 8.F.A.3 • Set 2 • FORWARD 1 • Q4/5

60

QUESTION ID

S2-F1-Q4

TYPE

Short answer

CORRECT ANSWER

No. y increases by 3, then 5, then 7, so the rate of change is not constant.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-F1-Q4

Grade 8 • Functions • 8.F.A.3 • Set 2 • FORWARD 1 • Q4/5

True or false?

A table with x-values 0, 1, 2 and y-values 7, 9, 11 shows a proportional relationship.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-F1-Q5

Grade 8 • Functions • 8.F.A.3 • Set 2 • FORWARD 1 • Q5/5

61

QUESTION ID

S2-F1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE It is linear, but y is not 0 when x is 0.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-F1-Q5

Grade 8 • Functions • 8.F.A.3 • Set 2 • FORWARD 1 • Q5/5

SET 2 • 7.RP.A.2.a

FORWARD 2

Recognize a quantity that changes at a constant rate per unit interval • Grade 9 (High School) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.1.b

True or false?

In the table, y grows by equal differences over equal intervals of x.

x	0	2	4	6
y	5	11	17	23

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-F2-Q1

Grade 9 (High School) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.1.b • Set 2 • FORWARD 2 • Q1/5

63

QUESTION ID

S2-F2-Q1

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-F2-Q1

Grade 9 (High School) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.1.b • Set 2 • FORWARD 2 • Q1/5

A tank holds 80 gallons and drains 6 gallons every minute.

What is the change in the amount of water per minute?

Change per minute: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-F2-Q2

Grade 9 (High School) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.1.b • Set 2 • FORWARD 2 • Q2/5

64

QUESTION ID

S2-F2-Q2

TYPE

Short answer

CORRECT ANSWER

−6 gallons per minute

GRADING NOTE "It decreases by 6 gallons each minute" is also correct.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-F2-Q2

Grade 9 (High School) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.1.b • Set 2 • FORWARD 2 • Q2/5

Which quantity changes at a constant rate per unit interval?

A. A price increases 5% each year.

B. A population doubles every year.

C. A bacteria count triples each hour.

D. A plant grows 2 cm every week.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-F2-Q3

Grade 9 (High School) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.1.b • Set 2 • FORWARD 2 • Q3/5

65

QUESTION ID

S2-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. A plant grows 2 cm every week.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-F2-Q3

Grade 9 (High School) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.1.b • Set 2 • FORWARD 2 • Q3/5

The table shows a linear function.

By how much does y change for each increase of 1 in x?

x	0	3	6	9
y	10	16	22	28

Change in y: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-F2-Q4

Grade 9 (High School) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.1.b • Set 2 • FORWARD 2 • Q4/5

66

QUESTION ID

S2-F2-Q4

TYPE

Short answer

CORRECT ANSWER

2

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-F2-Q4

Grade 9 (High School) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.1.b • Set 2 • FORWARD 2 • Q4/5

True or false?

An account that earns 4% of its current balance each year grows by the same number of dollars each year.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-F2-Q5

Grade 9 (High School) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.1.b • Set 2 • FORWARD 2 • Q5/5

67

QUESTION ID

S2-F2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-F2-Q5

Grade 9 (High School) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.1.b • Set 2 • FORWARD 2 • Q5/5

SET 3 • 7.RP.A.2.a • RATIOS & PROPORTIONAL RELATIONSHIPS

Decide whether a graph shows a proportional relationship

MAIN	Grade 7 main question	**7.RP.A.2.a**
BACKWARD 1	Plot the pairs from a ratio table	**6.RP.A.3.a**
BACKWARD 2	Find the coordinates of points in all four quadrants	**6.NS.C.6.c**
FORWARD 1	Recognize linear functions whose graphs do not pass through the…	**8.F.A.3**
FORWARD 2	Interpret the intercepts of a graph in context	**HSF-IF.B.4**

SET 3 • 7.RP.A.2.a

MAIN

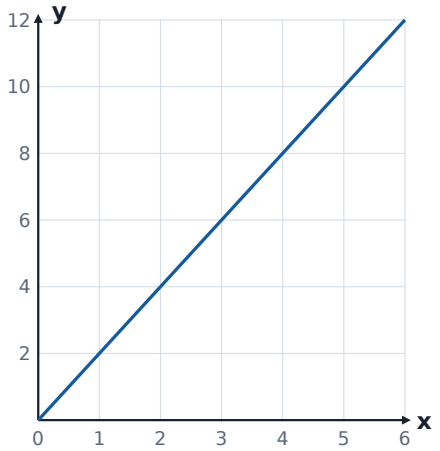
Grade 7 main question • Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.a

True or false?

The graph shows a proportional relationship between x and y.

A. True

B. False



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-M-Q1

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.a • Set 3 • MAIN • Q1/5

70

QUESTION ID

S3-M-Q1

TYPE

True or false

CORRECT ANSWER

A. True

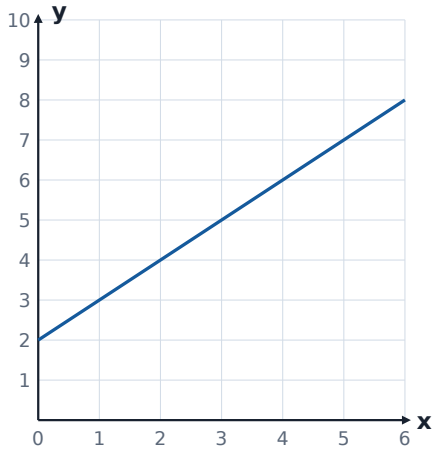
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ID S3-M-Q1

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.a • Set 3 • MAIN • Q1/5

Is the relationship in the graph proportional? Explain.

Answer: _____



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-M-Q2

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.a • Set 3 • MAIN • Q2/5

71

QUESTION ID

S3-M-Q2

TYPE

Short answer

CORRECT ANSWER

No. The line does not pass through the origin (0, 0).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-M-Q2

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.a • Set 3 • MAIN • Q2/5

Which graph shows a proportional relationship?

A. A straight line that crosses the y-axis at 3

B. Points that do not lie on one line

C. A curve through the origin

D. A straight line through the origin

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-M-Q3

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.a • Set 3 • MAIN • Q3/5

72

QUESTION ID

S3-M-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. A straight line through the origin

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

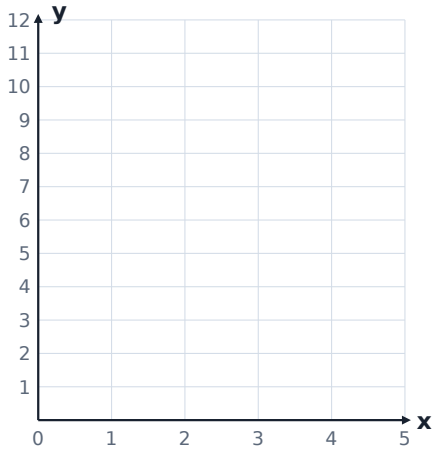
ID S3-M-Q3

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.a • Set 3 • MAIN • Q3/5

Plot the points (1, 2.5), (2, 5), and (4, 10).

Do the points show a proportional relationship? Explain.

Answer: _____



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-M-Q4

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.a • Set 3 • MAIN • Q4/5

73

QUESTION ID

S3-M-Q4

TYPE

Drawing + written answer

CORRECT ANSWER

Drawing: Points at (1, 2.5), (2, 5), and (4, 10)

Written: Yes. The points lie on a straight line through the origin; y ÷ x = 2.5 for every point.

GRADING NOTE Grade both: the three points plotted correctly, and "yes" with a reason (same ratio, or a line through the origin).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-M-Q4

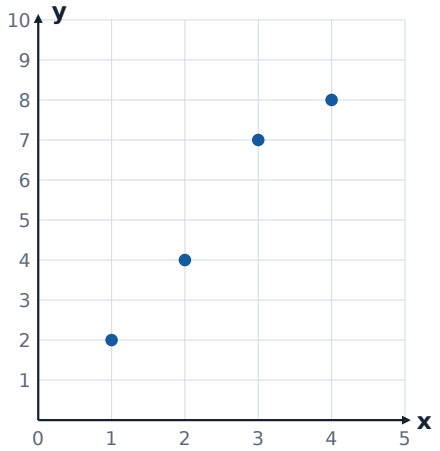
Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.a • Set 3 • MAIN • Q4/5

True or false?

The points in the graph show a proportional relationship.

A. True

B. False



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-M-Q5

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.a • Set 3 • MAIN • Q5/5

74

QUESTION ID

S3-M-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-M-Q5

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.a • Set 3 • MAIN • Q5/5

SET 3 • 7.RP.A.2.a

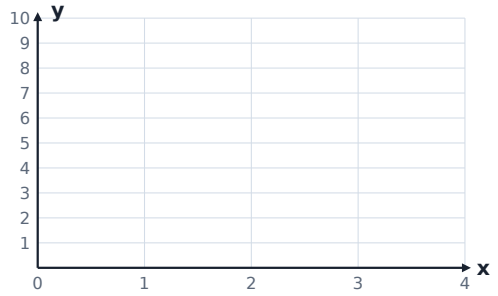
BACKWARD 1

Plot the pairs from a ratio table • Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.a

The table shows equivalent ratios.

Plot each pair (x, y) on the grid.

x	1	2	3
y	3	6	9



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-B1-Q1

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.a • Set 3 • BACKWARD 1 • Q1/5

76

QUESTION ID

S3-B1-Q1

TYPE

Drawing / plotting

CORRECT ANSWER

Points at (1, 3), (2, 6), and (3, 9)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-B1-Q1

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.a • Set 3 • BACKWARD 1 • Q1/5

A ratio table has the pairs (2, 5), (4, 10), and (6, ?).

What is the missing y-value?

y = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-B1-Q2

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.a • Set 3 • BACKWARD 1 • Q2/5

77

QUESTION ID

S3-B1-Q2

TYPE

Short answer

CORRECT ANSWER

15

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-B1-Q2

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.a • Set 3 • BACKWARD 1 • Q2/5

Which pair belongs in the same ratio table as (3, 4)?

A. (4, 5)

B. (9, 10)

C. (6, 7)

D. (6, 8)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-B1-Q3

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.a • Set 3 • BACKWARD 1 • Q3/5

78

QUESTION ID

S3-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. (6, 8)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-B1-Q3

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.a • Set 3 • BACKWARD 1 • Q3/5

True or false?

The pairs (2, 6), (4, 12), and (5, 15) can all come from the same ratio table.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-B1-Q4

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.a • Set 3 • BACKWARD 1 • Q4/5

79

QUESTION ID

S3-B1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-B1-Q4

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.a • Set 3 • BACKWARD 1 • Q4/5

True or false?

To plot the pair (3, 7), you start at the origin, move 7 units right, and then move 3 units up.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-B1-Q5

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.a • Set 3 • BACKWARD 1 • Q5/5

80

QUESTION ID

S3-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-B1-Q5

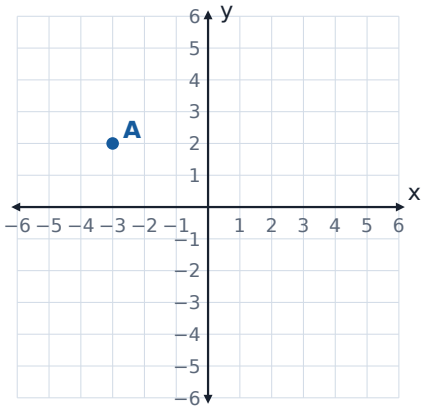
Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.a • Set 3 • BACKWARD 1 • Q5/5

SET 3 • 7.RP.A.2.a

BACKWARD 2

Find the coordinates of points in all four quadrants • Grade 6 • The Number System • 6.NS.C.6.c

What are the coordinates of point A?



Coordinates: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-B2-Q1

Grade 6 • The Number System • 6.NS.C.6.c • Set 3 • BACKWARD 2 • Q1/5

82

QUESTION ID

S3-B2-Q1

TYPE

Short answer

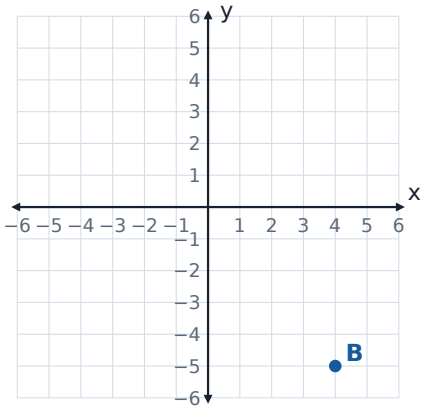
CORRECT ANSWER

(−3, 2)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-B2-Q1

Grade 6 • The Number System • 6.NS.C.6.c • Set 3 • BACKWARD 2 • Q1/5

What are the coordinates of point B?



Coordinates: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-B2-Q2

Grade 6 • The Number System • 6.NS.C.6.c • Set 3 • BACKWARD 2 • Q2/5

83

QUESTION ID

S3-B2-Q2

TYPE

Short answer

CORRECT ANSWER

(4, −5)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-B2-Q2

Grade 6 • The Number System • 6.NS.C.6.c • Set 3 • BACKWARD 2 • Q2/5

Which point is at the origin?

A. (1, 1)

B. (0, 1)

C. (1, 0)

D. (0, 0)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-B2-Q3

Grade 6 • The Number System • 6.NS.C.6.c • Set 3 • BACKWARD 2 • Q3/5

84

QUESTION ID

S3-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. (0, 0)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-B2-Q3

Grade 6 • The Number System • 6.NS.C.6.c • Set 3 • BACKWARD 2 • Q3/5

True or false?

The point (3, 0) lies on the x-axis.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-B2-Q4

Grade 6 • The Number System • 6.NS.C.6.c • Set 3 • BACKWARD 2 • Q4/5

85

QUESTION ID

S3-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-B2-Q4

Grade 6 • The Number System • 6.NS.C.6.c • Set 3 • BACKWARD 2 • Q4/5

True or false?

The point (0, 5) lies on the x-axis.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-B2-Q5

Grade 6 • The Number System • 6.NS.C.6.c • Set 3 • BACKWARD 2 • Q5/5

86

QUESTION ID

S3-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-B2-Q5

Grade 6 • The Number System • 6.NS.C.6.c • Set 3 • BACKWARD 2 • Q5/5

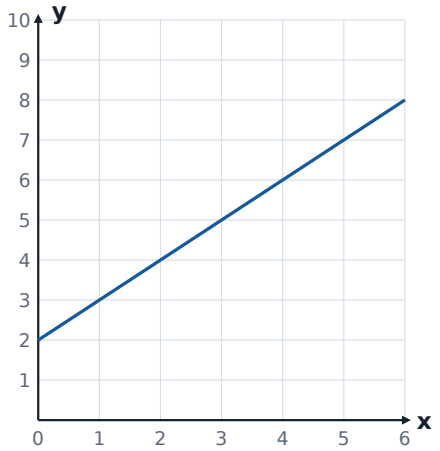
SET 3 • 7.RP.A.2.a

FORWARD 1

Recognize linear functions whose graphs do not pass through the origin • Grade 8 • Functions • 8.F.A.3

The graph shows a line.

Write its equation in the form $y = mx + b$.



Equation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-F1-Q1

Grade 8 • Functions • 8.F.A.3 • Set 3 • FORWARD 1 • Q1/5

88

QUESTION ID

S3-F1-Q1

TYPE

Short answer

CORRECT ANSWER

y = x + 2

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-F1-Q1

Grade 8 • Functions • 8.F.A.3 • Set 3 • FORWARD 1 • Q1/5

Which equation is linear but NOT proportional?

A. $y = x^2$

B. $y = \frac{3}{x}$

C. $y = 3x$

D. $y = 3x + 1$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-F1-Q2

Grade 8 • Functions • 8.F.A.3 • Set 3 • FORWARD 1 • Q2/5

89

QUESTION ID

S3-F1-Q2

TYPE

Multiple choice

CORRECT ANSWER

D. y = 3x + 1

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-F1-Q2

Grade 8 • Functions • 8.F.A.3 • Set 3 • FORWARD 1 • Q2/5

True or false?

The graph of y = 4x is a line through the origin.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-F1-Q3

Grade 8 • Functions • 8.F.A.3 • Set 3 • FORWARD 1 • Q3/5

90

QUESTION ID

S3-F1-Q3

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-F1-Q3

Grade 8 • Functions • 8.F.A.3 • Set 3 • FORWARD 1 • Q3/5

A line passes through (0, 3) and (2, 7).

Is the relationship proportional? Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-F1-Q4

Grade 8 • Functions • 8.F.A.3 • Set 3 • FORWARD 1 • Q4/5

91

QUESTION ID

S3-F1-Q4

TYPE

Short answer

CORRECT ANSWER

No. The line is linear, but it crosses the y-axis at 3, not at the origin.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-F1-Q4

Grade 8 • Functions • 8.F.A.3 • Set 3 • FORWARD 1 • Q4/5

True or false?

Every linear function is a proportional relationship.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-F1-Q5

Grade 8 • Functions • 8.F.A.3 • Set 3 • FORWARD 1 • Q5/5

92

QUESTION ID

S3-F1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-F1-Q5

Grade 8 • Functions • 8.F.A.3 • Set 3 • FORWARD 1 • Q5/5

SET 3 • 7.RP.A.2.a

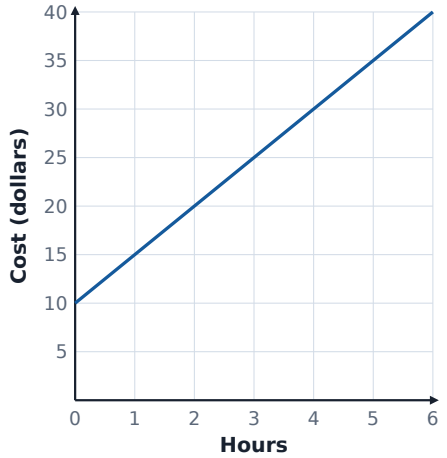
FORWARD 2

Interpret the intercepts of a graph in context • Grade 9 (High School) • Functions: Interpreting Functions • HSF-IF.B.4

The graph shows the cost of renting a kayak.

What does the point where the line meets the vertical axis mean?

Meaning: _____



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-F2-Q1

Grade 9 (High School) • Functions: Interpreting Functions • HSF-IF.B.4 • Set 3 • FORWARD 2 • Q1/5

94

QUESTION ID

S3-F2-Q1

TYPE

Short answer

CORRECT ANSWER

There is a $10 cost at 0 hours (a starting fee).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-F2-Q1

Grade 9 (High School) • Functions: Interpreting Functions • HSF-IF.B.4 • Set 3 • FORWARD 2 • Q1/5

The height of a candle after t hours is h = 24 − 3t centimeters.

After how many hours does the candle burn out (h = 0)?

Hours: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-F2-Q2

Grade 9 (High School) • Functions: Interpreting Functions • HSF-IF.B.4 • Set 3 • FORWARD 2 • Q2/5

95

QUESTION ID

S3-F2-Q2

TYPE

Short answer

CORRECT ANSWER

8

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-F2-Q2

Grade 9 (High School) • Functions: Interpreting Functions • HSF-IF.B.4 • Set 3 • FORWARD 2 • Q2/5

The water in a tank after m minutes is w = 50 − 5m gallons.

What does the w-intercept, 50, mean?

A. The tank starts with 50 gallons.

B. The tank drains 50 gallons per minute.

C. The tank is empty after 50 minutes.

D. The tank drains 5 gallons in 50 minutes.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-F2-Q3

Grade 9 (High School) • Functions: Interpreting Functions • HSF-IF.B.4 • Set 3 • FORWARD 2 • Q3/5

96

QUESTION ID

S3-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. The tank starts with 50 gallons.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-F2-Q3

Grade 9 (High School) • Functions: Interpreting Functions • HSF-IF.B.4 • Set 3 • FORWARD 2 • Q3/5

True or false?

The cost of a rental is C = 15 + 2h dollars for h hours. The C-intercept, 15, is the cost when h = 0.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-F2-Q4

Grade 9 (High School) • Functions: Interpreting Functions • HSF-IF.B.4 • Set 3 • FORWARD 2 • Q4/5

97

QUESTION ID

S3-F2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-F2-Q4

Grade 9 (High School) • Functions: Interpreting Functions • HSF-IF.B.4 • Set 3 • FORWARD 2 • Q4/5

True or false?

For h = 24 − 3t, the h-intercept is 3.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-F2-Q5

Grade 9 (High School) • Functions: Interpreting Functions • HSF-IF.B.4 • Set 3 • FORWARD 2 • Q5/5

98

QUESTION ID

S3-F2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE The h-intercept is 24.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-F2-Q5

Grade 9 (High School) • Functions: Interpreting Functions • HSF-IF.B.4 • Set 3 • FORWARD 2 • Q5/5

SET 4 • 7.RP.A.2.b • RATIOS & PROPORTIONAL RELATIONSHIPS

Identify the constant of proportionality in tables, graphs, equations, and descriptions

MAIN	Grade 7 main question	**7.RP.A.2.b**
BACKWARD 1	Describe a unit rate	**6.RP.A.2**
BACKWARD 2	Write an equation from a table of values	**6.EE.C.9**
FORWARD 1	Interpret the unit rate as the slope of a graph	**8.EE.B.5**
FORWARD 2	Write a linear function from two points or a description	**HSF-LE.A.2**

SET 4 • 7.RP.A.2.b

MAIN

Grade 7 main question • Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.b

The table shows a proportional relationship.

What is the constant of proportionality (y per x)?

x	2	4	6
y	5	10	15

k = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-M-Q1

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.b • Set 4 • MAIN • Q1/5

101

QUESTION ID

S4-M-Q1

TYPE

Short answer

CORRECT ANSWER

2.5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

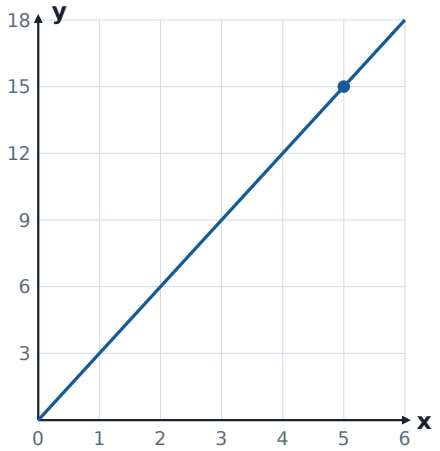
ID S4-M-Q1

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.b • Set 4 • MAIN • Q1/5

The graph shows a proportional relationship.

What is the constant of proportionality?

k = _____



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-M-Q2

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.b • Set 4 • MAIN • Q2/5

102

QUESTION ID

S4-M-Q2

TYPE

Short answer

CORRECT ANSWER

3

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-M-Q2

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.b • Set 4 • MAIN • Q2/5

What is the constant of proportionality in $y = \frac{2}{3}x$?

A. 2

B. $\frac{3}{2}$

C. 3

D. $\frac{2}{3}$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-M-Q3

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.b • Set 4 • MAIN • Q3/5

QUESTION ID

S4-M-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. 2/3

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-M-Q3

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.b • Set 4 • MAIN • Q3/5

True or false?

6 pencils cost $2.10. The constant of proportionality, in dollars per pencil, is 0.35.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-M-Q4

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.b • Set 4 • MAIN • Q4/5

104

QUESTION ID

S4-M-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-M-Q4

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.b • Set 4 • MAIN • Q4/5

Kim types 216 words in 4 minutes at a steady rate.

What is the constant of proportionality in words per minute?

k = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-M-Q5

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.b • Set 4 • MAIN • Q5/5

105

QUESTION ID

S4-M-Q5

TYPE

Short answer

CORRECT ANSWER

54

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-M-Q5

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.b • Set 4 • MAIN • Q5/5

SET 4 • 7.RP.A.2.b

BACKWARD 1

Describe a unit rate • Grade 6 • Ratios & Proportional Relationships • 6.RP.A.2

A pack of 8 juice boxes costs $6.

What is the cost per juice box?

Cost: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-B1-Q1

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.2 • Set 4 • BACKWARD 1 • Q1/5

107

QUESTION ID

S4-B1-Q1

TYPE

Short answer

CORRECT ANSWER

$0.75

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-B1-Q1

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.2 • Set 4 • BACKWARD 1 • Q1/5

A recipe uses 4 cups of flour for 2 loaves.

Which is the unit rate?

A. 6 cups of flour per loaf

B. 8 cups of flour per loaf

C. $\frac{1}{2}$ cup of flour per loaf

D. 2 cups of flour per loaf

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-B1-Q2

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.2 • Set 4 • BACKWARD 1 • Q2/5

108

QUESTION ID

S4-B1-Q2

TYPE

Multiple choice

CORRECT ANSWER

D. 2 cups of flour per loaf

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-B1-Q2

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.2 • Set 4 • BACKWARD 1 • Q2/5

Lena swims 600 meters in 12 minutes at a steady pace.

How many meters does she swim per minute?

Meters per minute: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-B1-Q3

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.2 • Set 4 • BACKWARD 1 • Q3/5

109

QUESTION ID

S4-B1-Q3

TYPE

Short answer

CORRECT ANSWER

50

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-B1-Q3

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.2 • Set 4 • BACKWARD 1 • Q3/5

True or false?

"$3 per pound" is a unit rate.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-B1-Q4

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.2 • Set 4 • BACKWARD 1 • Q4/5

110

QUESTION ID

S4-B1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-B1-Q4

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.2 • Set 4 • BACKWARD 1 • Q4/5

True or false?

5 apples cost $4. The unit rate is $1.25 per apple.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-B1-Q5

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.2 • Set 4 • BACKWARD 1 • Q5/5

111

QUESTION ID

S4-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE The unit rate is $0.80 per apple.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-B1-Q5

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.2 • Set 4 • BACKWARD 1 • Q5/5

SET 4 • 7.RP.A.2.b

BACKWARD 2

Write an equation from a table of values • Grade 6 • Expressions & Equations • 6.EE.C.9

In the table, y is always 4 times x.

Write an equation for y in terms of x.

x	1	2	3
y	4	8	12

Equation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-B2-Q1

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 4 • BACKWARD 2 • Q1/5

113

QUESTION ID

S4-B2-Q1

TYPE

Short answer

CORRECT ANSWER

y = 4x

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-B2-Q1

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 4 • BACKWARD 2 • Q1/5

Each ticket costs $7.

Which equation gives the cost c of t tickets?

A. $t = 7c$

B. $c = \frac{t}{7}$

C. $c = 7t$

D. $c = t + 7$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-B2-Q2

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 4 • BACKWARD 2 • Q2/5

114

QUESTION ID

S4-B2-Q2

TYPE

Multiple choice

CORRECT ANSWER

C. c = 7t

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-B2-Q2

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 4 • BACKWARD 2 • Q2/5

Write an equation for y in terms of x.

x	2	3	5
y	10	15	25

Equation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-B2-Q3

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 4 • BACKWARD 2 • Q3/5

115

QUESTION ID

S4-B2-Q3

TYPE

Short answer

CORRECT ANSWER

y = 5x

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-B2-Q3

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 4 • BACKWARD 2 • Q3/5

True or false?

For y = 6x, when x = 3, y = 18.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-B2-Q4

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 4 • BACKWARD 2 • Q4/5

116

QUESTION ID

S4-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-B2-Q4

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 4 • BACKWARD 2 • Q4/5

True or false?

In d = 50t, d is the independent variable.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-B2-Q5

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 4 • BACKWARD 2 • Q5/5

117

QUESTION ID

S4-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE t is the independent variable; d depends on t.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-B2-Q5

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 4 • BACKWARD 2 • Q5/5

SET 4 • 7.RP.A.2.b

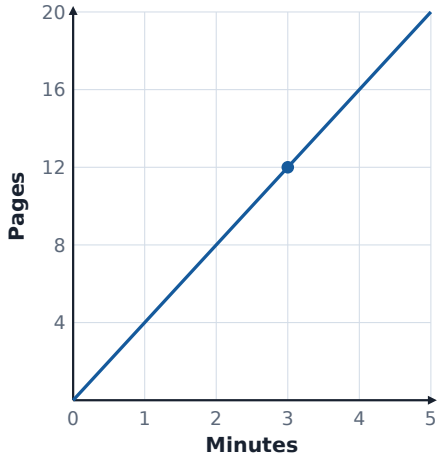
FORWARD 1

Interpret the unit rate as the slope of a graph • Grade 8 • Expressions & Equations • 8.EE.B.5

The graph shows a proportional relationship between minutes and pages read.

What is the slope of the line?

Slope: _____



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-F1-Q1

Grade 8 • Expressions & Equations • 8.EE.B.5 • Set 4 • FORWARD 1 • Q1/5

119

QUESTION ID

S4-F1-Q1

TYPE

Short answer

CORRECT ANSWER

4

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-F1-Q1

Grade 8 • Expressions & Equations • 8.EE.B.5 • Set 4 • FORWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

A line passes through (0, 0) and (8, 6).

What is its slope?

Slope: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-F1-Q2

Grade 8 • Expressions & Equations • 8.EE.B.5 • Set 4 • FORWARD 1 • Q2/5

120

QUESTION ID

S4-F1-Q2

TYPE

Short answer

CORRECT ANSWER

3/4

GRADING NOTE 0.75 is also correct.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-F1-Q2

Grade 8 • Expressions & Equations • 8.EE.B.5 • Set 4 • FORWARD 1 • Q2/5

The graph of a proportional relationship has a slope of 1.5 liters per minute.

Which point is on the graph?

A. (4, 6)

B. (6, 4)

C. (1.5, 1)

D. (4, 5.5)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-F1-Q3

Grade 8 • Expressions & Equations • 8.EE.B.5 • Set 4 • FORWARD 1 • Q3/5

121

QUESTION ID

S4-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. (4, 6)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-F1-Q3

Grade 8 • Expressions & Equations • 8.EE.B.5 • Set 4 • FORWARD 1 • Q3/5

True or false?

The slope of the graph of y = 0.8x is 0.8.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-F1-Q4

Grade 8 • Expressions & Equations • 8.EE.B.5 • Set 4 • FORWARD 1 • Q4/5

122

QUESTION ID

S4-F1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-F1-Q4

Grade 8 • Expressions & Equations • 8.EE.B.5 • Set 4 • FORWARD 1 • Q4/5

True or false?

A line through the origin and (2, 9) has a slope of $\frac{2}{9}$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-F1-Q5

Grade 8 • Expressions & Equations • 8.EE.B.5 • Set 4 • FORWARD 1 • Q5/5

123

QUESTION ID

S4-F1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE The slope is 9/2.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-F1-Q5

Grade 8 • Expressions & Equations • 8.EE.B.5 • Set 4 • FORWARD 1 • Q5/5

SET 4 • 7.RP.A.2.b

FORWARD 2

Write a linear function from two points or a description • Grade 9 (High School) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.2

A linear function passes through (1, 5) and (3, 11).

Write the function.

f(x) = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-F2-Q1

Grade 9 (High School) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.2 • Set 4 • FORWARD 2 • Q1/5

125

QUESTION ID

S4-F2-Q1

TYPE

Short answer

CORRECT ANSWER

3x + 2

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-F2-Q1

Grade 9 (High School) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.2 • Set 4 • FORWARD 2 • Q1/5

A gym charges a $20 joining fee plus $15 each month.

Write a function for the total cost C after m months.

C(m) = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-F2-Q2

Grade 9 (High School) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.2 • Set 4 • FORWARD 2 • Q2/5

126

QUESTION ID

S4-F2-Q2

TYPE

Short answer

CORRECT ANSWER

15m + 20

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S4-F2-Q2**

Grade 9 (High School) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.2 • Set 4 • FORWARD 2 • Q2/5

Which linear function passes through (0, −4) and (2, 2)?

A. f(x) = 3x + 2

B. f(x) = −4x + 3

C. f(x) = 3x − 4

D. f(x) = 2x − 4

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-F2-Q3

Grade 9 (High School) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.2 • Set 4 • FORWARD 2 • Q3/5

127

QUESTION ID

S4-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. f(x) = 3x − 4

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-F2-Q3

Grade 9 (High School) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.2 • Set 4 • FORWARD 2 • Q3/5

True or false?

The linear function through (2, 7) and (4, 11) is f(x) = 2x + 3.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-F2-Q4

Grade 9 (High School) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.2 • Set 4 • FORWARD 2 • Q4/5

128

QUESTION ID

S4-F2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-F2-Q4

Grade 9 (High School) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.2 • Set 4 • FORWARD 2 • Q4/5

True or false?

The linear function through (0, 1) and (5, 11) is f(x) = 5x + 1.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-F2-Q5

Grade 9 (High School) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.2 • Set 4 • FORWARD 2 • Q5/5

129

QUESTION ID

S4-F2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE The slope is 2, so f(x) = 2x + 1.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-F2-Q5

Grade 9 (High School) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.2 • Set 4 • FORWARD 2 • Q5/5

Represent a proportional relationship with an equation

MAIN	Grade 7 main question	**7.RP.A.2.c**
BACKWARD 1	Write an equation for a dependent variable	**6.EE.C.9**
BACKWARD 2	Solve unit rate problems	**6.RP.A.3.b**
FORWARD 1	Construct a linear function y = mx + b from a situation	**8.F.B.4**
FORWARD 2	Write an equation in two variables for a situation	**HSA-CED.A.2**

SET 5 • 7.RP.A.2.c

MAIN

Grade 7 main question • Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.c

Peaches cost $2.40 per pound.

Write an equation for the cost c of p pounds.

Equation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-M-Q1

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.c • Set 5 • MAIN • Q1/5

132

QUESTION ID

S5-M-Q1

TYPE

Short answer

CORRECT ANSWER

c = 2.4p

GRADING NOTE c = 2.40p is also correct.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-M-Q1

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.c • Set 5 • MAIN • Q1/5

A car travels 29 miles per gallon of gas.

Which equation gives the miles m for g gallons?

A. $m = 29g$

B. $m = g + 29$

C. $g = 29m$

D. $m = \frac{g}{29}$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-M-Q2

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.c • Set 5 • MAIN • Q2/5

133

QUESTION ID

S5-M-Q2

TYPE

Multiple choice

CORRECT ANSWER

A. m = 29g

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-M-Q2

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.c • Set 5 • MAIN • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

The table shows a proportional relationship.

Write an equation for y in terms of x.

x	4	8	12
y	3	6	9

Equation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-M-Q3

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.c • Set 5 • MAIN • Q3/5

134

QUESTION ID

S5-M-Q3

TYPE

Short answer

CORRECT ANSWER

y = 0.75x

GRADING NOTE y = (3/4)x is also correct.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-M-Q3

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.c • Set 5 • MAIN • Q3/5

True or false?

Notebooks cost $1.75 each. The equation c = 1.75n gives the cost c of n notebooks.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-M-Q4

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.c • Set 5 • MAIN • Q4/5

135

QUESTION ID

S5-M-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-M-Q4

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.c • Set 5 • MAIN • Q4/5

True or false?

A printer prints 12 pages every 3 minutes. The equation p = 12m gives the pages p printed in m minutes.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-M-Q5

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.c • Set 5 • MAIN • Q5/5

136

QUESTION ID

S5-M-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE The equation is p = 4m.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-M-Q5

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.c • Set 5 • MAIN • Q5/5

SET 5 • 7.RP.A.2.c

BACKWARD 1

Write an equation for a dependent variable • Grade 6 • Expressions & Equations • 6.EE.C.9

A worker earns $18 per hour.

Write an equation for the earnings e after h hours.

Equation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-B1-Q1

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 5 • BACKWARD 1 • Q1/5

138

QUESTION ID

S5-B1-Q1

TYPE

Short answer

CORRECT ANSWER

e = 18h

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-B1-Q1

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 5 • BACKWARD 1 • Q1/5

Which equation matches the table?

x	1	2	3
y	9	18	27

A. y = x + 8

B. y = 3x

C. x = 9y

D. y = 9x

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-B1-Q2

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 5 • BACKWARD 1 • Q2/5

139

QUESTION ID

S5-B1-Q2

TYPE

Multiple choice

CORRECT ANSWER

D. y = 9x

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-B1-Q2

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 5 • BACKWARD 1 • Q2/5

Water fills a tank at 6 liters per minute.

Write an equation for the liters L after t minutes.

Equation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-B1-Q3

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 5 • BACKWARD 1 • Q3/5

140

QUESTION ID

S5-B1-Q3

TYPE

Short answer

CORRECT ANSWER

L = 6t

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-B1-Q3

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 5 • BACKWARD 1 • Q3/5

True or false?

In e = 18h, the value of e depends on the value of h.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-B1-Q4

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 5 • BACKWARD 1 • Q4/5

141

QUESTION ID

S5-B1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-B1-Q4

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 5 • BACKWARD 1 • Q4/5

True or false?

The point (2, 7) is on the graph of y = 5x.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-B1-Q5

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 5 • BACKWARD 1 • Q5/5

142

QUESTION ID

S5-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-B1-Q5

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 5 • BACKWARD 1 • Q5/5

SET 5 • 7.RP.A.2.c

BACKWARD 2

Solve unit rate problems • Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.b

5 bags of rice cost $15.

At the same rate, how much do 8 bags cost?

Cost: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-B2-Q1

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.b • Set 5 • BACKWARD 2 • Q1/5

144

QUESTION ID

S5-B2-Q1

TYPE

Short answer

CORRECT ANSWER

$24

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-B2-Q1

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.b • Set 5 • BACKWARD 2 • Q1/5

A machine makes 140 parts in 7 hours.

At the same rate, how many parts does it make in 10 hours?

Parts: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-B2-Q2

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.b • Set 5 • BACKWARD 2 • Q2/5

145

QUESTION ID

S5-B2-Q2

TYPE

Short answer

CORRECT ANSWER

200

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-B2-Q2

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.b • Set 5 • BACKWARD 2 • Q2/5

A bus travels at 45 miles per hour.

How long does it take to travel 135 miles?

A. 3 hours

B. 2 hours

C. 4 hours

D. 6 hours

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-B2-Q3

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.b • Set 5 • BACKWARD 2 • Q3/5

146

QUESTION ID

S5-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. 3 hours

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-B2-Q3

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.b • Set 5 • BACKWARD 2 • Q3/5

True or false?

At $4 per pound, 6 pounds cost $24.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-B2-Q4

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.b • Set 5 • BACKWARD 2 • Q4/5

147

QUESTION ID

S5-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-B2-Q4

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.b • Set 5 • BACKWARD 2 • Q4/5

True or false?

A printer prints 12 pages per minute. It takes 6 minutes to print 60 pages.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-B2-Q5

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.b • Set 5 • BACKWARD 2 • Q5/5

148

QUESTION ID

S5-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE It takes 5 minutes.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-B2-Q5

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.b • Set 5 • BACKWARD 2 • Q5/5

SET 5 • 7.RP.A.2.c

FORWARD 1

Construct a linear function y = mx + b from a situation • Grade 8 • Functions • 8.F.B.4

A plumber charges $60 for a visit plus $45 per hour.

Write an equation for the cost y after x hours.

Equation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-F1-Q1

Grade 8 • Functions • 8.F.B.4 • Set 5 • FORWARD 1 • Q1/5

150

QUESTION ID

S5-F1-Q1

TYPE

Short answer

CORRECT ANSWER

y = 45x + 60

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-F1-Q1

Grade 8 • Functions • 8.F.B.4 • Set 5 • FORWARD 1 • Q1/5

The table shows a linear function.

Write its equation in the form y = mx + b.

x	0	1	2
y	7	10	13

Equation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-F1-Q2

Grade 8 • Functions • 8.F.B.4 • Set 5 • FORWARD 1 • Q2/5

151

QUESTION ID

S5-F1-Q2

TYPE

Short answer

CORRECT ANSWER

y = 3x + 7

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-F1-Q2

Grade 8 • Functions • 8.F.B.4 • Set 5 • FORWARD 1 • Q2/5

A line has a slope of 2 and passes through (0, −5).

Which is its equation?

A. y = −5x + 2

B. y = −2x − 5

C. y = 2x − 5

D. y = 2x + 5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-F1-Q3

Grade 8 • Functions • 8.F.B.4 • Set 5 • FORWARD 1 • Q3/5

152

QUESTION ID

S5-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. y = 2x − 5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-F1-Q3

Grade 8 • Functions • 8.F.B.4 • Set 5 • FORWARD 1 • Q3/5

True or false?

A pool has 300 gallons and fills at 20 gallons per minute. The equation y = 20x + 300 gives the gallons y after x minutes.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-F1-Q4

Grade 8 • Functions • 8.F.B.4 • Set 5 • FORWARD 1 • Q4/5

153

QUESTION ID

S5-F1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-F1-Q4

Grade 8 • Functions • 8.F.B.4 • Set 5 • FORWARD 1 • Q4/5

True or false?

A candle 15 cm tall burns 2 cm per hour. Its height after x hours is y = 15x − 2.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-F1-Q5

Grade 8 • Functions • 8.F.B.4 • Set 5 • FORWARD 1 • Q5/5

154

QUESTION ID

S5-F1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE The height is y = −2x + 15.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-F1-Q5

Grade 8 • Functions • 8.F.B.4 • Set 5 • FORWARD 1 • Q5/5

SET 5 • 7.RP.A.2.c

FORWARD 2

Write an equation in two variables for a situation • Grade 9 (High School) • Algebra: Creating Equations • HSA-CED.A.2

Pens cost $2 each and markers cost $3 each. Kai spends exactly $30.

Write an equation using p for pens and m for markers.

Equation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-F2-Q1

Grade 9 (High School) • Algebra: Creating Equations • HSA-CED.A.2 • Set 5 • FORWARD 2 • Q1/5

156

QUESTION ID

S5-F2-Q1

TYPE

Short answer

CORRECT ANSWER

2p + 3m = 30

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-F2-Q1

Grade 9 (High School) • Algebra: Creating Equations • HSA-CED.A.2 • Set 5 • FORWARD 2 • Q1/5

Adult tickets cost $12 and child tickets cost $8. A group pays $96.

Write an equation using a for adult tickets and c for child tickets.

Equation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-F2-Q2

Grade 9 (High School) • Algebra: Creating Equations • HSA-CED.A.2 • Set 5 • FORWARD 2 • Q2/5

157

QUESTION ID

S5-F2-Q2

TYPE

Short answer

CORRECT ANSWER

12a + 8c = 96

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-F2-Q2

Grade 9 (High School) • Algebra: Creating Equations • HSA-CED.A.2 • Set 5 • FORWARD 2 • Q2/5

A farm has 40 animals. Every animal is a chicken or a goat.

Which equation relates the number of chickens c and goats g?

A. cg = 40

B. c + g = 40

C. 2c + 4g = 40

D. c = 40g

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-F2-Q3

Grade 9 (High School) • Algebra: Creating Equations • HSA-CED.A.2 • Set 5 • FORWARD 2 • Q3/5

158

QUESTION ID

S5-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. c + g = 40

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-F2-Q3

Grade 9 (High School) • Algebra: Creating Equations • HSA-CED.A.2 • Set 5 • FORWARD 2 • Q3/5

True or false?

The equation 5x + 2y = 50 can describe buying x items at $5 each and y items at $2 each for a total of $50.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-F2-Q4

Grade 9 (High School) • Algebra: Creating Equations • HSA-CED.A.2 • Set 5 • FORWARD 2 • Q4/5

159

QUESTION ID

S5-F2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-F2-Q4

Grade 9 (High School) • Algebra: Creating Equations • HSA-CED.A.2 • Set 5 • FORWARD 2 • Q4/5

True or false?

p = 6 and m = 4 is a solution of 2p + 3m = 30.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-F2-Q5

Grade 9 (High School) • Algebra: Creating Equations • HSA-CED.A.2 • Set 5 • FORWARD 2 • Q5/5

160

QUESTION ID

S5-F2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-F2-Q5

Grade 9 (High School) • Algebra: Creating Equations • HSA-CED.A.2 • Set 5 • FORWARD 2 • Q5/5

Explain what a point on the graph of a proportional relationship means

MAIN	Grade 7 main question	**7.RP.A.2.d**
BACKWARD 1	Explain a pair in a table of equivalent ratios	**6.RP.A.3.a**
BACKWARD 2	Read values from the graph of an equation	**6.EE.C.9**
FORWARD 1	Interpret the rate of change and initial value of a linear function	**8.F.B.4**
FORWARD 2	Calculate the average rate of change over an interval	**HSF-IF.B.6**

SET 6 • 7.RP.A.2.d

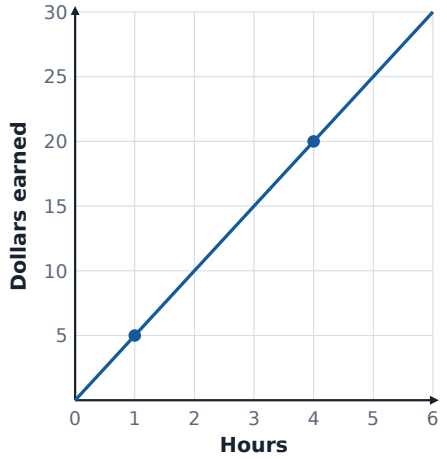
MAIN

Grade 7 main question • Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.d

The graph shows the money Raj earns.

What does the point (4, 20) mean in this situation?

Meaning: _____



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-M-Q1

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.d • Set 6 • MAIN • Q1/5

163

QUESTION ID

S6-M-Q1

TYPE

Short answer

CORRECT ANSWER

Raj earns $20 for working 4 hours.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

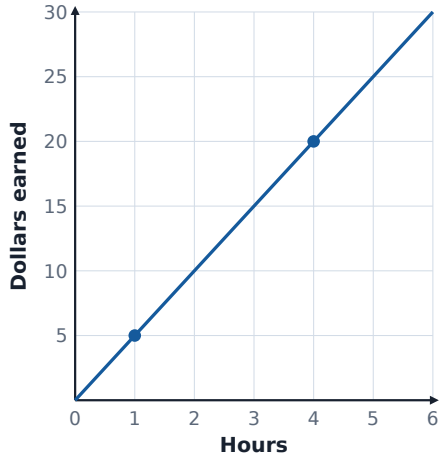
ID S6-M-Q1

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.d • Set 6 • MAIN • Q1/5

The graph shows the money Raj earns.

What does the point (1, 5) tell you?

Meaning: _____



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-M-Q2

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.d • Set 6 • MAIN • Q2/5

164

QUESTION ID

S6-M-Q2

TYPE

Short answer

CORRECT ANSWER

Raj earns $5 per hour. This is the unit rate.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-M-Q2

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.d • Set 6 • MAIN • Q2/5

The graph shows the money Raj earns.

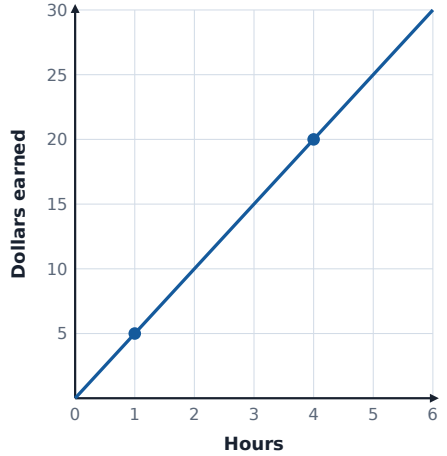
What does the point (0, 0) mean?

A. Raj earns $0 when he works 0 hours.

B. Raj earns $5 per hour.

C. Raj works 0 hours to earn $5.

D. Raj starts with $5.



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-M-Q3

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.d • Set 6 • MAIN • Q3/5

165

QUESTION ID

S6-M-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. Raj earns $0 when he works 0 hours.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-M-Q3

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.d • Set 6 • MAIN • Q3/5

True or false?

A proportional graph shows cups of rice (x) and cups of water (y). The point (1, 3.5) means 3.5 cups of water are used for each cup of rice.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-M-Q4

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.d • Set 6 • MAIN • Q4/5

166

QUESTION ID

S6-M-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-M-Q4

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.d • Set 6 • MAIN • Q4/5

True or false?

A car's distance graph passes through (2, 110), where x is hours and y is miles. This means the car travels 110 miles per hour.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-M-Q5

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.d • Set 6 • MAIN • Q5/5

167

QUESTION ID

S6-M-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE It means 110 miles in 2 hours, or 55 miles per hour.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-M-Q5

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.d • Set 6 • MAIN • Q5/5

SET 6 • 7.RP.A.2.d

BACKWARD 1

Explain a pair in a table of equivalent ratios • Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.a

The table shows the cost of apples.

What does the pair (2, 6) mean?

Pounds	1	2	3
Cost ($)	3	6	9

Meaning: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-B1-Q1

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.a • Set 6 • BACKWARD 1 • Q1/5

169

QUESTION ID

S6-B1-Q1

TYPE

Short answer

CORRECT ANSWER

2 pounds of apples cost $6.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-B1-Q1

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.a • Set 6 • BACKWARD 1 • Q1/5

The table shows the time a swimmer takes.

What does the pair (4, 10) mean?

Laps	2	4	6
Minutes	5	10	15

A. Each lap takes 4 minutes.

B. 4 laps take 10 minutes.

C. 10 laps take 4 minutes.

D. Each lap takes 10 minutes.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-B1-Q2

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.a • Set 6 • BACKWARD 1 • Q2/5

170

QUESTION ID

S6-B1-Q2

TYPE

Multiple choice

CORRECT ANSWER

B. 4 laps take 10 minutes.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-B1-Q2

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.a • Set 6 • BACKWARD 1 • Q2/5

True or false?

A table shows hours (x) and miles walked (y). The pair (3, 12) means 12 miles are walked in 3 hours.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-B1-Q3

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.a • Set 6 • BACKWARD 1 • Q3/5

171

QUESTION ID

S6-B1-Q3

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-B1-Q3

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.a • Set 6 • BACKWARD 1 • Q3/5

One ticket costs $8.

What pair (tickets, cost) shows the cost of 5 tickets?

Pair: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-B1-Q4

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.a • Set 6 • BACKWARD 1 • Q4/5

172

QUESTION ID

S6-B1-Q4

TYPE

Short answer

CORRECT ANSWER

(5, 40)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-B1-Q4

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.a • Set 6 • BACKWARD 1 • Q4/5

True or false?

A table shows boxes (x) and pencils (y). The pair (2, 24) means each box holds 2 pencils.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-B1-Q5

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.a • Set 6 • BACKWARD 1 • Q5/5

173

QUESTION ID

S6-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE It means 2 boxes hold 24 pencils.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-B1-Q5

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.a • Set 6 • BACKWARD 1 • Q5/5

SET 6 • 7.RP.A.2.d

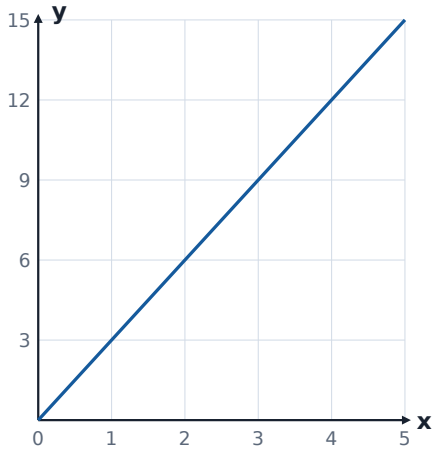
BACKWARD 2

Read values from the graph of an equation • Grade 6 • Expressions & Equations • 6.EE.C.9

The graph shows y = 3x.

What is y when x = 4?

y = _____



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-B2-Q1

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 6 • BACKWARD 2 • Q1/5

175

QUESTION ID

S6-B2-Q1

TYPE

Short answer

CORRECT ANSWER

12

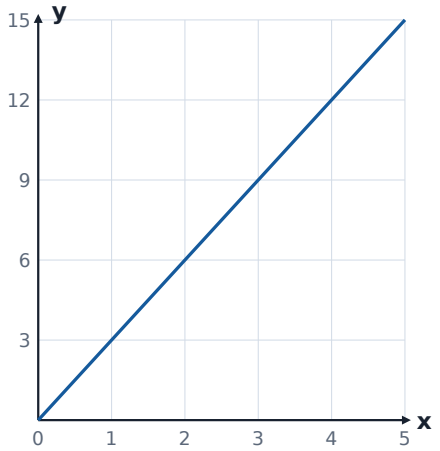
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-B2-Q1

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 6 • BACKWARD 2 • Q1/5

The graph shows y = 3x.

What is x when y = 6?



x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-B2-Q2

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 6 • BACKWARD 2 • Q2/5

176

QUESTION ID

S6-B2-Q2

TYPE

Short answer

CORRECT ANSWER

2

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-B2-Q2

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 6 • BACKWARD 2 • Q2/5

Which point is on the graph of y = 2x?

A. (6, 3)

B. (4, 6)

C. (3, 6)

D. (2, 2)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-B2-Q3

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 6 • BACKWARD 2 • Q3/5

177

QUESTION ID

S6-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. (3, 6)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-B2-Q3

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 6 • BACKWARD 2 • Q3/5

True or false?

The point (5, 15) is on the graph of y = 3x.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-B2-Q4

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 6 • BACKWARD 2 • Q4/5

178

QUESTION ID

S6-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-B2-Q4

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 6 • BACKWARD 2 • Q4/5

True or false?

The point (1, 4) is on the graph of y = 3x.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-B2-Q5

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 6 • BACKWARD 2 • Q5/5

179

QUESTION ID

S6-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-B2-Q5

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 6 • BACKWARD 2 • Q5/5

SET 6 • 7.RP.A.2.d

FORWARD 1

Interpret the rate of change and initial value of a linear function • Grade 8 • Functions • 8.F.B.4

A taxi ride costs y = 2.5x + 4 dollars for x miles.

What does the 4 mean?

Meaning: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-F1-Q1

Grade 8 • Functions • 8.F.B.4 • Set 6 • FORWARD 1 • Q1/5

181

QUESTION ID

S6-F1-Q1

TYPE

Short answer

CORRECT ANSWER

A $4 starting fee (the cost at 0 miles).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-F1-Q1

Grade 8 • Functions • 8.F.B.4 • Set 6 • FORWARD 1 • Q1/5

A taxi ride costs y = 2.5x + 4 dollars for x miles.

What does the 2.5 mean?

Meaning: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-F1-Q2

Grade 8 • Functions • 8.F.B.4 • Set 6 • FORWARD 1 • Q2/5

182

QUESTION ID

S6-F1-Q2

TYPE

Short answer

CORRECT ANSWER

The cost increases by $2.50 for each mile.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-F1-Q2

Grade 8 • Functions • 8.F.B.4 • Set 6 • FORWARD 1 • Q2/5

A tank holds $y = 200 - 8x$ gallons after x minutes.

What is the rate of change?

A. −8 gallons per minute

B. 200 gallons per minute

C. 8 gallons

D. −200 gallons

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-F1-Q3

Grade 8 • Functions • 8.F.B.4 • Set 6 • FORWARD 1 • Q3/5

183

QUESTION ID

S6-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. −8 gallons per minute

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-F1-Q3

Grade 8 • Functions • 8.F.B.4 • Set 6 • FORWARD 1 • Q3/5

True or false?

A plant's height after x weeks is y = 0.5x + 3 centimeters. The plant was 3 cm tall at the start.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-F1-Q4

Grade 8 • Functions • 8.F.B.4 • Set 6 • FORWARD 1 • Q4/5

184

QUESTION ID

S6-F1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-F1-Q4

Grade 8 • Functions • 8.F.B.4 • Set 6 • FORWARD 1 • Q4/5

True or false?

In y = 12x + 30, the initial value is 12.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-F1-Q5

Grade 8 • Functions • 8.F.B.4 • Set 6 • FORWARD 1 • Q5/5

185

QUESTION ID

S6-F1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE The initial value is 30.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-F1-Q5

Grade 8 • Functions • 8.F.B.4 • Set 6 • FORWARD 1 • Q5/5

SET 6 • 7.RP.A.2.d

FORWARD 2

Calculate the average rate of change over an interval • Grade 9 (High School) • Functions: Interpreting Functions • HSF-IF.B.6

$f(x) = x^2$

What is the average rate of change of f from x = 1 to x = 3?

Average rate: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-F2-Q1

Grade 9 (High School) • Functions: Interpreting Functions • HSF-IF.B.6 • Set 6 • FORWARD 2 • Q1/5

187

QUESTION ID

S6-F2-Q1

TYPE

Short answer

CORRECT ANSWER

4

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-F2-Q1

Grade 9 (High School) • Functions: Interpreting Functions • HSF-IF.B.6 • Set 6 • FORWARD 2 • Q1/5

The table shows the height of a balloon.

What is the average rate of change from 2 seconds to 4 seconds?

Time (s)	0	2	4
Height (m)	0	40	60

Average rate: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-F2-Q2

Grade 9 (High School) • Functions: Interpreting Functions • HSF-IF.B.6 • Set 6 • FORWARD 2 • Q2/5

188

QUESTION ID

S6-F2-Q2

TYPE

Short answer

CORRECT ANSWER

10 meters per second

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-F2-Q2

Grade 9 (High School) • Functions: Interpreting Functions • HSF-IF.B.6 • Set 6 • FORWARD 2 • Q2/5

A car is 30 miles from home at 1 p.m. and 150 miles from home at 3 p.m.

What is its average rate of change in distance?

A. 75 miles per hour

B. 60 miles per hour

C. 50 miles per hour

D. 120 miles per hour

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-F2-Q3

Grade 9 (High School) • Functions: Interpreting Functions • HSF-IF.B.6 • Set 6 • FORWARD 2 • Q3/5

189

QUESTION ID

S6-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. 60 miles per hour

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-F2-Q3

Grade 9 (High School) • Functions: Interpreting Functions • HSF-IF.B.6 • Set 6 • FORWARD 2 • Q3/5

True or false?

For f(x) = 2x + 1, the average rate of change over any interval is 2.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-F2-Q4

Grade 9 (High School) • Functions: Interpreting Functions • HSF-IF.B.6 • Set 6 • FORWARD 2 • Q4/5

190

QUESTION ID

S6-F2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-F2-Q4

Grade 9 (High School) • Functions: Interpreting Functions • HSF-IF.B.6 • Set 6 • FORWARD 2 • Q4/5

True or false?

For $f(x) = x^2$, the average rate of change from $x = 0$ to $x = 2$ equals the average rate of change from $x = 2$ to $x = 4$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-F2-Q5

Grade 9 (High School) • Functions: Interpreting Functions • HSF-IF.B.6 • Set 6 • FORWARD 2 • Q5/5

191

QUESTION ID

S6-F2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE They are 2 and 6.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-F2-Q5

Grade 9 (High School) • Functions: Interpreting Functions • HSF-IF.B.6 • Set 6 • FORWARD 2 • Q5/5

SET 7 • 7.RP.A.3 • RATIOS & PROPORTIONAL RELATIONSHIPS

Solve multistep ratio problems

MAIN	Grade 7 main question	**7.RP.A.3**
BACKWARD 1	Use ratio language to describe two quantities	**6.RP.A.1**
BACKWARD 2	Use a table of equivalent ratios to solve a problem	**6.RP.A.3.a**
FORWARD 1	Solve linear equations with rational coefficients	**8.EE.C.7.b**
FORWARD 2	Create one-variable equations to solve ratio problems	**HSA-CED.A.1**

SET 7 • 7.RP.A.3

MAIN

Grade 7 main question • Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3

A recipe uses flour and sugar in a ratio of 3 : 2. Lee uses $7\frac{1}{2}$ cups of flour.

How many cups of sugar does she need?

Cups of sugar: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-M-Q1

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3 • Set 7 • MAIN • Q1/5

194

QUESTION ID

S7-M-Q1

TYPE

Short answer

CORRECT ANSWER

5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-M-Q1

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3 • Set 7 • MAIN • Q1/5

A club has boys and girls in a ratio of 4 : 5. There are 36 members.

How many girls are in the club?

Girls: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-M-Q2

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3 • Set 7 • MAIN • Q2/5

195

QUESTION ID

S7-M-Q2

TYPE

Short answer

CORRECT ANSWER

20

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-M-Q2

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3 • Set 7 • MAIN • Q2/5

Ali mixes blue and yellow paint in a ratio of 2 : 3. He makes 15 liters of paint.

How many liters are blue?

A. 10 liters

B. 9 liters

C. 7.5 liters

D. 6 liters

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-M-Q3

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3 • Set 7 • MAIN • Q3/5

196

QUESTION ID

S7-M-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. 6 liters

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-M-Q3

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3 • Set 7 • MAIN • Q3/5

True or false?

On a map, 2 cm represents 15 km. A distance of 7 cm on the map represents 52.5 km.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-M-Q4

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3 • Set 7 • MAIN • Q4/5

197

QUESTION ID

S7-M-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-M-Q4

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3 • Set 7 • MAIN • Q4/5

True or false?

Juice concentrate and water are mixed in a ratio of 1 : 4. To make 6 liters of juice, you need 6 liters of water.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-M-Q5

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3 • Set 7 • MAIN • Q5/5

198

QUESTION ID

S7-M-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE You need 4.8 liters of water.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-M-Q5

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3 • Set 7 • MAIN • Q5/5

SET 7 • 7.RP.A.3

BACKWARD 1

Use ratio language to describe two quantities • Grade 6 • Ratios & Proportional Relationships • 6.RP.A.1

ANSWER KEY Reduced copy of the question page, with its answer

A class has 12 girls and 15 boys.

Write the ratio of boys to girls.

Ratio: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-B1-Q1

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.1 • Set 7 • BACKWARD 1 • Q1/5

200

QUESTION ID

S7-B1-Q1

TYPE

Short answer

CORRECT ANSWER

15 : 12

GRADING NOTE 5 : 4 is also correct.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-B1-Q1

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.1 • Set 7 • BACKWARD 1 • Q1/5

For every 3 red tiles, there are 5 blue tiles.

What is the ratio of blue tiles to all tiles?

A. 5 : 8

B. 5 : 3

C. 8 : 5

D. 3 : 8

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-B1-Q2

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.1 • Set 7 • BACKWARD 1 • Q2/5

201

QUESTION ID

S7-B1-Q2

TYPE

Multiple choice

CORRECT ANSWER

A. 5 : 8

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-B1-Q2

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.1 • Set 7 • BACKWARD 1 • Q2/5

A team won 9 games and lost 6 games. There were no ties.

Write the ratio of wins to games played.

Ratio: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-B1-Q3

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.1 • Set 7 • BACKWARD 1 • Q3/5

202

QUESTION ID

S7-B1-Q3

TYPE

Short answer

CORRECT ANSWER

9 : 15

GRADING NOTE 3 : 5 is also correct.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-B1-Q3

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.1 • Set 7 • BACKWARD 1 • Q3/5

True or false?

There are 2 cats for every 7 dogs. The ratio of dogs to cats is 7 : 2.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-B1-Q4

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.1 • Set 7 • BACKWARD 1 • Q4/5

203

QUESTION ID

S7-B1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-B1-Q4

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.1 • Set 7 • BACKWARD 1 • Q4/5

True or false?

A bowl has 4 apples and 9 oranges and no other fruit. The ratio of apples to all the fruit is 4 : 9.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-B1-Q5

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.1 • Set 7 • BACKWARD 1 • Q5/5

204

QUESTION ID

S7-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE The ratio is 4 : 13.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-B1-Q5

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.1 • Set 7 • BACKWARD 1 • Q5/5

SET 7 • 7.RP.A.3

BACKWARD 2

Use a table of equivalent ratios to solve a problem • Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.a

Lemonade uses 2 cups of mix for every 5 cups of water.

How many cups of water are needed for 8 cups of mix?

Cups of water: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-B2-Q1

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.a • Set 7 • BACKWARD 2 • Q1/5

206

QUESTION ID

S7-B2-Q1

TYPE

Short answer

CORRECT ANSWER

20

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-B2-Q1

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.a • Set 7 • BACKWARD 2 • Q1/5

A store sells 3 notebooks for $4.

At the same rate, how much do 12 notebooks cost?

Cost: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-B2-Q2

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.a • Set 7 • BACKWARD 2 • Q2/5

207

QUESTION ID

S7-B2-Q2

TYPE

Short answer

CORRECT ANSWER

$16

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-B2-Q2

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.a • Set 7 • BACKWARD 2 • Q2/5

Which pair belongs in a table of ratios equivalent to 3 : 4?

A. 9 : 16

B. 12 : 15

C. 6 : 7

D. 15 : 20

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-B2-Q3

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.a • Set 7 • BACKWARD 2 • Q3/5

208

QUESTION ID

S7-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. 15 : 20

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-B2-Q3

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.a • Set 7 • BACKWARD 2 • Q3/5

True or false?

A table of ratios equivalent to 5 : 2 can include 20 : 8.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-B2-Q4

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.a • Set 7 • BACKWARD 2 • Q4/5

209

QUESTION ID

S7-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-B2-Q4

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.a • Set 7 • BACKWARD 2 • Q4/5

True or false?

A table of ratios equivalent to 2 : 3 can include 8 : 10.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-B2-Q5

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.a • Set 7 • BACKWARD 2 • Q5/5

210

QUESTION ID

S7-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-B2-Q5

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.a • Set 7 • BACKWARD 2 • Q5/5

SET 7 • 7.RP.A.3

FORWARD 1

Solve linear equations with rational coefficients • Grade 8 • Expressions & Equations • 8.EE.C.7.b

Solve for b.

$5b = 3(b + 6)$

b = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-F1-Q1

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 7 • FORWARD 1 • Q1/5

212

QUESTION ID

S7-F1-Q1

TYPE

Short answer

CORRECT ANSWER

9

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-F1-Q1

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 7 • FORWARD 1 • Q1/5

Solve for x.

$\frac{2}{3}x + 4 = 10$

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-F1-Q2

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 7 • FORWARD 1 • Q2/5

213

QUESTION ID

S7-F1-Q2

TYPE

Short answer

CORRECT ANSWER

9

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-F1-Q2

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 7 • FORWARD 1 • Q2/5

Solve for x.

$0.4(x - 5) = 2$

A. $x = 0.5$

B. $x = 3$

C. $x = 10$

D. $x = 5.8$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-F1-Q3

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 7 • FORWARD 1 • Q3/5

214

QUESTION ID

S7-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. x = 10

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-F1-Q3

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 7 • FORWARD 1 • Q3/5

True or false?

$x = 12$ is the solution of $\frac{3}{4}x - 2 = 7$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-F1-Q4

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 7 • FORWARD 1 • Q4/5

215

QUESTION ID

S7-F1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-F1-Q4

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 7 • FORWARD 1 • Q4/5

True or false?

$x = 4$ is the solution of $2(x + 3) = 5x - 4$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-F1-Q5

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 7 • FORWARD 1 • Q5/5

216

QUESTION ID

S7-F1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-F1-Q5

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 7 • FORWARD 1 • Q5/5

SET 7 • 7.RP.A.3

FORWARD 2

Create one-variable equations to solve ratio problems • Grade 9 (High School) • Algebra: Creating Equations • HSA-CED.A.1

The ratio of red marbles to green marbles is 3 : 5. There are 14 more green marbles than red marbles.

Write and solve an equation to find the number of red marbles.

Red marbles: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-F2-Q1

Grade 9 (High School) • Algebra: Creating Equations • HSA-CED.A.1 • Set 7 • FORWARD 2 • Q1/5

218

QUESTION ID

S7-F2-Q1

TYPE

Short answer

CORRECT ANSWER

21

GRADING NOTE Example equation: 5x − 3x = 14, so x = 7 and there are 3(7) = 21 red marbles. Any correct equation earns credit.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-F2-Q1

Grade 9 (High School) • Algebra: Creating Equations • HSA-CED.A.1 • Set 7 • FORWARD 2 • Q1/5

A 60-meter rope is cut into two pieces in a ratio of 2 : 3.

Write and solve an equation to find the length of the longer piece.

Longer piece: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-F2-Q2

Grade 9 (High School) • Algebra: Creating Equations • HSA-CED.A.1 • Set 7 • FORWARD 2 • Q2/5

219

QUESTION ID

S7-F2-Q2

TYPE

Short answer

CORRECT ANSWER

36 meters

GRADING NOTE Example equation: 2x + 3x = 60, so x = 12. Any correct equation earns credit.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-F2-Q2

Grade 9 (High School) • Algebra: Creating Equations • HSA-CED.A.1 • Set 7 • FORWARD 2 • Q2/5

The ages of two cousins are in a ratio of 4 : 7. The sum of their ages is 66.

Which equation can find x, where the ages are 4x and 7x?

A. $4x \cdot 7x = 66$

B. $4x + 7x = 66$

C. $4 + 7x = 66$

D. $7x - 4x = 66$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-F2-Q3

Grade 9 (High School) • Algebra: Creating Equations • HSA-CED.A.1 • Set 7 • FORWARD 2 • Q3/5

220

QUESTION ID

S7-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. 4x + 7x = 66

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-F2-Q3

Grade 9 (High School) • Algebra: Creating Equations • HSA-CED.A.1 • Set 7 • FORWARD 2 • Q3/5

True or false?

The ratio of a to b is 2 : 5, and a + b = 35. Then a = 10.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-F2-Q4

Grade 9 (High School) • Algebra: Creating Equations • HSA-CED.A.1 • Set 7 • FORWARD 2 • Q4/5

221

QUESTION ID

S7-F2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-F2-Q4

Grade 9 (High School) • Algebra: Creating Equations • HSA-CED.A.1 • Set 7 • FORWARD 2 • Q4/5

True or false?

The ratio of a to b is 1 : 3, and b − a = 12. Then a = 4.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-F2-Q5

Grade 9 (High School) • Algebra: Creating Equations • HSA-CED.A.1 • Set 7 • FORWARD 2 • Q5/5

222

QUESTION ID

S7-F2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE a = 6.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-F2-Q5

Grade 9 (High School) • Algebra: Creating Equations • HSA-CED.A.1 • Set 7 • FORWARD 2 • Q5/5

Find a percent increase or a percent decrease

MAIN	Grade 7 main question	**7.RP.A.3**
BACKWARD 1	Find a percent of a quantity	**6.RP.A.3.c**
BACKWARD 2	Find the whole from a part and a percent	**6.RP.A.3.c**
FORWARD 1	Compare quantities written in scientific notation — nearest relate…	**8.EE.A.3**
FORWARD 2	Model a quantity that changes by a constant percent rate	**HSF-LE.A.1.c**

SET 8 • 7.RP.A.3

MAIN

Grade 7 main question • Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3

A town's population grew from 2,500 to 3,000.

What is the percent increase?

Percent increase: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-M-Q1

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3 • Set 8 • MAIN • Q1/5

225

QUESTION ID

S8-M-Q1

TYPE

Short answer

CORRECT ANSWER

20%

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-M-Q1

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3 • Set 8 • MAIN • Q1/5

The price of a jacket dropped from $80 to $60.

What is the percent decrease?

Percent decrease: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-M-Q2

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3 • Set 8 • MAIN • Q2/5

226

QUESTION ID

S8-M-Q2

TYPE

Short answer

CORRECT ANSWER

25%

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-M-Q2

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3 • Set 8 • MAIN • Q2/5

A plant grew from 40 cm to 46 cm tall.

What is the percent increase?

A. 15%

B. 13%

C. 87%

D. 6%

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-M-Q3

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3 • Set 8 • MAIN • Q3/5

227

QUESTION ID

S8-M-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. 15%

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-M-Q3

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3 • Set 8 • MAIN • Q3/5

True or false?

A score that falls from 50 to 40 is a 20% decrease.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-M-Q4

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3 • Set 8 • MAIN • Q4/5

228

QUESTION ID

S8-M-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-M-Q4

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3 • Set 8 • MAIN • Q4/5

True or false?

A price that rises from $60 to $75 is a 15% increase.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-M-Q5

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3 • Set 8 • MAIN • Q5/5

229

QUESTION ID

S8-M-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE It is a 25% increase.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-M-Q5

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3 • Set 8 • MAIN • Q5/5

SET 8 • 7.RP.A.3

BACKWARD 1

Find a percent of a quantity • Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c

What is 30% of 70?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-B1-Q1

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c • Set 8 • BACKWARD 1 • Q1/5

231

QUESTION ID

S8-B1-Q1

TYPE

Short answer

CORRECT ANSWER

21

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-B1-Q1

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c • Set 8 • BACKWARD 1 • Q1/5

What is 15% of $80?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-B1-Q2

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c • Set 8 • BACKWARD 1 • Q2/5

232

QUESTION ID

S8-B1-Q2

TYPE

Short answer

CORRECT ANSWER

$12

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-B1-Q2

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c • Set 8 • BACKWARD 1 • Q2/5

What is 4% of 250?

A. 40

B. 100

C. 1

D. 10

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-B1-Q3

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c • Set 8 • BACKWARD 1 • Q3/5

233

QUESTION ID

S8-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. 10

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-B1-Q3

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c • Set 8 • BACKWARD 1 • Q3/5

True or false?

75% of 40 is 30.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-B1-Q4

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c • Set 8 • BACKWARD 1 • Q4/5

234

QUESTION ID

S8-B1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-B1-Q4

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c • Set 8 • BACKWARD 1 • Q4/5

True or false?

20% of 45 is 8.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-B1-Q5

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c • Set 8 • BACKWARD 1 • Q5/5

235

QUESTION ID

S8-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE 20% of 45 is 9.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-B1-Q5

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c • Set 8 • BACKWARD 1 • Q5/5

SET 8 • 7.RP.A.3

BACKWARD 2

Find the whole from a part and a percent • Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c

12 is 40% of what number?

Number: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-B2-Q1

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c • Set 8 • BACKWARD 2 • Q1/5

237

QUESTION ID

S8-B2-Q1

TYPE

Short answer

CORRECT ANSWER

30

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-B2-Q1

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c • Set 8 • BACKWARD 2 • Q1/5

9 students in a class walk to school. That is 25% of the class.

How many students are in the class?

Students: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-B2-Q2

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c • Set 8 • BACKWARD 2 • Q2/5

238

QUESTION ID

S8-B2-Q2

TYPE

Short answer

CORRECT ANSWER

36

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-B2-Q2

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c • Set 8 • BACKWARD 2 • Q2/5

18 is 60% of what number?

A. 48

B. 12

C. 10.8

D. 30

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-B2-Q3

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c • Set 8 • BACKWARD 2 • Q3/5

239

QUESTION ID

S8-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. 30

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-B2-Q3

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c • Set 8 • BACKWARD 2 • Q3/5

True or false?

15 is 50% of 30.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-B2-Q4

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c • Set 8 • BACKWARD 2 • Q4/5

240

QUESTION ID

S8-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-B2-Q4

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c • Set 8 • BACKWARD 2 • Q4/5

True or false?

8 is 20% of 16.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-B2-Q5

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c • Set 8 • BACKWARD 2 • Q5/5

241

QUESTION ID

S8-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE 8 is 50% of 16.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-B2-Q5

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c • Set 8 • BACKWARD 2 • Q5/5

SET 8 • 7.RP.A.3

FORWARD 1

Compare quantities written in scientific notation • Grade 8 • Expressions & Equations • 8.EE.A.3

⚑ No Grade 8 standard directly continues this skill. This branch uses the nearest related Grade 8 standard.

City A has about 3×10^6 people. Town B has about 6×10^4 people.

How many times as many people live in City A as in Town B?

Times as many: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-F1-Q1

Grade 8 • Expressions & Equations • 8.EE.A.3 • Set 8 • FORWARD 1 • Q1/5 • NEAREST RELATED

243

QUESTION ID

S8-F1-Q1

TYPE

Short answer

CORRECT ANSWER

50

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-F1-Q1

Grade 8 • Expressions & Equations • 8.EE.A.3 • Set 8 • FORWARD 1 • Q1/5 • NEAREST RELATED

Write 4,500,000 in scientific notation.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-F1-Q2

Grade 8 • Expressions & Equations • 8.EE.A.3 • Set 8 • FORWARD 1 • Q2/5 • NEAREST RELATED

244

QUESTION ID

S8-F1-Q2

TYPE

Short answer

CORRECT ANSWER

4.5×10^6

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-F1-Q2

Grade 8 • Expressions & Equations • 8.EE.A.3 • Set 8 • FORWARD 1 • Q2/5 • NEAREST RELATED

Which number is the greatest?

A. 9×10^6

B. 8×10^5

C. 2×10^7

D. 1×10^6

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-F1-Q3

Grade 8 • Expressions & Equations • 8.EE.A.3 • Set 8 • FORWARD 1 • Q3/5 • NEAREST RELATED

245

QUESTION ID

S8-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. 2×10^7

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-F1-Q3

Grade 8 • Expressions & Equations • 8.EE.A.3 • Set 8 • FORWARD 1 • Q3/5 • NEAREST RELATED

True or false?

8×10^8 is 100 times as large as 8×10^6.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-F1-Q4

Grade 8 • Expressions & Equations • 8.EE.A.3 • Set 8 • FORWARD 1 • Q4/5 • NEAREST RELATED

246

QUESTION ID

S8-F1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-F1-Q4

Grade 8 • Expressions & Equations • 8.EE.A.3 • Set 8 • FORWARD 1 • Q4/5 • NEAREST RELATED

True or false?

3×10^{-2} is greater than 3×10^{-1}.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-F1-Q5

Grade 8 • Expressions & Equations • 8.EE.A.3 • Set 8 • FORWARD 1 • Q5/5 • NEAREST RELATED

247

QUESTION ID

S8-F1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-F1-Q5

Grade 8 • Expressions & Equations • 8.EE.A.3 • Set 8 • FORWARD 1 • Q5/5 • NEAREST RELATED

SET 8 • 7.RP.A.3

FORWARD 2

Model a quantity that changes by a constant percent rate • Grade 9 (High School) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.1.c

A town of 10,000 people grows by 5% each year.

What is the population after 2 years?

Population: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-F2-Q1

Grade 9 (High School) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.1.c • Set 8 • FORWARD 2 • Q1/5

249

QUESTION ID

S8-F2-Q1

TYPE

Short answer

CORRECT ANSWER

11,025

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-F2-Q1

Grade 9 (High School) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.1.c • Set 8 • FORWARD 2 • Q1/5

Which situation shows growth by a constant percent rate?

A. A plant grows 2 cm each week.

B. An account earns 3% of its current balance each year.

C. A worker earns $15 more each year.

D. A tank fills 4 gallons each minute.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-F2-Q2

Grade 9 (High School) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.1.c • Set 8 • FORWARD 2 • Q2/5

250

QUESTION ID

S8-F2-Q2

TYPE

Multiple choice

CORRECT ANSWER

B. An account earns 3% of its current balance each year.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-F2-Q2

Grade 9 (High School) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.1.c • Set 8 • FORWARD 2 • Q2/5

A car worth $20,000 loses 10% of its value each year.

What is it worth after 2 years?

Value: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-F2-Q3

Grade 9 (High School) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.1.c • Set 8 • FORWARD 2 • Q3/5

251

QUESTION ID

S8-F2-Q3

TYPE

Short answer

CORRECT ANSWER

$16,200

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-F2-Q3

Grade 9 (High School) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.1.c • Set 8 • FORWARD 2 • Q3/5

True or false?

A quantity that is multiplied by 1.08 each year grows by 8% per year.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-F2-Q4

Grade 9 (High School) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.1.c • Set 8 • FORWARD 2 • Q4/5

252

QUESTION ID

S8-F2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-F2-Q4

Grade 9 (High School) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.1.c • Set 8 • FORWARD 2 • Q4/5

True or false?

A quantity that decreases by 10% each year loses the same number of units each year.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-F2-Q5

Grade 9 (High School) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.1.c • Set 8 • FORWARD 2 • Q5/5

253

QUESTION ID

S8-F2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-F2-Q5

Grade 9 (High School) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.1.c • Set 8 • FORWARD 2 • Q5/5

Solve problems with markups, discounts, sales tax, and tips

MAIN	Grade 7 main question	**7.RP.A.3**
BACKWARD 1	Multiply decimals	**6.NS.B.3**
BACKWARD 2	Find a percent of an amount of money	**6.RP.A.3.c**
FORWARD 1	Solve equations to find an original price	**8.EE.C.7.b**
FORWARD 2	Interpret the parts of an expression in context	**HSA-SSE.A.1.a**

SET 9 • 7.RP.A.3

MAIN

Grade 7 main question • Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3

A meal costs $45. Kai leaves a 20% tip.

What is the total cost of the meal with the tip?

Total: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-M-Q1

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3 • Set 9 • MAIN • Q1/5

256

QUESTION ID

S9-M-Q1

TYPE

Short answer

CORRECT ANSWER

$54

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-M-Q1

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3 • Set 9 • MAIN • Q1/5

A bike costs $120. It is on sale for 25% off. Then 8% sales tax is added to the sale price.

What is the final cost?

Final cost: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-M-Q2

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3 • Set 9 • MAIN • Q2/5

257

QUESTION ID

S9-M-Q2

TYPE

Short answer

CORRECT ANSWER

$97.20

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-M-Q2

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3 • Set 9 • MAIN • Q2/5

A store buys a lamp for $30 and marks the price up by 60%.

What is the selling price?

A. $36

B. $18

C. $90

D. $48

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-M-Q3

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3 • Set 9 • MAIN • Q3/5

258

QUESTION ID

S9-M-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. $48

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-M-Q3

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3 • Set 9 • MAIN • Q3/5

True or false?

A pair of shoes costs $64. At 15% off, the sale price is $54.40.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-M-Q4

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3 • Set 9 • MAIN • Q4/5

259

QUESTION ID

S9-M-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-M-Q4

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3 • Set 9 • MAIN • Q4/5

True or false?

A shirt costs $35. With 6% sales tax, the total is $37.20.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-M-Q5

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3 • Set 9 • MAIN • Q5/5

260

QUESTION ID

S9-M-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE The total is $37.10.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-M-Q5

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3 • Set 9 • MAIN • Q5/5

SET 9 • 7.RP.A.3

BACKWARD 1

Multiply decimals • Grade 6 • The Number System • 6.NS.B.3

Multiply.

0.15 × 80

Product: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-B1-Q1

Grade 6 • The Number System • 6.NS.B.3 • Set 9 • BACKWARD 1 • Q1/5

262

QUESTION ID

S9-B1-Q1

TYPE

Short answer

CORRECT ANSWER

12

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-B1-Q1

Grade 6 • The Number System • 6.NS.B.3 • Set 9 • BACKWARD 1 • Q1/5

Multiply.

1.08 × 25

Product: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-B1-Q2

Grade 6 • The Number System • 6.NS.B.3 • Set 9 • BACKWARD 1 • Q2/5

263

QUESTION ID

S9-B1-Q2

TYPE

Short answer

CORRECT ANSWER

27

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-B1-Q2

Grade 6 • The Number System • 6.NS.B.3 • Set 9 • BACKWARD 1 • Q2/5

Multiply.

0.06×45

A. 270

B. 2.7

C. 0.27

D. 27

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-B1-Q3

Grade 6 • The Number System • 6.NS.B.3 • Set 9 • BACKWARD 1 • Q3/5

264

QUESTION ID

S9-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. 2.7

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-B1-Q3

Grade 6 • The Number System • 6.NS.B.3 • Set 9 • BACKWARD 1 • Q3/5

True or false?

$0.2 \times 36.5 = 7.3$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-B1-Q4

Grade 6 • The Number System • 6.NS.B.3 • Set 9 • BACKWARD 1 • Q4/5

265

QUESTION ID

S9-B1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-B1-Q4

Grade 6 • The Number System • 6.NS.B.3 • Set 9 • BACKWARD 1 • Q4/5

True or false?

$1.5 \times 4.2 = 6.2$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-B1-Q5

Grade 6 • The Number System • 6.NS.B.3 • Set 9 • BACKWARD 1 • Q5/5

266

QUESTION ID

S9-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-B1-Q5

Grade 6 • The Number System • 6.NS.B.3 • Set 9 • BACKWARD 1 • Q5/5

SET 9 • 7.RP.A.3

BACKWARD 2

Find a percent of an amount of money • Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c

What is 10% of $56?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-B2-Q1

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c • Set 9 • BACKWARD 2 • Q1/5

268

QUESTION ID

S9-B2-Q1

TYPE

Short answer

CORRECT ANSWER

$5.60

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-B2-Q1

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c • Set 9 • BACKWARD 2 • Q1/5

What is 25% of $48?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-B2-Q2

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c • Set 9 • BACKWARD 2 • Q2/5

269

QUESTION ID

S9-B2-Q2

TYPE

Short answer

CORRECT ANSWER

$12

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-B2-Q2

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c • Set 9 • BACKWARD 2 • Q2/5

What is 8% of $50?

A. $0.40

B. $4

C. $8

D. $40

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-B2-Q3

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c • Set 9 • BACKWARD 2 • Q3/5

270

QUESTION ID

S9-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. $4

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-B2-Q3

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c • Set 9 • BACKWARD 2 • Q3/5

True or false?

20% of $35 is $7.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-B2-Q4

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c • Set 9 • BACKWARD 2 • Q4/5

271

QUESTION ID

S9-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-B2-Q4

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c • Set 9 • BACKWARD 2 • Q4/5

True or false?

5% of $60 is $30.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-B2-Q5

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c • Set 9 • BACKWARD 2 • Q5/5

272

QUESTION ID

S9-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-B2-Q5

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c • Set 9 • BACKWARD 2 • Q5/5

SET 9 • 7.RP.A.3

FORWARD 1

Solve equations to find an original price • Grade 8 • Expressions & Equations • 8.EE.C.7.b

After a 20% discount, a game costs $36. The equation 0.8p = 36 models this.

What was the original price p?

p = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-F1-Q1

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 9 • FORWARD 1 • Q1/5

274

QUESTION ID

S9-F1-Q1

TYPE

Short answer

CORRECT ANSWER

$45

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-F1-Q1

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 9 • FORWARD 1 • Q1/5

A meal plus a 15% tip cost $46. The equation $1.15m = 46$ models this.

What was the cost m of the meal?

m = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-F1-Q2

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 9 • FORWARD 1 • Q2/5

275

QUESTION ID

S9-F1-Q2

TYPE

Short answer

CORRECT ANSWER

$40

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-F1-Q2

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 9 • FORWARD 1 • Q2/5

Solve for x.

1.06x = 53

A. x = 5

B. x = 51.94

C. x = 50

D. x = 56.18

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-F1-Q3

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 9 • FORWARD 1 • Q3/5

276

QUESTION ID

S9-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. x = 50

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-F1-Q3

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 9 • FORWARD 1 • Q3/5

True or false?

If 0.75p = 30, then p = 40.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-F1-Q4

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 9 • FORWARD 1 • Q4/5

277

QUESTION ID

S9-F1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-F1-Q4

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 9 • FORWARD 1 • Q4/5

True or false?

If 1.25p = 50, then p = 62.5.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-F1-Q5

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 9 • FORWARD 1 • Q5/5

278

QUESTION ID

S9-F1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE p = 40.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-F1-Q5

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 9 • FORWARD 1 • Q5/5

SET 9 • 7.RP.A.3

FORWARD 2

Interpret the parts of an expression in context • Grade 9 (High School) • Algebra: Seeing Structure in Expressions • HSA-SSE.A.1.a

A price p is discounted 10%. Then 7% sales tax is added. The final cost is 1.07(0.9p).

What does 0.9p represent?

Meaning: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-F2-Q1

Grade 9 (High School) • Algebra: Seeing Structure in Expressions • HSA-SSE.A.1.a • Set 9 • FORWARD 2 • Q1/5

280

QUESTION ID

S9-F2-Q1

TYPE

Short answer

CORRECT ANSWER

The price after the 10% discount (90% of p).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-F2-Q1

Grade 9 (High School) • Algebra: Seeing Structure in Expressions • HSA-SSE.A.1.a • Set 9 • FORWARD 2 • Q1/5

The final cost of an item is 1.07(0.9p).

What does the factor 1.07 represent?

Meaning: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-F2-Q2

Grade 9 (High School) • Algebra: Seeing Structure in Expressions • HSA-SSE.A.1.a • Set 9 • FORWARD 2 • Q2/5

281

QUESTION ID

S9-F2-Q2

TYPE

Short answer

CORRECT ANSWER

Adding 7% sales tax (paying 107% of the discounted price).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-F2-Q2

Grade 9 (High School) • Algebra: Seeing Structure in Expressions • HSA-SSE.A.1.a • Set 9 • FORWARD 2 • Q2/5

A store's selling price is p + 0.4p, where p is what the store paid.

What does 0.4p represent?

A. The selling price

B. The amount of a 40% discount

C. The price the store paid

D. The amount of the 40% markup

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-F2-Q3

Grade 9 (High School) • Algebra: Seeing Structure in Expressions • HSA-SSE.A.1.a • Set 9 • FORWARD 2 • Q3/5

282

QUESTION ID

S9-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. The amount of the 40% markup

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-F2-Q3

Grade 9 (High School) • Algebra: Seeing Structure in Expressions • HSA-SSE.A.1.a • Set 9 • FORWARD 2 • Q3/5

True or false?

The expression 0.85p gives the price after a 15% discount on price p.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-F2-Q4

Grade 9 (High School) • Algebra: Seeing Structure in Expressions • HSA-SSE.A.1.a • Set 9 • FORWARD 2 • Q4/5

283

QUESTION ID

S9-F2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-F2-Q4

Grade 9 (High School) • Algebra: Seeing Structure in Expressions • HSA-SSE.A.1.a • Set 9 • FORWARD 2 • Q4/5

True or false?

The expression 1.2p gives the price after a 20% discount on price p.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-F2-Q5

Grade 9 (High School) • Algebra: Seeing Structure in Expressions • HSA-SSE.A.1.a • Set 9 • FORWARD 2 • Q5/5

284

QUESTION ID

S9-F2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE 1.2p is a 20% increase.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-F2-Q5

Grade 9 (High School) • Algebra: Seeing Structure in Expressions • HSA-SSE.A.1.a • Set 9 • FORWARD 2 • Q5/5

SET 10 • 7.RP.A.3 • RATIOS & PROPORTIONAL RELATIONSHIPS

Solve problems with simple interest, commissions, and fees

MAIN	Grade 7 main question	**7.RP.A.3**
BACKWARD 1	Evaluate a formula	**6.EE.A.2.c**
BACKWARD 2	Write a percent as a rate per 100 and as a decimal	**6.RP.A.3.c**
FORWARD 1	Model a growing balance with a linear function	**8.F.B.4**
FORWARD 2	Compare simple interest with compound interest	**HSF-LE.A.1.a**

SET 10 • 7.RP.A.3

MAIN

Grade 7 main question • Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3

Ella deposits $800 in an account that earns 3% simple interest per year.

How much interest does she earn in 4 years?

Interest: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-M-Q1

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3 • Set 10 • MAIN • Q1/5

287

QUESTION ID

S10-M-Q1

TYPE

Short answer

CORRECT ANSWER

$96

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-M-Q1

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3 • Set 10 • MAIN • Q1/5

A salesperson earns a 6% commission. She sells $2,450 of products.

What is her commission?

Commission: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-M-Q2

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3 • Set 10 • MAIN • Q2/5

288

QUESTION ID

S10-M-Q2

TYPE

Short answer

CORRECT ANSWER

$147

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-M-Q2

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3 • Set 10 • MAIN • Q2/5

A ticket website charges a 2.5% fee on a $600 order.

What is the fee?

A. $150

B. $15

C. $25

D. $1.50

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-M-Q3

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3 • Set 10 • MAIN • Q3/5

289

QUESTION ID

S10-M-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. $15

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-M-Q3

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3 • Set 10 • MAIN • Q3/5

True or false?

$1,500 borrowed at 5% simple interest per year for 2 years has $150 in interest.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-M-Q4

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3 • Set 10 • MAIN • Q4/5

290

QUESTION ID

S10-M-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-M-Q4

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3 • Set 10 • MAIN • Q4/5

True or false?

A 4% fee on $250 is $1.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-M-Q5

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3 • Set 10 • MAIN • Q5/5

291

QUESTION ID

S10-M-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE The fee is $10.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-M-Q5

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3 • Set 10 • MAIN • Q5/5

SET 10 • 7.RP.A.3

BACKWARD 1

Evaluate a formula • Grade 6 • Expressions & Equations • 6.EE.A.2.c

Use I = prt.

Find I when p = 200, r = 0.05, and t = 3.

I = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-B1-Q1

Grade 6 • Expressions & Equations • 6.EE.A.2.c • Set 10 • BACKWARD 1 • Q1/5

293

QUESTION ID

S10-B1-Q1

TYPE

Short answer

CORRECT ANSWER

30

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-B1-Q1

Grade 6 • Expressions & Equations • 6.EE.A.2.c • Set 10 • BACKWARD 1 • Q1/5

Use A = lw.

Find A when l = 12 and w = 7.5.

A = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-B1-Q2

Grade 6 • Expressions & Equations • 6.EE.A.2.c • Set 10 • BACKWARD 1 • Q2/5

294

QUESTION ID

S10-B1-Q2

TYPE

Short answer

CORRECT ANSWER

90

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-B1-Q2

Grade 6 • Expressions & Equations • 6.EE.A.2.c • Set 10 • BACKWARD 1 • Q2/5

Use d = rt.

Find d when r = 55 and t = 4.

A. 13.75

B. 220

C. 51

D. 59

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-B1-Q3

Grade 6 • Expressions & Equations • 6.EE.A.2.c • Set 10 • BACKWARD 1 • Q3/5

295

QUESTION ID

S10-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. 220

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-B1-Q3

Grade 6 • Expressions & Equations • 6.EE.A.2.c • Set 10 • BACKWARD 1 • Q3/5

True or false?

For P = 2l + 2w, when l = 6 and w = 4, P = 20.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-B1-Q4

Grade 6 • Expressions & Equations • 6.EE.A.2.c • Set 10 • BACKWARD 1 • Q4/5

296

QUESTION ID

S10-B1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-B1-Q4

Grade 6 • Expressions & Equations • 6.EE.A.2.c • Set 10 • BACKWARD 1 • Q4/5

True or false?

For C = 5n + 3, when n = 4, C = 32.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-B1-Q5

Grade 6 • Expressions & Equations • 6.EE.A.2.c • Set 10 • BACKWARD 1 • Q5/5

297

QUESTION ID

S10-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE C = 23.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-B1-Q5

Grade 6 • Expressions & Equations • 6.EE.A.2.c • Set 10 • BACKWARD 1 • Q5/5

SET 10 • 7.RP.A.3

BACKWARD 2

Write a percent as a rate per 100 and as a decimal • Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c

Write 6% as a decimal.

Decimal: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-B2-Q1

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c • Set 10 • BACKWARD 2 • Q1/5

299

QUESTION ID

S10-B2-Q1

TYPE

Short answer

CORRECT ANSWER

0.06

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-B2-Q1

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c • Set 10 • BACKWARD 2 • Q1/5

Write 2.5% as a decimal.

Decimal: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-B2-Q2

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c • Set 10 • BACKWARD 2 • Q2/5

300

QUESTION ID

S10-B2-Q2

TYPE

Short answer

CORRECT ANSWER

0.025

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-B2-Q2

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c • Set 10 • BACKWARD 2 • Q2/5

Which decimal is equal to 45%?

A. 4.5

B. 0.45

C. 45

D. 0.045

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-B2-Q3

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c • Set 10 • BACKWARD 2 • Q3/5

301

QUESTION ID

S10-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. 0.45

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-B2-Q3

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c • Set 10 • BACKWARD 2 • Q3/5

True or false?

$3\% = \frac{3}{100} = 0.03$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-B2-Q4

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c • Set 10 • BACKWARD 2 • Q4/5

302

QUESTION ID

S10-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-B2-Q4

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c • Set 10 • BACKWARD 2 • Q4/5

True or false?

8% = 0.8

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-B2-Q5

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c • Set 10 • BACKWARD 2 • Q5/5

303

QUESTION ID

S10-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-B2-Q5

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c • Set 10 • BACKWARD 2 • Q5/5

SET 10 • 7.RP.A.3

FORWARD 1

Model a growing balance with a linear function • Grade 8 • Functions • 8.F.B.4

An account starts with $500 and earns $20 in simple interest each year.

Write a function for the balance B after t years.

B = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-F1-Q1

Grade 8 • Functions • 8.F.B.4 • Set 10 • FORWARD 1 • Q1/5

305

QUESTION ID

S10-F1-Q1

TYPE

Short answer

CORRECT ANSWER

20t + 500

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-F1-Q1

Grade 8 • Functions • 8.F.B.4 • Set 10 • FORWARD 1 • Q1/5

A salesperson earns $300 a week plus a 5% commission on sales s.

Write a function for the weekly pay P.

P = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-F1-Q2

Grade 8 • Functions • 8.F.B.4 • Set 10 • FORWARD 1 • Q2/5

306

QUESTION ID

S10-F1-Q2

TYPE

Short answer

CORRECT ANSWER

0.05s + 300

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-F1-Q2

Grade 8 • Functions • 8.F.B.4 • Set 10 • FORWARD 1 • Q2/5

A loan balance after t years is B = 1,200 + 36t.

What is the rate of change?

A. $36 per year

B. $1,236 per year

C. $1,200 per year

D. 3% per year

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-F1-Q3

Grade 8 • Functions • 8.F.B.4 • Set 10 • FORWARD 1 • Q3/5

QUESTION ID

S10-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. $36 per year

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-F1-Q3

Grade 8 • Functions • 8.F.B.4 • Set 10 • FORWARD 1 • Q3/5

True or false?

For B = 20t + 500, the balance after 3 years is $560.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-F1-Q4

Grade 8 • Functions • 8.F.B.4 • Set 10 • FORWARD 1 • Q4/5

308

QUESTION ID

S10-F1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-F1-Q4

Grade 8 • Functions • 8.F.B.4 • Set 10 • FORWARD 1 • Q4/5

True or false?

For P = 0.05s + 300, the initial value is 0.05.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-F1-Q5

Grade 8 • Functions • 8.F.B.4 • Set 10 • FORWARD 1 • Q5/5

309

QUESTION ID

S10-F1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE The initial value is 300.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-F1-Q5

Grade 8 • Functions • 8.F.B.4 • Set 10 • FORWARD 1 • Q5/5

SET 10 • 7.RP.A.3

FORWARD 2

Compare simple interest with compound interest • Grade 9 (High School) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.1.a

$1,000 earns 5% simple interest per year.

What is the balance after 2 years?

Balance: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-F2-Q1

Grade 9 (High School) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.1.a • Set 10 • FORWARD 2 • Q1/5

311

QUESTION ID

S10-F2-Q1

TYPE

Short answer

CORRECT ANSWER

$1,100

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-F2-Q1

Grade 9 (High School) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.1.a • Set 10 • FORWARD 2 • Q1/5

$1,000 earns 5% interest compounded once each year.

What is the balance after 2 years?

Balance: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-F2-Q2

Grade 9 (High School) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.1.a • Set 10 • FORWARD 2 • Q2/5

312

QUESTION ID

S10-F2-Q2

TYPE

Short answer

CORRECT ANSWER

$1,102.50

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-F2-Q2

Grade 9 (High School) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.1.a • Set 10 • FORWARD 2 • Q2/5

Which kind of interest makes a balance grow by equal differences over equal time intervals?

A. Both kinds

B. Compound interest

C. Simple interest

D. Neither kind

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-F2-Q3

Grade 9 (High School) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.1.a • Set 10 • FORWARD 2 • Q3/5

313

QUESTION ID

S10-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. Simple interest

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-F2-Q3

Grade 9 (High School) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.1.a • Set 10 • FORWARD 2 • Q3/5

True or false?

With compound interest, a balance grows by equal factors over equal time intervals.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-F2-Q4

Grade 9 (High School) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.1.a • Set 10 • FORWARD 2 • Q4/5

314

QUESTION ID

S10-F2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-F2-Q4

Grade 9 (High School) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.1.a • Set 10 • FORWARD 2 • Q4/5

True or false?

After 1 year, $500 at 4% compound interest (compounded yearly) earns more than $500 at 4% simple interest.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-F2-Q5

Grade 9 (High School) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.1.a • Set 10 • FORWARD 2 • Q5/5

315

QUESTION ID

S10-F2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE After 1 year, both earn $20.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-F2-Q5

Grade 9 (High School) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.1.a • Set 10 • FORWARD 2 • Q5/5

SET 11 • 7.RP.A.3 • RATIOS & PROPORTIONAL RELATIONSHIPS

Find the percent error of an estimate or a measurement

MAIN	Grade 7 main question	**7.RP.A.3**
BACKWARD 1	Find the absolute value of a number	**6.NS.C.7.c**
BACKWARD 2	Write a fraction as a percent	**6.RP.A.3.c**
FORWARD 1	Compute with numbers in scientific notation — nearest related (n…	**8.EE.A.4**
FORWARD 2	Choose a level of accuracy that fits a measurement	**HSN-Q.A.3**

SET 11 • 7.RP.A.3

MAIN

Grade 7 main question • Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3

Ty estimates that 50 people are at a meeting. The actual number is 40.

What is the percent error?

Percent error: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S11-M-Q1

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3 • Set 11 • MAIN • Q1/5

318

QUESTION ID

S11-M-Q1

TYPE

Short answer

CORRECT ANSWER

25%

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S11-M-Q1

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3 • Set 11 • MAIN • Q1/5

A scale shows 4.8 kg for a bag whose actual mass is 5 kg.

What is the percent error?

Percent error: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S11-M-Q2

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3 • Set 11 • MAIN • Q2/5

319

QUESTION ID

S11-M-Q2

TYPE

Short answer

CORRECT ANSWER

4%

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S11-M-Q2

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3 • Set 11 • MAIN • Q2/5

A student measures a line that is actually 20 cm long as 19 cm.

What is the percent error?

A. 95%

B. 5%

C. 5.3%

D. 1%

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S11-M-Q3

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3 • Set 11 • MAIN • Q3/5

320

QUESTION ID

S11-M-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. 5%

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S11-M-Q3

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3 • Set 11 • MAIN • Q3/5

True or false?

A guess of 90 for an actual value of 100 has a percent error of 10%.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S11-M-Q4

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3 • Set 11 • MAIN • Q4/5

321

QUESTION ID

S11-M-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S11-M-Q4

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3 • Set 11 • MAIN • Q4/5

True or false?

To find percent error, you divide the size of the error by the estimate.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-M-Q5

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3 • Set 11 • MAIN • Q5/5

322

QUESTION ID

S11-M-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE You divide by the actual value.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-M-Q5

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3 • Set 11 • MAIN • Q5/5

SET 11 • 7.RP.A.3

BACKWARD 1

Find the absolute value of a number • Grade 6 • The Number System • 6.NS.C.7.c

Find the value.

$|-12|$

Value: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S11-B1-Q1

Grade 6 • The Number System • 6.NS.C.7.c • Set 11 • BACKWARD 1 • Q1/5

QUESTION ID

S11-B1-Q1

TYPE

Short answer

CORRECT ANSWER

12

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S11-B1-Q1

Grade 6 • The Number System • 6.NS.C.7.c • Set 11 • BACKWARD 1 • Q1/5

Find the value.

$|-4.5|$

Value: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S11-B1-Q2

Grade 6 • The Number System • 6.NS.C.7.c • Set 11 • BACKWARD 1 • Q2/5

325

QUESTION ID

S11-B1-Q2

TYPE

Short answer

CORRECT ANSWER

4.5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S11-B1-Q2

Grade 6 • The Number System • 6.NS.C.7.c • Set 11 • BACKWARD 1 • Q2/5

Which number has the greatest absolute value?

A. −3

B. 8

C. −9

D. 6

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-B1-Q3

Grade 6 • The Number System • 6.NS.C.7.c • Set 11 • BACKWARD 1 • Q3/5

326

QUESTION ID

S11-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. −9

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-B1-Q3

Grade 6 • The Number System • 6.NS.C.7.c • Set 11 • BACKWARD 1 • Q3/5

True or false?

$|-15| = 15$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S11-B1-Q4

Grade 6 • The Number System • 6.NS.C.7.c • Set 11 • BACKWARD 1 • Q4/5

327

QUESTION ID

S11-B1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S11-B1-Q4

Grade 6 • The Number System • 6.NS.C.7.c • Set 11 • BACKWARD 1 • Q4/5

True or false?

$|-2| < |1|$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S11-B1-Q5

Grade 6 • The Number System • 6.NS.C.7.c • Set 11 • BACKWARD 1 • Q5/5

328

QUESTION ID

S11-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S11-B1-Q5

Grade 6 • The Number System • 6.NS.C.7.c • Set 11 • BACKWARD 1 • Q5/5

SET 11 • 7.RP.A.3

BACKWARD 2

Write a fraction as a percent • Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c

Write $\frac{3}{20}$ as a percent.

Percent: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-B2-Q1

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c • Set 11 • BACKWARD 2 • Q1/5

330

QUESTION ID

S11-B2-Q1

TYPE

Short answer

CORRECT ANSWER

15%

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-B2-Q1

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c • Set 11 • BACKWARD 2 • Q1/5

Write $\frac{7}{25}$ as a percent.

Percent: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-B2-Q2

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c • Set 11 • BACKWARD 2 • Q2/5

331

QUESTION ID

S11-B2-Q2

TYPE

Short answer

CORRECT ANSWER

28%

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-B2-Q2

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c • Set 11 • BACKWARD 2 • Q2/5

Which percent is equal to $\frac{9}{12}$?

A. 75%

B. 80%

C. 9%

D. 12%

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S11-B2-Q3

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c • Set 11 • BACKWARD 2 • Q3/5

332

QUESTION ID

S11-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. 75%

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S11-B2-Q3

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c • Set 11 • BACKWARD 2 • Q3/5

True or false?

$\frac{1}{8} = 12.5\%$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-B2-Q4

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c • Set 11 • BACKWARD 2 • Q4/5

333

QUESTION ID

S11-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-B2-Q4

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c • Set 11 • BACKWARD 2 • Q4/5

True or false?

$\frac{2}{5} = 25\%$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-B2-Q5

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c • Set 11 • BACKWARD 2 • Q5/5

334

QUESTION ID

S11-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-B2-Q5

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c • Set 11 • BACKWARD 2 • Q5/5

SET 11 • 7.RP.A.3

FORWARD 1

Compute with numbers in scientific notation • Grade 8 • Expressions & Equations • 8.EE.A.4

⚑ No Grade 8 standard directly continues this skill. This branch uses the nearest related Grade 8 standard.

Multiply. Write the answer in scientific notation.

$(3 \times 10^4) \times (2 \times 10^3)$

Product: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-F1-Q1

Grade 8 • Expressions & Equations • 8.EE.A.4 • Set 11 • FORWARD 1 • Q1/5 • NEAREST RELATED

336

QUESTION ID

S11-F1-Q1

TYPE

Short answer

CORRECT ANSWER

6×10^7

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-F1-Q1

Grade 8 • Expressions & Equations • 8.EE.A.4 • Set 11 • FORWARD 1 • Q1/5 • NEAREST RELATED

Divide. Write the answer in scientific notation.

$(8 \times 10^{9}) \div (4 \times 10^{3})$

Quotient: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-F1-Q2

Grade 8 • Expressions & Equations • 8.EE.A.4 • Set 11 • FORWARD 1 • Q2/5 • NEAREST RELATED

337

QUESTION ID

S11-F1-Q2

TYPE

Short answer

CORRECT ANSWER

2×10^{6}

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-F1-Q2

Grade 8 • Expressions & Equations • 8.EE.A.4 • Set 11 • FORWARD 1 • Q2/5 • NEAREST RELATED

Add.

$(2.5 \times 10^5) + (4 \times 10^4)$

A. 6.5×10^5

B. 2.9×10^5

C. 2.9×10^4

D. 6.5×10^9

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-F1-Q3

Grade 8 • Expressions & Equations • 8.EE.A.4 • Set 11 • FORWARD 1 • Q3/5 • NEAREST RELATED

338

QUESTION ID

S11-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. 2.9×10^5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-F1-Q3

Grade 8 • Expressions & Equations • 8.EE.A.4 • Set 11 • FORWARD 1 • Q3/5 • NEAREST RELATED

True or false?

$(5 \times 10^3) \times (3 \times 10^2) = 1.5 \times 10^6$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-F1-Q4

Grade 8 • Expressions & Equations • 8.EE.A.4 • Set 11 • FORWARD 1 • Q4/5 • NEAREST RELATED

339

QUESTION ID

S11-F1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-F1-Q4

Grade 8 • Expressions & Equations • 8.EE.A.4 • Set 11 • FORWARD 1 • Q4/5 • NEAREST RELATED

True or false?

$(6 \times 10^8) \div (2 \times 10^2) = 3 \times 10^4$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-F1-Q5

Grade 8 • Expressions & Equations • 8.EE.A.4 • Set 11 • FORWARD 1 • Q5/5 • NEAREST RELATED

340

QUESTION ID

S11-F1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE The quotient is 3×10^6.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-F1-Q5

Grade 8 • Expressions & Equations • 8.EE.A.4 • Set 11 • FORWARD 1 • Q5/5 • NEAREST RELATED

SET 11 • 7.RP.A.3

FORWARD 2

Choose a level of accuracy that fits a measurement • Grade 9 (High School) • Number & Quantity: Quantities • HSN-Q.A.3

A ruler is marked in millimeters. A length is read as 12.4 cm.

To what place value is this measurement given?

Place: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-F2-Q1

Grade 9 (High School) • Number & Quantity: Quantities • HSN-Q.A.3 • Set 11 • FORWARD 2 • Q1/5

342

QUESTION ID

S11-F2-Q1

TYPE

Short answer

CORRECT ANSWER

Tenths of a centimeter (the nearest millimeter)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-F2-Q1

Grade 9 (High School) • Number & Quantity: Quantities • HSN-Q.A.3 • Set 11 • FORWARD 2 • Q1/5

A scale measures mass to the nearest 0.1 kg.

Which reading is reported to an accuracy that fits this scale?

A. 3 kg

B. 3.4 kg

C. 3.4172 kg

D. 3.41 kg

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S11-F2-Q2

Grade 9 (High School) • Number & Quantity: Quantities • HSN-Q.A.3 • Set 11 • FORWARD 2 • Q2/5

343

QUESTION ID

S11-F2-Q2

TYPE

Multiple choice

CORRECT ANSWER

B. 3.4 kg

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S11-F2-Q2

Grade 9 (High School) • Number & Quantity: Quantities • HSN-Q.A.3 • Set 11 • FORWARD 2 • Q2/5

A track is measured as 400 m. The measurement could be off by as much as 2 m.

What is the greatest possible percent error?

Percent error: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-F2-Q3

Grade 9 (High School) • Number & Quantity: Quantities • HSN-Q.A.3 • Set 11 • FORWARD 2 • Q3/5

344

QUESTION ID

S11-F2-Q3

TYPE

Short answer

CORRECT ANSWER

0.5%

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-F2-Q3

Grade 9 (High School) • Number & Quantity: Quantities • HSN-Q.A.3 • Set 11 • FORWARD 2 • Q3/5

True or false?

A length measured with a meter stick marked only in centimeters should not be reported as 1.2347 m.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-F2-Q4

Grade 9 (High School) • Number & Quantity: Quantities • HSN-Q.A.3 • Set 11 • FORWARD 2 • Q4/5

345

QUESTION ID

S11-F2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-F2-Q4

Grade 9 (High School) • Number & Quantity: Quantities • HSN-Q.A.3 • Set 11 • FORWARD 2 • Q4/5

True or false?

Writing more decimal places always makes a measurement more accurate.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-F2-Q5

Grade 9 (High School) • Number & Quantity: Quantities • HSN-Q.A.3 • Set 11 • FORWARD 2 • Q5/5

346

QUESTION ID

S11-F2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-F2-Q5

Grade 9 (High School) • Number & Quantity: Quantities • HSN-Q.A.3 • Set 11 • FORWARD 2 • Q5/5

SET 12 • 7.NS.A.1.a • THE NUMBER SYSTEM

Describe situations in which opposite quantities combine to make 0

MAIN	Grade 7 main question	**7.NS.A.1.a**
BACKWARD 1	Use positive and negative numbers to represent quantities	**6.NS.C.5**
BACKWARD 2	Recognize opposite numbers	**6.NS.C.6.a**
FORWARD 1	Use opposite exponents with the properties of integer exponents…	**8.EE.A.1**
FORWARD 2	Explain the steps used to solve an equation	**HSA-REI.A.1**

SET 12 • 7.NS.A.1.a

MAIN

Grade 7 main question • Grade 7 • The Number System • 7.NS.A.1.a

Ari has $15 in his wallet. He spends all $15.

Write an addition expression for this situation and find its value.

Expression and value: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-M-Q1

Grade 7 • The Number System • 7.NS.A.1.a • Set 12 • MAIN • Q1/5

349

QUESTION ID

S12-M-Q1

TYPE

Short answer

CORRECT ANSWER

15 + (−15) = 0

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-M-Q1

Grade 7 • The Number System • 7.NS.A.1.a • Set 12 • MAIN • Q1/5

Which situation combines to make 0?

A. Gaining 5 points, then gaining 5 more points

B. A temperature drop of 6°, then another drop of 6°

C. Climbing up 8 floors, then down 8 floors

D. Walking 3 miles east, then 2 miles west

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-M-Q2

Grade 7 • The Number System • 7.NS.A.1.a • Set 12 • MAIN • Q2/5

350

QUESTION ID

S12-M-Q2

TYPE

Multiple choice

CORRECT ANSWER

C. Climbing up 8 floors, then down 8 floors

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-M-Q2

Grade 7 • The Number System • 7.NS.A.1.a • Set 12 • MAIN • Q2/5

A diver is 12 feet below the surface of the water (−12 feet).

How many feet must she rise to be at the surface (0 feet)?

Feet: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-M-Q3

Grade 7 • The Number System • 7.NS.A.1.a • Set 12 • MAIN • Q3/5

351

QUESTION ID

S12-M-Q3

TYPE

Short answer

CORRECT ANSWER

12

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-M-Q3

Grade 7 • The Number System • 7.NS.A.1.a • Set 12 • MAIN • Q3/5

True or false?

A gain of 9 yards followed by a loss of 9 yards is a total change of 0 yards.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-M-Q4

Grade 7 • The Number System • 7.NS.A.1.a • Set 12 • MAIN • Q4/5

352

QUESTION ID

S12-M-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-M-Q4

Grade 7 • The Number System • 7.NS.A.1.a • Set 12 • MAIN • Q4/5

True or false?

A deposit of $40 followed by a withdrawal of $30 leaves an account balance unchanged.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-M-Q5

Grade 7 • The Number System • 7.NS.A.1.a • Set 12 • MAIN • Q5/5

353

QUESTION ID

S12-M-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-M-Q5

Grade 7 • The Number System • 7.NS.A.1.a • Set 12 • MAIN • Q5/5

SET 12 • 7.NS.A.1.a

BACKWARD 1

Use positive and negative numbers to represent quantities • Grade 6 • The Number System • 6.NS.C.5

Write an integer for 20 feet below sea level.

Integer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-B1-Q1

Grade 6 • The Number System • 6.NS.C.5 • Set 12 • BACKWARD 1 • Q1/5

355

QUESTION ID

S12-B1-Q1

TYPE

Short answer

CORRECT ANSWER

−20

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-B1-Q1

Grade 6 • The Number System • 6.NS.C.5 • Set 12 • BACKWARD 1 • Q1/5

Write an integer for a gain of 7 yards.

Integer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-B1-Q2

Grade 6 • The Number System • 6.NS.C.5 • Set 12 • BACKWARD 1 • Q2/5

356

QUESTION ID

S12-B1-Q2

TYPE

Short answer

CORRECT ANSWER

7

GRADING NOTE +7 is also correct.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-B1-Q2

Grade 6 • The Number System • 6.NS.C.5 • Set 12 • BACKWARD 1 • Q2/5

Which integer represents a withdrawal of $35?

A. 35

B. −35

C. −3.5

D. 0

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-B1-Q3

Grade 6 • The Number System • 6.NS.C.5 • Set 12 • BACKWARD 1 • Q3/5

357

QUESTION ID

S12-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. −35

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-B1-Q3

Grade 6 • The Number System • 6.NS.C.5 • Set 12 • BACKWARD 1 • Q3/5

True or false?

A temperature of 4 degrees below zero can be written as $-4°$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-B1-Q4

Grade 6 • The Number System • 6.NS.C.5 • Set 12 • BACKWARD 1 • Q4/5

358

QUESTION ID

S12-B1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-B1-Q4

Grade 6 • The Number System • 6.NS.C.5 • Set 12 • BACKWARD 1 • Q4/5

True or false?

The integer −10 represents a deposit of $10.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-B1-Q5

Grade 6 • The Number System • 6.NS.C.5 • Set 12 • BACKWARD 1 • Q5/5

359

QUESTION ID

S12-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-B1-Q5

Grade 6 • The Number System • 6.NS.C.5 • Set 12 • BACKWARD 1 • Q5/5

SET 12 • 7.NS.A.1.a

BACKWARD 2

Recognize opposite numbers • Grade 6 • The Number System • 6.NS.C.6.a

What is the opposite of 8?

Opposite: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-B2-Q1

Grade 6 • The Number System • 6.NS.C.6.a • Set 12 • BACKWARD 2 • Q1/5

361

QUESTION ID

S12-B2-Q1

TYPE

Short answer

CORRECT ANSWER

−8

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-B2-Q1

Grade 6 • The Number System • 6.NS.C.6.a • Set 12 • BACKWARD 2 • Q1/5

What is the opposite of −13?

Opposite: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-B2-Q2

Grade 6 • The Number System • 6.NS.C.6.a • Set 12 • BACKWARD 2 • Q2/5

362

QUESTION ID

S12-B2-Q2

TYPE

Short answer

CORRECT ANSWER

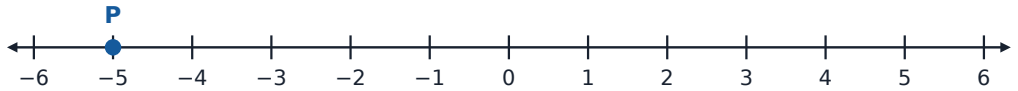
13

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-B2-Q2

Grade 6 • The Number System • 6.NS.C.6.a • Set 12 • BACKWARD 2 • Q2/5

Point P is at −5.

Which number is the same distance from 0 on the other side of 0?



A. 5 **B.** −10 **C.** 10 **D.** 0

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-B2-Q3

Grade 6 • The Number System • 6.NS.C.6.a • Set 12 • BACKWARD 2 • Q3/5

363

QUESTION ID

S12-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. 5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-B2-Q3

Grade 6 • The Number System • 6.NS.C.6.a • Set 12 • BACKWARD 2 • Q3/5

True or false?

The opposite of the opposite of 6 is 6.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-B2-Q4

Grade 6 • The Number System • 6.NS.C.6.a • Set 12 • BACKWARD 2 • Q4/5

364

QUESTION ID

S12-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-B2-Q4

Grade 6 • The Number System • 6.NS.C.6.a • Set 12 • BACKWARD 2 • Q4/5

True or false?

0 and 1 are opposites.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-B2-Q5

Grade 6 • The Number System • 6.NS.C.6.a • Set 12 • BACKWARD 2 • Q5/5

365

QUESTION ID

S12-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-B2-Q5

Grade 6 • The Number System • 6.NS.C.6.a • Set 12 • BACKWARD 2 • Q5/5

SET 12 • 7.NS.A.1.a

FORWARD 1

Use opposite exponents with the properties of integer exponents • Grade 8 • Expressions & Equations • 8.EE.A.1

⚑ No Grade 8 standard directly continues this skill. This branch uses the nearest related Grade 8 standard.

Find the value.

$5^{3} \times 5^{-3}$

Value: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-F1-Q1

Grade 8 • Expressions & Equations • 8.EE.A.1 • Set 12 • FORWARD 1 • Q1/5 • NEAREST RELATED

367

QUESTION ID

S12-F1-Q1

TYPE

Short answer

CORRECT ANSWER

1

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-F1-Q1

Grade 8 • Expressions & Equations • 8.EE.A.1 • Set 12 • FORWARD 1 • Q1/5 • NEAREST RELATED

Write as a single power. Then find its value.

$2^{4} \times 2^{-6}$

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-F1-Q2

Grade 8 • Expressions & Equations • 8.EE.A.1 • Set 12 • FORWARD 1 • Q2/5 • NEAREST RELATED

368

QUESTION ID

S12-F1-Q2

TYPE

Short answer

CORRECT ANSWER

$2^{-2} = 1/4$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-F1-Q2

Grade 8 • Expressions & Equations • 8.EE.A.1 • Set 12 • FORWARD 1 • Q2/5 • NEAREST RELATED

Which expression is equal to 1?

A. $7^2 \times 7^2$

B. $7^{-2} \times 7^{-2}$

C. $7^{-2} + 7^2$

D. $7^{-2} \times 7^2$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-F1-Q3

Grade 8 • Expressions & Equations • 8.EE.A.1 • Set 12 • FORWARD 1 • Q3/5 • NEAREST RELATED

369

QUESTION ID

S12-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. $7^{-2} \times 7^2$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-F1-Q3

Grade 8 • Expressions & Equations • 8.EE.A.1 • Set 12 • FORWARD 1 • Q3/5 • NEAREST RELATED

True or false?

$10^{-2} = \frac{1}{100}$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-F1-Q4

Grade 8 • Expressions & Equations • 8.EE.A.1 • Set 12 • FORWARD 1 • Q4/5 • NEAREST RELATED

370

QUESTION ID

S12-F1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-F1-Q4

Grade 8 • Expressions & Equations • 8.EE.A.1 • Set 12 • FORWARD 1 • Q4/5 • NEAREST RELATED

True or false?

$3^{-2} = -9$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-F1-Q5

Grade 8 • Expressions & Equations • 8.EE.A.1 • Set 12 • FORWARD 1 • Q5/5 • NEAREST RELATED

371

QUESTION ID

S12-F1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE $3^{-2} = 1/9$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-F1-Q5

Grade 8 • Expressions & Equations • 8.EE.A.1 • Set 12 • FORWARD 1 • Q5/5 • NEAREST RELATED

SET 12 • 7.NS.A.1.a

FORWARD 2

Explain the steps used to solve an equation • Grade 9 (High School) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.1

To solve $x - 9 = 4$, Lia adds 9 to both sides.

What property of equality justifies this step?

Property: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-F2-Q1

Grade 9 (High School) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.1 • Set 12 • FORWARD 2 • Q1/5

373

QUESTION ID

S12-F2-Q1

TYPE

Short answer

CORRECT ANSWER

The addition property of equality

GRADING NOTE Any answer saying that adding the same number to both sides keeps the equation true.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-F2-Q1

Grade 9 (High School) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.1 • Set 12 • FORWARD 2 • Q1/5

Solve x + 15 = 6.

What number do you add to both sides, and what is x?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-F2-Q2

Grade 9 (High School) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.1 • Set 12 • FORWARD 2 • Q2/5

374

QUESTION ID

S12-F2-Q2

TYPE

Short answer

CORRECT ANSWER

Add −15 to both sides; x = −9

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-F2-Q2

Grade 9 (High School) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.1 • Set 12 • FORWARD 2 • Q2/5

Which step removes the 5 when solving 3x + 5 = 20?

A. Multiply both sides by −5.

B. Divide both sides by 5.

C. Add −5 to both sides.

D. Add 5 to both sides.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-F2-Q3

Grade 9 (High School) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.1 • Set 12 • FORWARD 2 • Q3/5

375

QUESTION ID

S12-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. Add −5 to both sides.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-F2-Q3

Grade 9 (High School) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.1 • Set 12 • FORWARD 2 • Q3/5

True or false?

Adding −4 to both sides of x + 4 = 10 gives x = 6.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-F2-Q4

Grade 9 (High School) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.1 • Set 12 • FORWARD 2 • Q4/5

376

QUESTION ID

S12-F2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-F2-Q4

Grade 9 (High School) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.1 • Set 12 • FORWARD 2 • Q4/5

True or false?

Adding 8 to both sides of $x - 8 = 3$ gives $x = -5$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-F2-Q5

Grade 9 (High School) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.1 • Set 12 • FORWARD 2 • Q5/5

377

QUESTION ID

S12-F2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE It gives $x = 11$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-F2-Q5

Grade 9 (High School) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.1 • Set 12 • FORWARD 2 • Q5/5

SET 13 • 7.NS.A.1.b • THE NUMBER SYSTEM

Add rational numbers on a number line

MAIN	Grade 7 main question	**7.NS.A.1.b**
BACKWARD 1	Find and position integers on a number line	**6.NS.C.6.c**
BACKWARD 2	Add decimals	**5.NBT.B.7**
FORWARD 1	Locate irrational numbers on a number line	**8.NS.A.2**
FORWARD 2	Decide whether a sum is rational or irrational	**HSN-RN.B.3**

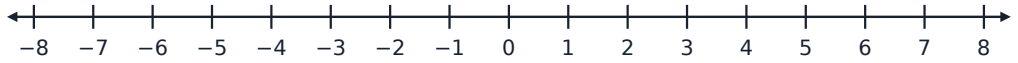
SET 13 • 7.NS.A.1.b

MAIN

Grade 7 main question • Grade 7 • The Number System • 7.NS.A.1.b

Use the number line to add.

$-3 + 7$



Sum: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-M-Q1

Grade 7 • The Number System • 7.NS.A.1.b • Set 13 • MAIN • Q1/5

380

QUESTION ID

S13-M-Q1

TYPE

Short answer

CORRECT ANSWER

4

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-M-Q1

Grade 7 • The Number System • 7.NS.A.1.b • Set 13 • MAIN • Q1/5

Add.

$-6 + (-4)$

Sum: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-M-Q2

Grade 7 • The Number System • 7.NS.A.1.b • Set 13 • MAIN • Q2/5

381

QUESTION ID

S13-M-Q2

TYPE

Short answer

CORRECT ANSWER

−10

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-M-Q2

Grade 7 • The Number System • 7.NS.A.1.b • Set 13 • MAIN • Q2/5

Start at 2 on a number line and move 5 units to the left.

Which sum does this show?

A. −5 + (−2)

B. −2 + 5

C. 2 + 5

D. 2 + (−5)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-M-Q3

Grade 7 • The Number System • 7.NS.A.1.b • Set 13 • MAIN • Q3/5

382

QUESTION ID

S13-M-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. 2 + (−5)

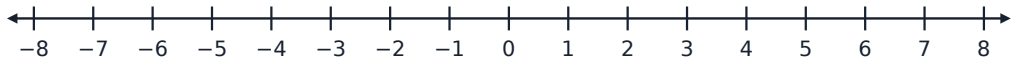
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-M-Q3

Grade 7 • The Number System • 7.NS.A.1.b • Set 13 • MAIN • Q3/5

Show −2 + 6 on the number line. Start at −2 and draw an arrow 6 units long.

What is the sum?



Sum: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-M-Q4

Grade 7 • The Number System • 7.NS.A.1.b • Set 13 • MAIN • Q4/5

383

QUESTION ID

S13-M-Q4

TYPE

Drawing + written answer

CORRECT ANSWER

Drawing: An arrow that starts at −2 and ends at 4 (6 units to the right)

Written: 4

GRADING NOTE Grade both: the arrow from −2 to 4, and the sum 4.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-M-Q4

Grade 7 • The Number System • 7.NS.A.1.b • Set 13 • MAIN • Q4/5

True or false?

$-4.5 + 2 = -2.5$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-M-Q5

Grade 7 • The Number System • 7.NS.A.1.b • Set 13 • MAIN • Q5/5

384

QUESTION ID

S13-M-Q5

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-M-Q5

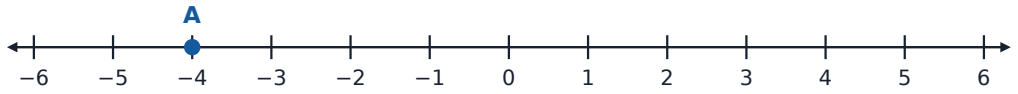
Grade 7 • The Number System • 7.NS.A.1.b • Set 13 • MAIN • Q5/5

SET 13 • 7.NS.A.1.b

BACKWARD 1

Find and position integers on a number line • Grade 6 • The Number System • 6.NS.C.6.c

What number is at point A?



Number: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-B1-Q1

Grade 6 • The Number System • 6.NS.C.6.c • Set 13 • BACKWARD 1 • Q1/5

386

QUESTION ID

S13-B1-Q1

TYPE

Short answer

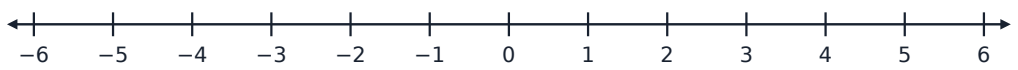
CORRECT ANSWER

−4

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-B1-Q1

Grade 6 • The Number System • 6.NS.C.6.c • Set 13 • BACKWARD 1 • Q1/5

Plot −3 and 5 on the number line.



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-B1-Q2

Grade 6 • The Number System • 6.NS.C.6.c • Set 13 • BACKWARD 1 • Q2/5

387

QUESTION ID

S13-B1-Q2

TYPE

Drawing / plotting

CORRECT ANSWER

Points at −3 and 5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-B1-Q2

Grade 6 • The Number System • 6.NS.C.6.c • Set 13 • BACKWARD 1 • Q2/5

Which number is farthest to the left on a number line?

A. −2

B. 0

C. 3

D. −7

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-B1-Q3

Grade 6 • The Number System • 6.NS.C.6.c • Set 13 • BACKWARD 1 • Q3/5

388

QUESTION ID

S13-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. −7

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-B1-Q3

Grade 6 • The Number System • 6.NS.C.6.c • Set 13 • BACKWARD 1 • Q3/5

True or false?

−1 is to the right of −5 on a number line.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-B1-Q4

Grade 6 • The Number System • 6.NS.C.6.c • Set 13 • BACKWARD 1 • Q4/5

389

QUESTION ID

S13-B1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-B1-Q4

Grade 6 • The Number System • 6.NS.C.6.c • Set 13 • BACKWARD 1 • Q4/5

True or false?

−6 is between −3 and 0 on a number line.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-B1-Q5

Grade 6 • The Number System • 6.NS.C.6.c • Set 13 • BACKWARD 1 • Q5/5

390

QUESTION ID

S13-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-B1-Q5

Grade 6 • The Number System • 6.NS.C.6.c • Set 13 • BACKWARD 1 • Q5/5

SET 13 • 7.NS.A.1.b

BACKWARD 2

Add decimals • Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7

Add.

3.75 + 2.5

Sum: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-B2-Q1

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7 • Set 13 • BACKWARD 2 • Q1/5

392

QUESTION ID

S13-B2-Q1

TYPE

Short answer

CORRECT ANSWER

6.25

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-B2-Q1

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7 • Set 13 • BACKWARD 2 • Q1/5

Add.

0.8 + 4.65

Sum: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-B2-Q2

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7 • Set 13 • BACKWARD 2 • Q2/5

393

QUESTION ID

S13-B2-Q2

TYPE

Short answer

CORRECT ANSWER

5.45

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-B2-Q2

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7 • Set 13 • BACKWARD 2 • Q2/5

Add.

12.4 + 3.06

A. 46.4

B. 15.10

C. 12.71

D. 15.46

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-B2-Q3

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7 • Set 13 • BACKWARD 2 • Q3/5

394

QUESTION ID

S13-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. 15.46

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-B2-Q3

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7 • Set 13 • BACKWARD 2 • Q3/5

True or false?

$2.7 + 1.3 = 4$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-B2-Q4

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7 • Set 13 • BACKWARD 2 • Q4/5

395

QUESTION ID

S13-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-B2-Q4

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7 • Set 13 • BACKWARD 2 • Q4/5

True or false?

5.6 + 0.04 = 6

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-B2-Q5

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7 • Set 13 • BACKWARD 2 • Q5/5

396

QUESTION ID

S13-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-B2-Q5

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7 • Set 13 • BACKWARD 2 • Q5/5

SET 13 • 7.NS.A.1.b

FORWARD 1

Locate irrational numbers on a number line • Grade 8 • The Number System • 8.NS.A.2

Between which two whole numbers is √20?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-F1-Q1

Grade 8 • The Number System • 8.NS.A.2 • Set 13 • FORWARD 1 • Q1/5

398

QUESTION ID

S13-F1-Q1

TYPE

Short answer

CORRECT ANSWER

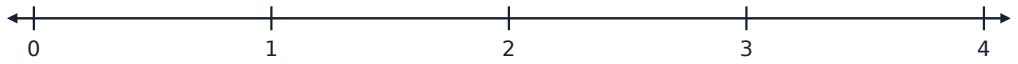
4 and 5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-F1-Q1

Grade 8 • The Number System • 8.NS.A.2 • Set 13 • FORWARD 1 • Q1/5

Plot an approximate location of √2 on the number line.



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-F1-Q2

Grade 8 • The Number System • 8.NS.A.2 • Set 13 • FORWARD 1 • Q2/5

399

QUESTION ID

S13-F1-Q2

TYPE

Drawing / plotting

CORRECT ANSWER

A point between 1 and 2, a little less than halfway (about 1.4)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-F1-Q2

Grade 8 • The Number System • 8.NS.A.2 • Set 13 • FORWARD 1 • Q2/5

Which is the best approximation of √50 to the nearest tenth?

A. 25

B. 7.1

C. 7.5

D. 7.0

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-F1-Q3

Grade 8 • The Number System • 8.NS.A.2 • Set 13 • FORWARD 1 • Q3/5

400

QUESTION ID

S13-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. 7.1

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-F1-Q3

Grade 8 • The Number System • 8.NS.A.2 • Set 13 • FORWARD 1 • Q3/5

True or false?

√10 is between 3 and 4.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-F1-Q4

Grade 8 • The Number System • 8.NS.A.2 • Set 13 • FORWARD 1 • Q4/5

401

QUESTION ID

S13-F1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-F1-Q4

Grade 8 • The Number System • 8.NS.A.2 • Set 13 • FORWARD 1 • Q4/5

True or false?

π is less than 3.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-F1-Q5

Grade 8 • The Number System • 8.NS.A.2 • Set 13 • FORWARD 1 • Q5/5

QUESTION ID

S13-F1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-F1-Q5

Grade 8 • The Number System • 8.NS.A.2 • Set 13 • FORWARD 1 • Q5/5

SET 13 • 7.NS.A.1.b

FORWARD 2

Decide whether a sum is rational or irrational • Grade 9 (High School) • Number & Quantity: The Real Number System • HSN-RN.B.3

Is 3 + √2 rational or irrational?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-F2-Q1

Grade 9 (High School) • Number & Quantity: The Real Number System • HSN-RN.B.3 • Set 13 • FORWARD 2 • Q1/5

404

QUESTION ID

S13-F2-Q1

TYPE

Short answer

CORRECT ANSWER

Irrational

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-F2-Q1

Grade 9 (High School) • Number & Quantity: The Real Number System • HSN-RN.B.3 • Set 13 • FORWARD 2 • Q1/5

Is $\frac{1}{2} + \frac{3}{4}$ rational or irrational? Give its value.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-F2-Q2

Grade 9 (High School) • Number & Quantity: The Real Number System • HSN-RN.B.3 • Set 13 • FORWARD 2 • Q2/5

405

QUESTION ID

S13-F2-Q2

TYPE

Short answer

CORRECT ANSWER

Rational; 5/4

GRADING NOTE 1 1/4 or 1.25 is also correct for the value.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-F2-Q2

Grade 9 (High School) • Number & Quantity: The Real Number System • HSN-RN.B.3 • Set 13 • FORWARD 2 • Q2/5

Which sum is irrational?

A. $\sqrt{9} + 4$

B. $0.5 + \frac{1}{2}$

C. $-2 + 2$

D. $\sqrt{3} + 4$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-F2-Q3

Grade 9 (High School) • Number & Quantity: The Real Number System • HSN-RN.B.3 • Set 13 • FORWARD 2 • Q3/5

406

QUESTION ID

S13-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. $\sqrt{3} + 4$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-F2-Q3

Grade 9 (High School) • Number & Quantity: The Real Number System • HSN-RN.B.3 • Set 13 • FORWARD 2 • Q3/5

True or false?

The sum of two rational numbers is always rational.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-F2-Q4

Grade 9 (High School) • Number & Quantity: The Real Number System • HSN-RN.B.3 • Set 13 • FORWARD 2 • Q4/5

407

QUESTION ID

S13-F2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-F2-Q4

Grade 9 (High School) • Number & Quantity: The Real Number System • HSN-RN.B.3 • Set 13 • FORWARD 2 • Q4/5

True or false?

The sum of a rational number and an irrational number can be rational.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-F2-Q5

Grade 9 (High School) • Number & Quantity: The Real Number System • HSN-RN.B.3 • Set 13 • FORWARD 2 • Q5/5

408

QUESTION ID

S13-F2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-F2-Q5

Grade 9 (High School) • Number & Quantity: The Real Number System • HSN-RN.B.3 • Set 13 • FORWARD 2 • Q5/5

SET 14 • 7.NS.A.1.b • THE NUMBER SYSTEM

Use additive inverses: a number plus its opposite is 0

MAIN	Grade 7 main question	**7.NS.A.1.b**
BACKWARD 1	Opposites are the same distance from 0	**6.NS.C.6.a**
BACKWARD 2	Interpret absolute value as distance from 0	**6.NS.C.7.c**
FORWARD 1	Tell whether a linear equation has one, no, or infinitely many solut…	**8.EE.C.7.a**
FORWARD 2	Solve linear equations in one variable	**HSA-REI.B.3**

SET 14 • 7.NS.A.1.b

MAIN

Grade 7 main question • Grade 7 • The Number System • 7.NS.A.1.b

What number added to −9.5 gives 0?

Number: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-M-Q1

Grade 7 • The Number System • 7.NS.A.1.b • Set 14 • MAIN • Q1/5

411

QUESTION ID

S14-M-Q1

TYPE

Short answer

CORRECT ANSWER

9.5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-M-Q1

Grade 7 • The Number System • 7.NS.A.1.b • Set 14 • MAIN • Q1/5

Fill in the blank.

$\frac{3}{4}$ + ___ = 0

Blank: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-M-Q2

Grade 7 • The Number System • 7.NS.A.1.b • Set 14 • MAIN • Q2/5

412

QUESTION ID

S14-M-Q2

TYPE

Short answer

CORRECT ANSWER

−3/4

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-M-Q2

Grade 7 • The Number System • 7.NS.A.1.b • Set 14 • MAIN • Q2/5

Which sum is equal to 0?

A. $-2.6 + (-2.6)$

B. $-2.6 + 2.6$

C. $2.6 + 0$

D. $-2.6 + 6.2$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-M-Q3

Grade 7 • The Number System • 7.NS.A.1.b • Set 14 • MAIN • Q3/5

413

QUESTION ID

S14-M-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. −2.6 + 2.6

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-M-Q3

Grade 7 • The Number System • 7.NS.A.1.b • Set 14 • MAIN • Q3/5

True or false?

For any rational number p, −p + p = 0.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-M-Q4

Grade 7 • The Number System • 7.NS.A.1.b • Set 14 • MAIN • Q4/5

414

QUESTION ID

S14-M-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-M-Q4

Grade 7 • The Number System • 7.NS.A.1.b • Set 14 • MAIN • Q4/5

True or false?

The additive inverse of −7 is −7.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-M-Q5

Grade 7 • The Number System • 7.NS.A.1.b • Set 14 • MAIN • Q5/5

415

QUESTION ID

S14-M-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE The additive inverse of −7 is 7.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-M-Q5

Grade 7 • The Number System • 7.NS.A.1.b • Set 14 • MAIN • Q5/5

SET 14 • 7.NS.A.1.b

BACKWARD 1

Opposites are the same distance from 0 • Grade 6 • The Number System • 6.NS.C.6.a

What is the opposite of 2.5?

Opposite: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-B1-Q1

Grade 6 • The Number System • 6.NS.C.6.a • Set 14 • BACKWARD 1 • Q1/5

417

QUESTION ID

S14-B1-Q1

TYPE

Short answer

CORRECT ANSWER

−2.5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-B1-Q1

Grade 6 • The Number System • 6.NS.C.6.a • Set 14 • BACKWARD 1 • Q1/5

Which pair of numbers are opposites?

A. −11 and 11

B. −11 and −11

C. 11 and $\frac{1}{11}$

D. 1 and −11

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-B1-Q2

Grade 6 • The Number System • 6.NS.C.6.a • Set 14 • BACKWARD 1 • Q2/5

418

QUESTION ID

S14-B1-Q2

TYPE

Multiple choice

CORRECT ANSWER

A. −11 and 11

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-B1-Q2

Grade 6 • The Number System • 6.NS.C.6.a • Set 14 • BACKWARD 1 • Q2/5

How far is −9 from 0 on a number line?

Distance: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-B1-Q3

Grade 6 • The Number System • 6.NS.C.6.a • Set 14 • BACKWARD 1 • Q3/5

419

QUESTION ID

S14-B1-Q3

TYPE

Short answer

CORRECT ANSWER

9 units

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-B1-Q3

Grade 6 • The Number System • 6.NS.C.6.a • Set 14 • BACKWARD 1 • Q3/5

True or false?

$-\frac{2}{3}$ and $\frac{2}{3}$ are the same distance from 0.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-B1-Q4

Grade 6 • The Number System • 6.NS.C.6.a • Set 14 • BACKWARD 1 • Q4/5

420

QUESTION ID

S14-B1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-B1-Q4

Grade 6 • The Number System • 6.NS.C.6.a • Set 14 • BACKWARD 1 • Q4/5

True or false?

The opposite of 0 is 1.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-B1-Q5

Grade 6 • The Number System • 6.NS.C.6.a • Set 14 • BACKWARD 1 • Q5/5

421

QUESTION ID

S14-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-B1-Q5

Grade 6 • The Number System • 6.NS.C.6.a • Set 14 • BACKWARD 1 • Q5/5

SET 14 • 7.NS.A.1.b

BACKWARD 2

Interpret absolute value as distance from 0 • Grade 6 • The Number System • 6.NS.C.7.c

Find the value.

$|-18|$

Value: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-B2-Q1

Grade 6 • The Number System • 6.NS.C.7.c • Set 14 • BACKWARD 2 • Q1/5

423

QUESTION ID

S14-B2-Q1

TYPE

Short answer

CORRECT ANSWER

18

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-B2-Q1

Grade 6 • The Number System • 6.NS.C.7.c • Set 14 • BACKWARD 2 • Q1/5

A diver is at −30 feet.

How far is she from the surface (0 feet)?

Distance: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-B2-Q2

Grade 6 • The Number System • 6.NS.C.7.c • Set 14 • BACKWARD 2 • Q2/5

424

QUESTION ID

S14-B2-Q2

TYPE

Short answer

CORRECT ANSWER

30 feet

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-B2-Q2

Grade 6 • The Number System • 6.NS.C.7.c • Set 14 • BACKWARD 2 • Q2/5

Which number has an absolute value of 6?

A. $\frac{1}{6}$

B. -6

C. $-\frac{1}{6}$

D. 0

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-B2-Q3

Grade 6 • The Number System • 6.NS.C.7.c • Set 14 • BACKWARD 2 • Q3/5

425

QUESTION ID

S14-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. −6

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-B2-Q3

Grade 6 • The Number System • 6.NS.C.7.c • Set 14 • BACKWARD 2 • Q3/5

True or false?

$|-3.5| = |3.5|$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-B2-Q4

Grade 6 • The Number System • 6.NS.C.7.c • Set 14 • BACKWARD 2 • Q4/5

426

QUESTION ID

S14-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-B2-Q4

Grade 6 • The Number System • 6.NS.C.7.c • Set 14 • BACKWARD 2 • Q4/5

True or false?

$|-20| = -20$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-B2-Q5

Grade 6 • The Number System • 6.NS.C.7.c • Set 14 • BACKWARD 2 • Q5/5

427

QUESTION ID

S14-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-B2-Q5

Grade 6 • The Number System • 6.NS.C.7.c • Set 14 • BACKWARD 2 • Q5/5

SET 14 • 7.NS.A.1.b

FORWARD 1

Tell whether a linear equation has one, no, or infinitely many solutions • Grade 8 • Expressions & Equations • 8.EE.C.7.a

How many solutions does this equation have?

$3x + 5 = 3x - 2$

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-F1-Q1

Grade 8 • Expressions & Equations • 8.EE.C.7.a • Set 14 • FORWARD 1 • Q1/5

429

QUESTION ID

S14-F1-Q1

TYPE

Short answer

CORRECT ANSWER

No solution

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-F1-Q1

Grade 8 • Expressions & Equations • 8.EE.C.7.a • Set 14 • FORWARD 1 • Q1/5

How many solutions does this equation have?

$2(x + 4) = 2x + 8$

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-F1-Q2

Grade 8 • Expressions & Equations • 8.EE.C.7.a • Set 14 • FORWARD 1 • Q2/5

430

QUESTION ID

S14-F1-Q2

TYPE

Short answer

CORRECT ANSWER

Infinitely many solutions

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-F1-Q2

Grade 8 • Expressions & Equations • 8.EE.C.7.a • Set 14 • FORWARD 1 • Q2/5

Which equation has exactly one solution?

A. $4x + 1 = 2x + 7$

B. $2(2x) = 4x$

C. $4x + 1 = 4x + 7$

D. $4x + 1 = 4x + 1$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-F1-Q3

Grade 8 • Expressions & Equations • 8.EE.C.7.a • Set 14 • FORWARD 1 • Q3/5

431

QUESTION ID

S14-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. $4x + 1 = 2x + 7$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-F1-Q3

Grade 8 • Expressions & Equations • 8.EE.C.7.a • Set 14 • FORWARD 1 • Q3/5

True or false?

$x + 6 = x - 6$ has no solution.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-F1-Q4

Grade 8 • Expressions & Equations • 8.EE.C.7.a • Set 14 • FORWARD 1 • Q4/5

QUESTION ID

S14-F1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-F1-Q4

Grade 8 • Expressions & Equations • 8.EE.C.7.a • Set 14 • FORWARD 1 • Q4/5

True or false?

$5x - 5x = 0$ has no solution.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-F1-Q5

Grade 8 • Expressions & Equations • 8.EE.C.7.a • Set 14 • FORWARD 1 • Q5/5

433

QUESTION ID

S14-F1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE It is true for every x, so it has infinitely many solutions.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-F1-Q5

Grade 8 • Expressions & Equations • 8.EE.C.7.a • Set 14 • FORWARD 1 • Q5/5

SET 14 • 7.NS.A.1.b

FORWARD 2

Solve linear equations in one variable • Grade 9 (High School) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3

Solve for x.

$4x - 7 = -2x + 11$

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-F2-Q1

Grade 9 (High School) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 14 • FORWARD 2 • Q1/5

435

QUESTION ID

S14-F2-Q1

TYPE

Short answer

CORRECT ANSWER

3

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-F2-Q1

Grade 9 (High School) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 14 • FORWARD 2 • Q1/5

Solve for x.

$-3(x - 2) = 15$

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-F2-Q2

Grade 9 (High School) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 14 • FORWARD 2 • Q2/5

436

QUESTION ID

S14-F2-Q2

TYPE

Short answer

CORRECT ANSWER

−3

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-F2-Q2

Grade 9 (High School) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 14 • FORWARD 2 • Q2/5

Solve for x.

$\frac{x}{5} + 3 = -1$

A. x = −20

B. x = −4

C. x = 10

D. x = 20

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-F2-Q3

Grade 9 (High School) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 14 • FORWARD 2 • Q3/5

437

QUESTION ID

S14-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. x = −20

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-F2-Q3

Grade 9 (High School) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 14 • FORWARD 2 • Q3/5

True or false?

$x = -2$ is the solution of $6 + 2x = 2$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-F2-Q4

Grade 9 (High School) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 14 • FORWARD 2 • Q4/5

438

QUESTION ID

S14-F2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-F2-Q4

Grade 9 (High School) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 14 • FORWARD 2 • Q4/5

True or false?

$x = 4$ is the solution of $7 - x = 11$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-F2-Q5

Grade 9 (High School) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 14 • FORWARD 2 • Q5/5

439

QUESTION ID

S14-F2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE The solution is $x = -4$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-F2-Q5

Grade 9 (High School) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 14 • FORWARD 2 • Q5/5

SET 15 • 7.NS.A.1.b • THE NUMBER SYSTEM

Interpret sums of rational numbers in real-world situations

MAIN	Grade 7 main question	**7.NS.A.1.b**
BACKWARD 1	Use signed numbers in a real-world situation	**6.NS.C.5**
BACKWARD 2	Compare signed numbers in a real-world situation	**6.NS.C.7.b**
FORWARD 1	Model a changing quantity with a linear function	**8.F.B.4**
FORWARD 2	Evaluate a function and interpret the value in context	**HSF-IF.A.2**

SET 15 • 7.NS.A.1.b

MAIN

Grade 7 main question • Grade 7 • The Number System • 7.NS.A.1.b

The temperature was −5°F at dawn. It rose 12°F by noon.

What was the temperature at noon?

Temperature: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-M-Q1

Grade 7 • The Number System • 7.NS.A.1.b • Set 15 • MAIN • Q1/5

442

QUESTION ID

S15-M-Q1

TYPE

Short answer

CORRECT ANSWER

7°F

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-M-Q1

Grade 7 • The Number System • 7.NS.A.1.b • Set 15 • MAIN • Q1/5

A submarine is at −240 feet. It rises 85 feet.

What is its new position?

Position: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-M-Q2

Grade 7 • The Number System • 7.NS.A.1.b • Set 15 • MAIN • Q2/5

443

QUESTION ID

S15-M-Q2

TYPE

Short answer

CORRECT ANSWER

−155 feet

GRADING NOTE "155 feet below sea level" is also correct.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-M-Q2

Grade 7 • The Number System • 7.NS.A.1.b • Set 15 • MAIN • Q2/5

Jo's account balance is −$18. She deposits $25.

Which sum gives her new balance?

A. −18 + (−25) = −43

B. 18 + 25 = 43

C. −18 + 25 = 7

D. 18 + (−25) = −7

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-M-Q3

Grade 7 • The Number System • 7.NS.A.1.b • Set 15 • MAIN • Q3/5

444

QUESTION ID

S15-M-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. −18 + 25 = 7

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-M-Q3

Grade 7 • The Number System • 7.NS.A.1.b • Set 15 • MAIN • Q3/5

True or false?

A football team loses 6 yards, then gains 4 yards. The total change is −2 yards.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-M-Q4

Grade 7 • The Number System • 7.NS.A.1.b • Set 15 • MAIN • Q4/5

445

QUESTION ID

S15-M-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-M-Q4

Grade 7 • The Number System • 7.NS.A.1.b • Set 15 • MAIN • Q4/5

True or false?

A hiker 350 feet below sea level climbs 400 feet. She ends 50 feet below sea level.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-M-Q5

Grade 7 • The Number System • 7.NS.A.1.b • Set 15 • MAIN • Q5/5

446

QUESTION ID

S15-M-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE She ends 50 feet above sea level.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-M-Q5

Grade 7 • The Number System • 7.NS.A.1.b • Set 15 • MAIN • Q5/5

SET 15 • 7.NS.A.1.b

BACKWARD 1

Use signed numbers in a real-world situation • Grade 6 • The Number System • 6.NS.C.5

A stock's price changed by −$3.25.

Was this a gain or a loss, and of how much?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-B1-Q1

Grade 6 • The Number System • 6.NS.C.5 • Set 15 • BACKWARD 1 • Q1/5

448

QUESTION ID

S15-B1-Q1

TYPE

Short answer

CORRECT ANSWER

A loss of $3.25

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-B1-Q1

Grade 6 • The Number System • 6.NS.C.5 • Set 15 • BACKWARD 1 • Q1/5

Which number describes 15 meters above sea level?

A. $\frac{1}{15}$

B. 15

C. 0

D. -15

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-B1-Q2

Grade 6 • The Number System • 6.NS.C.5 • Set 15 • BACKWARD 1 • Q2/5

449

QUESTION ID

S15-B1-Q2

TYPE

Multiple choice

CORRECT ANSWER

B. 15

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-B1-Q2

Grade 6 • The Number System • 6.NS.C.5 • Set 15 • BACKWARD 1 • Q2/5

Write a number for a temperature of 8 degrees below zero.

Number: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-B1-Q3

Grade 6 • The Number System • 6.NS.C.5 • Set 15 • BACKWARD 1 • Q3/5

450

QUESTION ID

S15-B1-Q3

TYPE

Short answer

CORRECT ANSWER

−8°

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-B1-Q3

Grade 6 • The Number System • 6.NS.C.5 • Set 15 • BACKWARD 1 • Q3/5

True or false?

In a bank account, a balance of −$50 can mean $50 is owed.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-B1-Q4

Grade 6 • The Number System • 6.NS.C.5 • Set 15 • BACKWARD 1 • Q4/5

451

QUESTION ID

S15-B1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-B1-Q4

Grade 6 • The Number System • 6.NS.C.5 • Set 15 • BACKWARD 1 • Q4/5

True or false?

A temperature of −2° is warmer than 0°.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-B1-Q5

Grade 6 • The Number System • 6.NS.C.5 • Set 15 • BACKWARD 1 • Q5/5

452

QUESTION ID

S15-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-B1-Q5

Grade 6 • The Number System • 6.NS.C.5 • Set 15 • BACKWARD 1 • Q5/5

SET 15 • 7.NS.A.1.b

BACKWARD 2

Compare signed numbers in a real-world situation • Grade 6 • The Number System • 6.NS.C.7.b

Which temperature is colder, −8°C or −3°C?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-B2-Q1

Grade 6 • The Number System • 6.NS.C.7.b • Set 15 • BACKWARD 2 • Q1/5

454

QUESTION ID

S15-B2-Q1

TYPE

Short answer

CORRECT ANSWER

−8°C

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-B2-Q1

Grade 6 • The Number System • 6.NS.C.7.b • Set 15 • BACKWARD 2 • Q1/5

Which statement is true?

A. $-7 < -12$

B. $-12 > -7$

C. $-12 < -7$

D. $-7 = -12$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-B2-Q2

Grade 6 • The Number System • 6.NS.C.7.b • Set 15 • BACKWARD 2 • Q2/5

455

QUESTION ID

S15-B2-Q2

TYPE

Multiple choice

CORRECT ANSWER

C. $-12 < -7$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-B2-Q2

Grade 6 • The Number System • 6.NS.C.7.b • Set 15 • BACKWARD 2 • Q2/5

Write an inequality to compare elevations of −15 feet and −9 feet.

Inequality: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-B2-Q3

Grade 6 • The Number System • 6.NS.C.7.b • Set 15 • BACKWARD 2 • Q3/5

456

QUESTION ID

S15-B2-Q3

TYPE

Short answer

CORRECT ANSWER

−15 < −9

GRADING NOTE −9 > −15 is also correct.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-B2-Q3

Grade 6 • The Number System • 6.NS.C.7.b • Set 15 • BACKWARD 2 • Q3/5

True or false?

A balance of −$20 is less than a balance of −$5.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-B2-Q4

Grade 6 • The Number System • 6.NS.C.7.b • Set 15 • BACKWARD 2 • Q4/5

457

QUESTION ID

S15-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-B2-Q4

Grade 6 • The Number System • 6.NS.C.7.b • Set 15 • BACKWARD 2 • Q4/5

True or false?

−4°F is colder than −10°F.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-B2-Q5

Grade 6 • The Number System • 6.NS.C.7.b • Set 15 • BACKWARD 2 • Q5/5

458

QUESTION ID

S15-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-B2-Q5

Grade 6 • The Number System • 6.NS.C.7.b • Set 15 • BACKWARD 2 • Q5/5

SET 15 • 7.NS.A.1.b

FORWARD 1

Model a changing quantity with a linear function • Grade 8 • Functions • 8.F.B.4

A diver starts at −10 feet and descends 4 feet per minute.

Write a function for her position d after t minutes.

d = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-F1-Q1

Grade 8 • Functions • 8.F.B.4 • Set 15 • FORWARD 1 • Q1/5

460

QUESTION ID

S15-F1-Q1

TYPE

Short answer

CORRECT ANSWER

−4t − 10

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-F1-Q1

Grade 8 • Functions • 8.F.B.4 • Set 15 • FORWARD 1 • Q1/5

The temperature is −6°F and rises 3°F per hour.

Write a function for the temperature T after h hours.

T = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-F1-Q2

Grade 8 • Functions • 8.F.B.4 • Set 15 • FORWARD 1 • Q2/5

461

QUESTION ID

S15-F1-Q2

TYPE

Short answer

CORRECT ANSWER

3h − 6

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-F1-Q2

Grade 8 • Functions • 8.F.B.4 • Set 15 • FORWARD 1 • Q2/5

An account balance starts at −$40 and goes up $15 per week.

What is the balance after 4 weeks?

A. $20

B. −$100

C. −$20

D. $60

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-F1-Q3

Grade 8 • Functions • 8.F.B.4 • Set 15 • FORWARD 1 • Q3/5

462

QUESTION ID

S15-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. $20

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-F1-Q3

Grade 8 • Functions • 8.F.B.4 • Set 15 • FORWARD 1 • Q3/5

True or false?

For T = 3h − 6, the temperature reaches 0°F after 2 hours.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-F1-Q4

Grade 8 • Functions • 8.F.B.4 • Set 15 • FORWARD 1 • Q4/5

463

QUESTION ID

S15-F1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-F1-Q4

Grade 8 • Functions • 8.F.B.4 • Set 15 • FORWARD 1 • Q4/5

True or false?

For d = −4t − 10, the starting position is −4 feet.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-F1-Q5

Grade 8 • Functions • 8.F.B.4 • Set 15 • FORWARD 1 • Q5/5

464

QUESTION ID

S15-F1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE The starting position is −10 feet.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-F1-Q5

Grade 8 • Functions • 8.F.B.4 • Set 15 • FORWARD 1 • Q5/5

SET 15 • 7.NS.A.1.b

FORWARD 2

Evaluate a function and interpret the value in context • Grade 9 (High School) • Functions: Interpreting Functions • HSF-IF.A.2

$h(t) = -16t^2 + 64t$ gives the height, in feet, of a ball after t seconds.

Find h(1).

h(1) = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-F2-Q1

Grade 9 (High School) • Functions: Interpreting Functions • HSF-IF.A.2 • Set 15 • FORWARD 2 • Q1/5

466

QUESTION ID

S15-F2-Q1

TYPE

Short answer

CORRECT ANSWER

48 feet

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-F2-Q1

Grade 9 (High School) • Functions: Interpreting Functions • HSF-IF.A.2 • Set 15 • FORWARD 2 • Q1/5

T(h) = −2 + 1.5h gives the temperature in °C after h hours.

Find T(6). What does it mean?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-F2-Q2

Grade 9 (High School) • Functions: Interpreting Functions • HSF-IF.A.2 • Set 15 • FORWARD 2 • Q2/5

467

QUESTION ID

S15-F2-Q2

TYPE

Short answer

CORRECT ANSWER

7; the temperature is 7°C after 6 hours.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-F2-Q2

Grade 9 (High School) • Functions: Interpreting Functions • HSF-IF.A.2 • Set 15 • FORWARD 2 • Q2/5

$f(x) = 3x - 8$

What is $f(-2)$?

A. 14

B. -14

C. -2

D. 2

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-F2-Q3

Grade 9 (High School) • Functions: Interpreting Functions • HSF-IF.A.2 • Set 15 • FORWARD 2 • Q3/5

468

QUESTION ID

S15-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. −14

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-F2-Q3

Grade 9 (High School) • Functions: Interpreting Functions • HSF-IF.A.2 • Set 15 • FORWARD 2 • Q3/5

True or false?

If $g(x) = -x + 5$, then $g(-3) = 8$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-F2-Q4

Grade 9 (High School) • Functions: Interpreting Functions • HSF-IF.A.2 • Set 15 • FORWARD 2 • Q4/5

469

QUESTION ID

S15-F2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-F2-Q4

Grade 9 (High School) • Functions: Interpreting Functions • HSF-IF.A.2 • Set 15 • FORWARD 2 • Q4/5

True or false?

If $f(x) = x^2 - 4$, then $f(-3) = -13$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-F2-Q5

Grade 9 (High School) • Functions: Interpreting Functions • HSF-IF.A.2 • Set 15 • FORWARD 2 • Q5/5

470

QUESTION ID

S15-F2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE $f(-3) = 5$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-F2-Q5

Grade 9 (High School) • Functions: Interpreting Functions • HSF-IF.A.2 • Set 15 • FORWARD 2 • Q5/5

SET 16 • 7.NS.A.1.c • THE NUMBER SYSTEM

Subtract rational numbers by adding the additive inverse

MAIN	Grade 7 main question	**7.NS.A.1.c**
BACKWARD 1	Subtract decimals	**5.NBT.B.7**
BACKWARD 2	Subtract fractions with unlike denominators	**5.NF.A.1**
FORWARD 1	Find a rate of change from two points with negative coordinates	**8.F.B.4**
FORWARD 2	Subtract polynomials	**HSA-APR.A.1**

SET 16 • 7.NS.A.1.c

MAIN

Grade 7 main question • Grade 7 • The Number System • 7.NS.A.1.c

Rewrite as an addition expression. Then find the value.

$4 - 9$

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-M-Q1

Grade 7 • The Number System • 7.NS.A.1.c • Set 16 • MAIN • Q1/5

473

QUESTION ID

S16-M-Q1

TYPE

Short answer

CORRECT ANSWER

$4 + (-9) = -5$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-M-Q1

Grade 7 • The Number System • 7.NS.A.1.c • Set 16 • MAIN • Q1/5

Subtract.

$-7 - (-12)$

Difference: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-M-Q2

Grade 7 • The Number System • 7.NS.A.1.c • Set 16 • MAIN • Q2/5

474

QUESTION ID

S16-M-Q2

TYPE

Short answer

CORRECT ANSWER

5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-M-Q2

Grade 7 • The Number System • 7.NS.A.1.c • Set 16 • MAIN • Q2/5

Which expression is equal to $-3 - 8$?

A. $3 + (-8)$

B. $3 + 8$

C. $-3 + 8$

D. $-3 + (-8)$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-M-Q3

Grade 7 • The Number System • 7.NS.A.1.c • Set 16 • MAIN • Q3/5

QUESTION ID

S16-M-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. −3 + (−8)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-M-Q3

Grade 7 • The Number System • 7.NS.A.1.c • Set 16 • MAIN • Q3/5

True or false?

$-2.5 - 4 = -6.5$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-M-Q4

Grade 7 • The Number System • 7.NS.A.1.c • Set 16 • MAIN • Q4/5

476

QUESTION ID

S16-M-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-M-Q4

Grade 7 • The Number System • 7.NS.A.1.c • Set 16 • MAIN • Q4/5

True or false?

$\frac{1}{2} - (-\frac{1}{4}) = \frac{1}{4}$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-M-Q5

Grade 7 • The Number System • 7.NS.A.1.c • Set 16 • MAIN • Q5/5

477

QUESTION ID

S16-M-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE The difference is 3/4.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-M-Q5

Grade 7 • The Number System • 7.NS.A.1.c • Set 16 • MAIN • Q5/5

SET 16 • 7.NS.A.1.c

BACKWARD 1

Subtract decimals • Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7

Subtract.

8.2 − 3.75

Difference: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-B1-Q1

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7 • Set 16 • BACKWARD 1 • Q1/5

479

QUESTION ID

S16-B1-Q1

TYPE

Short answer

CORRECT ANSWER

4.45

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-B1-Q1

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7 • Set 16 • BACKWARD 1 • Q1/5

Subtract.

10 − 6.38

Difference: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-B1-Q2

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7 • Set 16 • BACKWARD 1 • Q2/5

480

QUESTION ID

S16-B1-Q2

TYPE

Short answer

CORRECT ANSWER

3.62

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-B1-Q2

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7 • Set 16 • BACKWARD 1 • Q2/5

Subtract.

$5.4 - 2.9$

A. 8.3

B. 2.5

C. 3.5

D. 2.15

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-B1-Q3

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7 • Set 16 • BACKWARD 1 • Q3/5

481

QUESTION ID

S16-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. 2.5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-B1-Q3

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7 • Set 16 • BACKWARD 1 • Q3/5

True or false?

$7.06 - 2.5 = 4.56$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-B1-Q4

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7 • Set 16 • BACKWARD 1 • Q4/5

482

QUESTION ID

S16-B1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-B1-Q4

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7 • Set 16 • BACKWARD 1 • Q4/5

True or false?

$9.3 - 0.7 = 9.6$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-B1-Q5

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7 • Set 16 • BACKWARD 1 • Q5/5

483

QUESTION ID

S16-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-B1-Q5

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7 • Set 16 • BACKWARD 1 • Q5/5

SET 16 • 7.NS.A.1.c

BACKWARD 2

Subtract fractions with unlike denominators • Grade 5 • Number & Operations–Fractions • 5.NF.A.1

Subtract.

$\frac{5}{6} - \frac{1}{4}$

Difference: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-B2-Q1

Grade 5 • Number & Operations–Fractions • 5.NF.A.1 • Set 16 • BACKWARD 2 • Q1/5

485

QUESTION ID

S16-B2-Q1

TYPE

Short answer

CORRECT ANSWER

7/12

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-B2-Q1

Grade 5 • Number & Operations–Fractions • 5.NF.A.1 • Set 16 • BACKWARD 2 • Q1/5

Subtract.

$3\frac{1}{2} - 1\frac{2}{3}$

Difference: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-B2-Q2

Grade 5 • Number & Operations–Fractions • 5.NF.A.1 • Set 16 • BACKWARD 2 • Q2/5

486

QUESTION ID

S16-B2-Q2

TYPE

Short answer

CORRECT ANSWER

1 5/6

GRADING NOTE 11/6 is also correct.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-B2-Q2

Grade 5 • Number & Operations–Fractions • 5.NF.A.1 • Set 16 • BACKWARD 2 • Q2/5

Subtract.

$\frac{3}{4} - \frac{2}{5}$

A. $\frac{7}{20}$

B. $\frac{1}{20}$

C. $\frac{1}{2}$

D. $\frac{5}{9}$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-B2-Q3

Grade 5 • Number & Operations–Fractions • 5.NF.A.1 • Set 16 • BACKWARD 2 • Q3/5

487

QUESTION ID

S16-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. 7/20

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-B2-Q3

Grade 5 • Number & Operations–Fractions • 5.NF.A.1 • Set 16 • BACKWARD 2 • Q3/5

True or false?

$\frac{2}{3} - \frac{1}{6} = \frac{1}{2}$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-B2-Q4

Grade 5 • Number & Operations–Fractions • 5.NF.A.1 • Set 16 • BACKWARD 2 • Q4/5

488

QUESTION ID

S16-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-B2-Q4

Grade 5 • Number & Operations–Fractions • 5.NF.A.1 • Set 16 • BACKWARD 2 • Q4/5

True or false?

$\frac{4}{5} - \frac{1}{3} = \frac{3}{2}$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-B2-Q5

Grade 5 • Number & Operations–Fractions • 5.NF.A.1 • Set 16 • BACKWARD 2 • Q5/5

489

QUESTION ID

S16-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-B2-Q5

Grade 5 • Number & Operations–Fractions • 5.NF.A.1 • Set 16 • BACKWARD 2 • Q5/5

SET 16 • 7.NS.A.1.c

FORWARD 1

Find a rate of change from two points with negative coordinates • Grade 8 • Functions • 8.F.B.4

Find the rate of change of the line through (−2, 5) and (3, −5).

Rate of change: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-F1-Q1

Grade 8 • Functions • 8.F.B.4 • Set 16 • FORWARD 1 • Q1/5

491

QUESTION ID

S16-F1-Q1

TYPE

Short answer

CORRECT ANSWER

−2

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-F1-Q1

Grade 8 • Functions • 8.F.B.4 • Set 16 • FORWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Find the slope of the line through $(-4, -1)$ and $(2, 8)$.

Slope: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-F1-Q2

Grade 8 • Functions • 8.F.B.4 • Set 16 • FORWARD 1 • Q2/5

492

QUESTION ID

S16-F1-Q2

TYPE

Short answer

CORRECT ANSWER

3/2

GRADING NOTE 1.5 or 1 1/2 is also correct.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-F1-Q2

Grade 8 • Functions • 8.F.B.4 • Set 16 • FORWARD 1 • Q2/5

What is the slope of the line through (1, −3) and (−1, 5)?

A. 1

B. 4

C. −1

D. −4

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-F1-Q3

Grade 8 • Functions • 8.F.B.4 • Set 16 • FORWARD 1 • Q3/5

493

QUESTION ID

S16-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. −4

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-F1-Q3

Grade 8 • Functions • 8.F.B.4 • Set 16 • FORWARD 1 • Q3/5

True or false?

The slope of the line through (−3, −2) and (0, 4) is 2.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-F1-Q4

Grade 8 • Functions • 8.F.B.4 • Set 16 • FORWARD 1 • Q4/5

494

QUESTION ID

S16-F1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-F1-Q4

Grade 8 • Functions • 8.F.B.4 • Set 16 • FORWARD 1 • Q4/5

True or false?

The slope of the line through (2, −1) and (5, −7) is 2.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-F1-Q5

Grade 8 • Functions • 8.F.B.4 • Set 16 • FORWARD 1 • Q5/5

495

GRADING NOTE The slope is −2.

QUESTION ID

S16-F1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-F1-Q5

Grade 8 • Functions • 8.F.B.4 • Set 16 • FORWARD 1 • Q5/5

SET 16 • 7.NS.A.1.c

FORWARD 2

Subtract polynomials • Grade 9 (High School) • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1

Subtract.

$(5x^2 - 3x + 2) - (2x^2 - 7x - 4)$

Difference: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-F2-Q1

Grade 9 (High School) • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 16 • FORWARD 2 • Q1/5

497

QUESTION ID

S16-F2-Q1

TYPE

Short answer

CORRECT ANSWER

$3x^2 + 4x + 6$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-F2-Q1

Grade 9 (High School) • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 16 • FORWARD 2 • Q1/5

Subtract.

$(4x - 9) - (-x + 3)$

Difference: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-F2-Q2

Grade 9 (High School) • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 16 • FORWARD 2 • Q2/5

498

QUESTION ID

S16-F2-Q2

TYPE

Short answer

CORRECT ANSWER

5x − 12

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-F2-Q2

Grade 9 (High School) • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 16 • FORWARD 2 • Q2/5

Subtract.

$(x^2 + 2x) - (3x^2 - 5x)$

A. $-2x^2 + 7x$

B. $-2x^2 - 3x$

C. $-2x^2 + 3x$

D. $4x^2 + 7x$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-F2-Q3

Grade 9 (High School) • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 16 • FORWARD 2 • Q3/5

499

QUESTION ID

S16-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. $-2x^2 + 7x$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-F2-Q3

Grade 9 (High School) • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 16 • FORWARD 2 • Q3/5

True or false?

$(6x + 1) - (2x - 5) = 4x + 6$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-F2-Q4

Grade 9 (High School) • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 16 • FORWARD 2 • Q4/5

500

QUESTION ID

S16-F2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-F2-Q4

Grade 9 (High School) • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 16 • FORWARD 2 • Q4/5

True or false?

$(3x^2 - x) - (x^2 + x) = 2x^2$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-F2-Q5

Grade 9 (High School) • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 16 • FORWARD 2 • Q5/5

501

QUESTION ID

S16-F2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE The difference is $2x^2 - 2x$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-F2-Q5

Grade 9 (High School) • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 16 • FORWARD 2 • Q5/5

SET 17 • 7.NS.A.1.c • THE NUMBER SYSTEM

Find the distance between two rational numbers as the absolute value of their difference

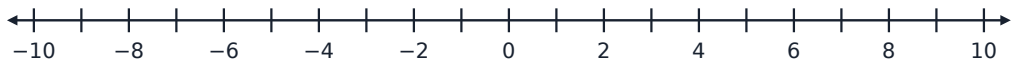
MAIN	Grade 7 main question	**7.NS.A.1.c**
BACKWARD 1	Find the distance between points with the same x- or y-coordinate	**6.NS.C.8**
BACKWARD 2	Position fractions and decimals on a number line	**6.NS.C.6.c**
FORWARD 1	Use the Pythagorean theorem to find the distance between two po…	**8.G.B.8**
FORWARD 2	Use coordinates to find the perimeter of a polygon	**HSG-GPE.B.7**

SET 17 • 7.NS.A.1.c

MAIN

Grade 7 main question • Grade 7 • The Number System • 7.NS.A.1.c

Find the distance between −6 and 9 on the number line.



Distance: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-M-Q1

Grade 7 • The Number System • 7.NS.A.1.c • Set 17 • MAIN • Q1/5

504

QUESTION ID

S17-M-Q1

TYPE

Short answer

CORRECT ANSWER

15

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-M-Q1

Grade 7 • The Number System • 7.NS.A.1.c • Set 17 • MAIN • Q1/5

Find the distance between −4.5 and −1.2 on a number line.

Distance: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-M-Q2

Grade 7 • The Number System • 7.NS.A.1.c • Set 17 • MAIN • Q2/5

505

QUESTION ID

S17-M-Q2

TYPE

Short answer

CORRECT ANSWER

3.3

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-M-Q2

Grade 7 • The Number System • 7.NS.A.1.c • Set 17 • MAIN • Q2/5

Which expression gives the distance between −3 and 8 on a number line?

A. −3 − 8

B. |−3 − 8|

C. −3 + 8

D. |−3 + 8|

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-M-Q3

Grade 7 • The Number System • 7.NS.A.1.c • Set 17 • MAIN • Q3/5

506

QUESTION ID

S17-M-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. |−3 − 8|

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-M-Q3

Grade 7 • The Number System • 7.NS.A.1.c • Set 17 • MAIN • Q3/5

True or false?

$|2 - (-5)| = 7$, so 2 and -5 are 7 units apart.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-M-Q4

Grade 7 • The Number System • 7.NS.A.1.c • Set 17 • MAIN • Q4/5

507

QUESTION ID

S17-M-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-M-Q4

Grade 7 • The Number System • 7.NS.A.1.c • Set 17 • MAIN • Q4/5

True or false?

The distance between −10 and −2 on a number line is 12.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-M-Q5

Grade 7 • The Number System • 7.NS.A.1.c • Set 17 • MAIN • Q5/5

508

QUESTION ID

S17-M-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE The distance is 8.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-M-Q5

Grade 7 • The Number System • 7.NS.A.1.c • Set 17 • MAIN • Q5/5

SET 17 • 7.NS.A.1.c

BACKWARD 1

Find the distance between points with the same x- or y-coordinate • Grade 6 • The Number System • 6.NS.C.8

Find the distance between (−3, 4) and (5, 4).

Distance: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-B1-Q1

Grade 6 • The Number System • 6.NS.C.8 • Set 17 • BACKWARD 1 • Q1/5

510

QUESTION ID

S17-B1-Q1

TYPE

Short answer

CORRECT ANSWER

8 units

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-B1-Q1

Grade 6 • The Number System • 6.NS.C.8 • Set 17 • BACKWARD 1 • Q1/5

Find the distance between (2, −6) and (2, 1).

Distance: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-B1-Q2

Grade 6 • The Number System • 6.NS.C.8 • Set 17 • BACKWARD 1 • Q2/5

511

QUESTION ID

S17-B1-Q2

TYPE

Short answer

CORRECT ANSWER

7 units

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-B1-Q2

Grade 6 • The Number System • 6.NS.C.8 • Set 17 • BACKWARD 1 • Q2/5

Which two points are 5 units apart?

A. (1, −2) and (1, 3)

B. (1, −2) and (6, 3)

C. (−2, 1) and (2, 1)

D. (1, −2) and (1, 2)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-B1-Q3

Grade 6 • The Number System • 6.NS.C.8 • Set 17 • BACKWARD 1 • Q3/5

QUESTION ID

S17-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. (1, −2) and (1, 3)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-B1-Q3

Grade 6 • The Number System • 6.NS.C.8 • Set 17 • BACKWARD 1 • Q3/5

True or false?

(−4, −1) and (−4, 6) are 7 units apart.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-B1-Q4

Grade 6 • The Number System • 6.NS.C.8 • Set 17 • BACKWARD 1 • Q4/5

513

QUESTION ID

S17-B1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-B1-Q4

Grade 6 • The Number System • 6.NS.C.8 • Set 17 • BACKWARD 1 • Q4/5

True or false?

(−5, 2) and (3, 2) are 2 units apart.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-B1-Q5

Grade 6 • The Number System • 6.NS.C.8 • Set 17 • BACKWARD 1 • Q5/5

514

QUESTION ID

S17-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE They are 8 units apart.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-B1-Q5

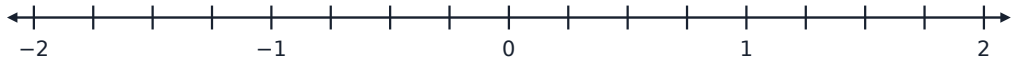
Grade 6 • The Number System • 6.NS.C.8 • Set 17 • BACKWARD 1 • Q5/5

SET 17 • 7.NS.A.1.c

BACKWARD 2

Position fractions and decimals on a number line • Grade 6 • The Number System • 6.NS.C.6.c

Plot -1.5 and $\frac{3}{4}$ on the number line.



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-B2-Q1

Grade 6 • The Number System • 6.NS.C.6.c • Set 17 • BACKWARD 2 • Q1/5

516

QUESTION ID

S17-B2-Q1

TYPE

Drawing / plotting

CORRECT ANSWER

Points at −1.5 and 3/4 (0.75)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-B2-Q1

Grade 6 • The Number System • 6.NS.C.6.c • Set 17 • BACKWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Name a number between −2 and −1.

Number: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-B2-Q2

Grade 6 • The Number System • 6.NS.C.6.c • Set 17 • BACKWARD 2 • Q2/5

517

QUESTION ID

S17-B2-Q2

TYPE

Short answer

CORRECT ANSWER

−1.5

GRADING NOTE Any number greater than −2 and less than −1 is correct.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-B2-Q2

Grade 6 • The Number System • 6.NS.C.6.c • Set 17 • BACKWARD 2 • Q2/5

Which number is between −0.5 and 0?

A. −0.75

B. −1

C. 0.25

D. −0.25

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-B2-Q3

Grade 6 • The Number System • 6.NS.C.6.c • Set 17 • BACKWARD 2 • Q3/5

518

QUESTION ID

S17-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. −0.25

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-B2-Q3

Grade 6 • The Number System • 6.NS.C.6.c • Set 17 • BACKWARD 2 • Q3/5

True or false?

$-\frac{1}{2}$ is to the left of 0 on a number line.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-B2-Q4

Grade 6 • The Number System • 6.NS.C.6.c • Set 17 • BACKWARD 2 • Q4/5

519

QUESTION ID

S17-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-B2-Q4

Grade 6 • The Number System • 6.NS.C.6.c • Set 17 • BACKWARD 2 • Q4/5

True or false?

−2.3 is to the right of −2 on a number line.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-B2-Q5

Grade 6 • The Number System • 6.NS.C.6.c • Set 17 • BACKWARD 2 • Q5/5

520

QUESTION ID

S17-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-B2-Q5

Grade 6 • The Number System • 6.NS.C.6.c • Set 17 • BACKWARD 2 • Q5/5

SET 17 • 7.NS.A.1.c

FORWARD 1

Use the Pythagorean theorem to find the distance between two points • Grade 8 • Geometry • 8.G.B.8

Find the distance between (1, 2) and (4, 6).

Distance: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-F1-Q1

Grade 8 • Geometry • 8.G.B.8 • Set 17 • FORWARD 1 • Q1/5

522

QUESTION ID

S17-F1-Q1

TYPE

Short answer

CORRECT ANSWER

5 units

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-F1-Q1

Grade 8 • Geometry • 8.G.B.8 • Set 17 • FORWARD 1 • Q1/5

Find the distance between (−2, −3) and (4, 5).

Distance: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-F1-Q2

Grade 8 • Geometry • 8.G.B.8 • Set 17 • FORWARD 1 • Q2/5

523

QUESTION ID

S17-F1-Q2

TYPE

Short answer

CORRECT ANSWER

10 units

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-F1-Q2

Grade 8 • Geometry • 8.G.B.8 • Set 17 • FORWARD 1 • Q2/5

What is the distance between (0, 0) and (5, 12)?

A. 13 units

B. 7 units

C. 60 units

D. 17 units

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-F1-Q3

Grade 8 • Geometry • 8.G.B.8 • Set 17 • FORWARD 1 • Q3/5

524

QUESTION ID

S17-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. 13 units

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-F1-Q3

Grade 8 • Geometry • 8.G.B.8 • Set 17 • FORWARD 1 • Q3/5

True or false?

The distance between (−1, 1) and (2, 5) is 5 units.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-F1-Q4

Grade 8 • Geometry • 8.G.B.8 • Set 17 • FORWARD 1 • Q4/5

525

QUESTION ID

S17-F1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-F1-Q4

Grade 8 • Geometry • 8.G.B.8 • Set 17 • FORWARD 1 • Q4/5

True or false?

The distance between (0, 0) and (3, 3) is 6 units.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-F1-Q5

Grade 8 • Geometry • 8.G.B.8 • Set 17 • FORWARD 1 • Q5/5

526

QUESTION ID

S17-F1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE The distance is √18, about 4.24 units.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-F1-Q5

Grade 8 • Geometry • 8.G.B.8 • Set 17 • FORWARD 1 • Q5/5

SET 17 • 7.NS.A.1.c

FORWARD 2

Use coordinates to find the perimeter of a polygon • Grade 9 (High School) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.7

A triangle has vertices (0, 0), (6, 0), and (6, 8).

Find its perimeter.

Perimeter: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-F2-Q1

Grade 9 (High School) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.7 • Set 17 • FORWARD 2 • Q1/5

528

QUESTION ID

S17-F2-Q1

TYPE

Short answer

CORRECT ANSWER

24 units

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-F2-Q1

Grade 9 (High School) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.7 • Set 17 • FORWARD 2 • Q1/5

A rectangle has vertices (−3, −2), (5, −2), (5, 4), and (−3, 4).

Find its perimeter.

Perimeter: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-F2-Q2

Grade 9 (High School) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.7 • Set 17 • FORWARD 2 • Q2/5

529

QUESTION ID

S17-F2-Q2

TYPE

Short answer

CORRECT ANSWER

28 units

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S17-F2-Q2**

Grade 9 (High School) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.7 • Set 17 • FORWARD 2 • Q2/5

A triangle has vertices (−2, 1), (2, 1), and (2, 4).

What is its perimeter?

A. 10 units

B. 6 units

C. 12 units

D. 7 units

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-F2-Q3

Grade 9 (High School) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.7 • Set 17 • FORWARD 2 • Q3/5

530

QUESTION ID

S17-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. 12 units

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-F2-Q3

Grade 9 (High School) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.7 • Set 17 • FORWARD 2 • Q3/5

True or false?

A square with vertices (1, 1), (4, 1), (4, 4), and (1, 4) has a perimeter of 12 units.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-F2-Q4

Grade 9 (High School) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.7 • Set 17 • FORWARD 2 • Q4/5

531

QUESTION ID

S17-F2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-F2-Q4

Grade 9 (High School) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.7 • Set 17 • FORWARD 2 • Q4/5

True or false?

A triangle with vertices (0, 0), (8, 0), and (0, 6) has a perimeter of 14 units.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-F2-Q5

Grade 9 (High School) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.7 • Set 17 • FORWARD 2 • Q5/5

532

QUESTION ID

S17-F2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE The perimeter is 24 units.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-F2-Q5

Grade 9 (High School) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.7 • Set 17 • FORWARD 2 • Q5/5

SET 18 • 7.NS.A.1.d • THE NUMBER SYSTEM

Use properties of operations to add and subtract rational numbers

MAIN	Grade 7 main question	**7.NS.A.1.d**
BACKWARD 1	Apply properties to write equivalent expressions	**6.EE.A.3**
BACKWARD 2	Add multi-digit decimals	**6.NS.B.3**
FORWARD 1	Solve equations by collecting like terms	**8.EE.C.7.b**
FORWARD 2	Use the structure of an expression to rewrite it	**HSA-SSE.A.2**

SET 18 • 7.NS.A.1.d

MAIN

Grade 7 main question • Grade 7 • The Number System • 7.NS.A.1.d

ANSWER KEY Reduced copy of the question page, with its answer

Find the sum by grouping numbers to make the work easier. Show your work.

$-8 + 13.5 + 8$

Sum: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-M-Q1

Grade 7 • The Number System • 7.NS.A.1.d • Set 18 • MAIN • Q1/5

535

QUESTION ID

S18-M-Q1

TYPE

Answer with required work

CORRECT ANSWER

13.5

GRADING NOTE REQUIRED WORK: Any grouping that uses the commutative and associative properties correctly. Example: $(-8 + 8) + 13.5 = 0 + 13.5$. Other valid groupings are also correct.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-M-Q1

Grade 7 • The Number System • 7.NS.A.1.d • Set 18 • MAIN • Q1/5

Find the sum.

$\frac{3}{4} + (-2) + \frac{1}{4}$

Sum: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-M-Q2

Grade 7 • The Number System • 7.NS.A.1.d • Set 18 • MAIN • Q2/5

536

QUESTION ID

S18-M-Q2

TYPE

Short answer

CORRECT ANSWER

−1

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-M-Q2

Grade 7 • The Number System • 7.NS.A.1.d • Set 18 • MAIN • Q2/5

Which property is shown?

$-5 + 2.7 = 2.7 + (-5)$

A. Associative property of addition

B. Additive inverse property

C. Commutative property of addition

D. Additive identity property

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-M-Q3

Grade 7 • The Number System • 7.NS.A.1.d • Set 18 • MAIN • Q3/5

537

QUESTION ID

S18-M-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. Commutative property of addition

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-M-Q3

Grade 7 • The Number System • 7.NS.A.1.d • Set 18 • MAIN • Q3/5

True or false?

$-6 + (4 + 6) = (-6 + 6) + 4$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-M-Q4

Grade 7 • The Number System • 7.NS.A.1.d • Set 18 • MAIN • Q4/5

538

QUESTION ID

S18-M-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-M-Q4

Grade 7 • The Number System • 7.NS.A.1.d • Set 18 • MAIN • Q4/5

True or false?

$7 - 3 = 3 - 7$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-M-Q5

Grade 7 • The Number System • 7.NS.A.1.d • Set 18 • MAIN • Q5/5

539

QUESTION ID

S18-M-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Subtraction is not commutative: $4 \neq -4$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-M-Q5

Grade 7 • The Number System • 7.NS.A.1.d • Set 18 • MAIN • Q5/5

SET 18 • 7.NS.A.1.d

BACKWARD 1

Apply properties to write equivalent expressions • Grade 6 • Expressions & Equations • 6.EE.A.3

Use the distributive property to rewrite 3(x + 5).

Expression: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-B1-Q1

Grade 6 • Expressions & Equations • 6.EE.A.3 • Set 18 • BACKWARD 1 • Q1/5

541

QUESTION ID

S18-B1-Q1

TYPE

Short answer

CORRECT ANSWER

3x + 15

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-B1-Q1

Grade 6 • Expressions & Equations • 6.EE.A.3 • Set 18 • BACKWARD 1 • Q1/5

Combine like terms.

4y + 2 + 6y

Expression: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-B1-Q2

Grade 6 • Expressions & Equations • 6.EE.A.3 • Set 18 • BACKWARD 1 • Q2/5

542

QUESTION ID

S18-B1-Q2

TYPE

Short answer

CORRECT ANSWER

10y + 2

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-B1-Q2

Grade 6 • Expressions & Equations • 6.EE.A.3 • Set 18 • BACKWARD 1 • Q2/5

Which expression is equivalent to 2(4 + n)?

A. 4 + 2n

B. 6 + n

C. 8 + 2n

D. 8 + n

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-B1-Q3

Grade 6 • Expressions & Equations • 6.EE.A.3 • Set 18 • BACKWARD 1 • Q3/5

543

QUESTION ID

S18-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. 8 + 2n

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-B1-Q3

Grade 6 • Expressions & Equations • 6.EE.A.3 • Set 18 • BACKWARD 1 • Q3/5

True or false?

$5 + (a + 7) = (5 + 7) + a$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-B1-Q4

Grade 6 • Expressions & Equations • 6.EE.A.3 • Set 18 • BACKWARD 1 • Q4/5

544

QUESTION ID

S18-B1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-B1-Q4

Grade 6 • Expressions & Equations • 6.EE.A.3 • Set 18 • BACKWARD 1 • Q4/5

True or false?

$6(x + 2) = 6x + 2$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-B1-Q5

Grade 6 • Expressions & Equations • 6.EE.A.3 • Set 18 • BACKWARD 1 • Q5/5

545

QUESTION ID

S18-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-B1-Q5

Grade 6 • Expressions & Equations • 6.EE.A.3 • Set 18 • BACKWARD 1 • Q5/5

SET 18 • 7.NS.A.1.d

BACKWARD 2

Add multi-digit decimals • Grade 6 • The Number System • 6.NS.B.3

Add.

14.62 + 7.9

Sum: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-B2-Q1

Grade 6 • The Number System • 6.NS.B.3 • Set 18 • BACKWARD 2 • Q1/5

547

QUESTION ID

S18-B2-Q1

TYPE

Short answer

CORRECT ANSWER

22.52

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-B2-Q1

Grade 6 • The Number System • 6.NS.B.3 • Set 18 • BACKWARD 2 • Q1/5

Add.

0.75 + 3.6 + 1.25

Sum: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-B2-Q2

Grade 6 • The Number System • 6.NS.B.3 • Set 18 • BACKWARD 2 • Q2/5

548

QUESTION ID

S18-B2-Q2

TYPE

Short answer

CORRECT ANSWER

5.6

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-B2-Q2

Grade 6 • The Number System • 6.NS.B.3 • Set 18 • BACKWARD 2 • Q2/5

Add.

9.08 + 2.5

A. 34.08

B. 11.58

C. 9.33

D. 11.13

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-B2-Q3

Grade 6 • The Number System • 6.NS.B.3 • Set 18 • BACKWARD 2 • Q3/5

549

QUESTION ID

S18-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. 11.58

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-B2-Q3

Grade 6 • The Number System • 6.NS.B.3 • Set 18 • BACKWARD 2 • Q3/5

True or false?

$2.25 + 3.75 = 6$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-B2-Q4

Grade 6 • The Number System • 6.NS.B.3 • Set 18 • BACKWARD 2 • Q4/5

550

QUESTION ID

S18-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-B2-Q4

Grade 6 • The Number System • 6.NS.B.3 • Set 18 • BACKWARD 2 • Q4/5

True or false?

$4.5 + 0.55 = 5.5$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-B2-Q5

Grade 6 • The Number System • 6.NS.B.3 • Set 18 • BACKWARD 2 • Q5/5

551

QUESTION ID

S18-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-B2-Q5

Grade 6 • The Number System • 6.NS.B.3 • Set 18 • BACKWARD 2 • Q5/5

SET 18 • 7.NS.A.1.d

FORWARD 1

Solve equations by collecting like terms • Grade 8 • Expressions & Equations • 8.EE.C.7.b

Solve for x.

$-2.5x + 4 + 1.5x = -3$

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-F1-Q1

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 18 • FORWARD 1 • Q1/5

553

QUESTION ID

S18-F1-Q1

TYPE

Short answer

CORRECT ANSWER

7

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-F1-Q1

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 18 • FORWARD 1 • Q1/5

Solve for x.

$\frac{1}{3}x - 5 + \frac{2}{3}x = 9$

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-F1-Q2

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 18 • FORWARD 1 • Q2/5

554

QUESTION ID

S18-F1-Q2

TYPE

Short answer

CORRECT ANSWER

14

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-F1-Q2

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 18 • FORWARD 1 • Q2/5

Solve for x.

$3x - 7 + x = 2x + 5$

A. $x = 12$

B. $x = 2$

C. $x = -1$

D. $x = 6$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-F1-Q3

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 18 • FORWARD 1 • Q3/5

555

QUESTION ID

S18-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. x = 6

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-F1-Q3

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 18 • FORWARD 1 • Q3/5

True or false?

$x = -4$ is the solution of $5x + 3 - 2x = -9$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-F1-Q4

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 18 • FORWARD 1 • Q4/5

556

QUESTION ID

S18-F1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-F1-Q4

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 18 • FORWARD 1 • Q4/5

True or false?

$x = 2$ is the solution of $-x + 6 + 3x = 8$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-F1-Q5

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 18 • FORWARD 1 • Q5/5

557

QUESTION ID

S18-F1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE The solution is $x = 1$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-F1-Q5

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 18 • FORWARD 1 • Q5/5

SET 18 • 7.NS.A.1.d

FORWARD 2

Use the structure of an expression to rewrite it • Grade 9 (High School) • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2

Rewrite $x^2 - 9$ as a product of two factors.

Product: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-F2-Q1

Grade 9 (High School) • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 18 • FORWARD 2 • Q1/5

559

QUESTION ID

S18-F2-Q1

TYPE

Short answer

CORRECT ANSWER

$(x + 3)(x - 3)$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-F2-Q1

Grade 9 (High School) • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 18 • FORWARD 2 • Q1/5

Factor out the greatest common factor.

$4x^2 + 12x$

Product: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-F2-Q2

Grade 9 (High School) • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 18 • FORWARD 2 • Q2/5

560

QUESTION ID

S18-F2-Q2

TYPE

Short answer

CORRECT ANSWER

4x(x + 3)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-F2-Q2

Grade 9 (High School) • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 18 • FORWARD 2 • Q2/5

Which expression is equivalent to $x^2 - 25$?

A. $(x - 5)(x + 5)$

B. $(x + 5)^2$

C. $(x - 5)^2$

D. $(x - 25)(x + 1)$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-F2-Q3

Grade 9 (High School) • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 18 • FORWARD 2 • Q3/5

561

QUESTION ID

S18-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. $(x - 5)(x + 5)$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-F2-Q3

Grade 9 (High School) • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 18 • FORWARD 2 • Q3/5

True or false?

$x^2 + 6x + 9 = (x + 3)^2$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-F2-Q4

Grade 9 (High School) • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 18 • FORWARD 2 • Q4/5

562

QUESTION ID

S18-F2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-F2-Q4

Grade 9 (High School) • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 18 • FORWARD 2 • Q4/5

True or false?

$x^2 - 16 = (x - 4)^2$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-F2-Q5

Grade 9 (High School) • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 18 • FORWARD 2 • Q5/5

563

QUESTION ID

S18-F2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE $x^2 - 16 = (x - 4)(x + 4)$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-F2-Q5

Grade 9 (High School) • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 18 • FORWARD 2 • Q5/5

SET 19 • 7.NS.A.2.a • THE NUMBER SYSTEM

Multiply rational numbers using the rules for signs

MAIN	Grade 7 main question	**7.NS.A.2.a**
BACKWARD 1	Multiply fractions	**5.NF.B.4.a**
BACKWARD 2	Multiply decimals	**5.NBT.B.7**
FORWARD 1	Find powers of negative numbers	**8.EE.A.1**
FORWARD 2	Multiply polynomials	**HSA-APR.A.1**

SET 19 • 7.NS.A.2.a

MAIN

Grade 7 main question • Grade 7 • The Number System • 7.NS.A.2.a

Multiply.

-6×7

Product: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-M-Q1

Grade 7 • The Number System • 7.NS.A.2.a • Set 19 • MAIN • Q1/5

566

QUESTION ID

S19-M-Q1

TYPE

Short answer

CORRECT ANSWER

−42

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-M-Q1

Grade 7 • The Number System • 7.NS.A.2.a • Set 19 • MAIN • Q1/5

Multiply.

$-2.5 \times (-4)$

Product: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-M-Q2

Grade 7 • The Number System • 7.NS.A.2.a • Set 19 • MAIN • Q2/5

QUESTION ID

S19-M-Q2

TYPE

Short answer

CORRECT ANSWER

10

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-M-Q2

Grade 7 • The Number System • 7.NS.A.2.a • Set 19 • MAIN • Q2/5

Multiply.

$-\frac{2}{3} \times \frac{9}{10}$

A. $-\frac{3}{5}$

B. $\frac{3}{5}$

C. $-\frac{27}{20}$

D. $-\frac{11}{13}$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-M-Q3

Grade 7 • The Number System • 7.NS.A.2.a • Set 19 • MAIN • Q3/5

568

QUESTION ID

S19-M-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. −3/5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-M-Q3

Grade 7 • The Number System • 7.NS.A.2.a • Set 19 • MAIN • Q3/5

True or false?

$(-1)(-1) = 1$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-M-Q4

Grade 7 • The Number System • 7.NS.A.2.a • Set 19 • MAIN • Q4/5

569

QUESTION ID

S19-M-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-M-Q4

Grade 7 • The Number System • 7.NS.A.2.a • Set 19 • MAIN • Q4/5

True or false?

The product of three negative numbers is positive.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-M-Q5

Grade 7 • The Number System • 7.NS.A.2.a • Set 19 • MAIN • Q5/5

570

QUESTION ID

S19-M-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-M-Q5

Grade 7 • The Number System • 7.NS.A.2.a • Set 19 • MAIN • Q5/5

SET 19 • 7.NS.A.2.a

BACKWARD 1

Multiply fractions • Grade 5 • Number & Operations–Fractions • 5.NF.B.4.a

Multiply.

$\frac{3}{5} \times \frac{10}{9}$

Product: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-B1-Q1

Grade 5 • Number & Operations–Fractions • 5.NF.B.4.a • Set 19 • BACKWARD 1 • Q1/5

572

QUESTION ID

S19-B1-Q1

TYPE

Short answer

CORRECT ANSWER

2/3

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-B1-Q1

Grade 5 • Number & Operations–Fractions • 5.NF.B.4.a • Set 19 • BACKWARD 1 • Q1/5

Multiply.

$\frac{4}{7} \times 14$

Product: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-B1-Q2

Grade 5 • Number & Operations–Fractions • 5.NF.B.4.a • Set 19 • BACKWARD 1 • Q2/5

573

QUESTION ID

S19-B1-Q2

TYPE

Short answer

CORRECT ANSWER

8

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-B1-Q2

Grade 5 • Number & Operations–Fractions • 5.NF.B.4.a • Set 19 • BACKWARD 1 • Q2/5

Multiply.

$\frac{5}{6} \times \frac{3}{4}$

A. $\frac{5}{8}$

B. $\frac{3}{8}$

C. $\frac{15}{10}$

D. $\frac{8}{10}$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-B1-Q3

Grade 5 • Number & Operations–Fractions • 5.NF.B.4.a • Set 19 • BACKWARD 1 • Q3/5

574

QUESTION ID

S19-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. 5/8

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-B1-Q3

Grade 5 • Number & Operations–Fractions • 5.NF.B.4.a • Set 19 • BACKWARD 1 • Q3/5

True or false?

$\frac{1}{2} \times \frac{2}{3} = \frac{1}{3}$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-B1-Q4

Grade 5 • Number & Operations–Fractions • 5.NF.B.4.a • Set 19 • BACKWARD 1 • Q4/5

575

QUESTION ID

S19-B1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-B1-Q4

Grade 5 • Number & Operations–Fractions • 5.NF.B.4.a • Set 19 • BACKWARD 1 • Q4/5

True or false?

$\frac{3}{4} \times 8 = 24$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-B1-Q5

Grade 5 • Number & Operations–Fractions • 5.NF.B.4.a • Set 19 • BACKWARD 1 • Q5/5

576

QUESTION ID

S19-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-B1-Q5

Grade 5 • Number & Operations–Fractions • 5.NF.B.4.a • Set 19 • BACKWARD 1 • Q5/5

SET 19 • 7.NS.A.2.a

BACKWARD 2

Multiply decimals • Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7

Multiply.

2.4 × 3.5

Product: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-B2-Q1

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7 • Set 19 • BACKWARD 2 • Q1/5

578

QUESTION ID

S19-B2-Q1

TYPE

Short answer

CORRECT ANSWER

8.4

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-B2-Q1

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7 • Set 19 • BACKWARD 2 • Q1/5

Multiply.

0.6×0.7

Product: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-B2-Q2

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7 • Set 19 • BACKWARD 2 • Q2/5

579

QUESTION ID

S19-B2-Q2

TYPE

Short answer

CORRECT ANSWER

0.42

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-B2-Q2

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7 • Set 19 • BACKWARD 2 • Q2/5

Multiply.

1.25 × 8

A. 100

B. 1

C. 9.25

D. 10

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-B2-Q3

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7 • Set 19 • BACKWARD 2 • Q3/5

580

QUESTION ID

S19-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. 10

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-B2-Q3

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7 • Set 19 • BACKWARD 2 • Q3/5

True or false?

$0.3 \times 0.3 = 0.09$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-B2-Q4

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7 • Set 19 • BACKWARD 2 • Q4/5

581

QUESTION ID

S19-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-B2-Q4

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7 • Set 19 • BACKWARD 2 • Q4/5

True or false?

$4.2 \times 10 = 4.20$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-B2-Q5

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7 • Set 19 • BACKWARD 2 • Q5/5

582

QUESTION ID

S19-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-B2-Q5

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7 • Set 19 • BACKWARD 2 • Q5/5

SET 19 • 7.NS.A.2.a

FORWARD 1

Find powers of negative numbers • Grade 8 • Expressions & Equations • 8.EE.A.1

Find the value.

$(-2)^5$

Value: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-F1-Q1

Grade 8 • Expressions & Equations • 8.EE.A.1 • Set 19 • FORWARD 1 • Q1/5

584

QUESTION ID

S19-F1-Q1

TYPE

Short answer

CORRECT ANSWER

−32

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-F1-Q1

Grade 8 • Expressions & Equations • 8.EE.A.1 • Set 19 • FORWARD 1 • Q1/5

Find the value.

$(-3)^2 \times (-3)$

Value: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-F1-Q2

Grade 8 • Expressions & Equations • 8.EE.A.1 • Set 19 • FORWARD 1 • Q2/5

585

QUESTION ID

S19-F1-Q2

TYPE

Short answer

CORRECT ANSWER

−27

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-F1-Q2

Grade 8 • Expressions & Equations • 8.EE.A.1 • Set 19 • FORWARD 1 • Q2/5

Which expression has a positive value?

A. -4^4

B. $(-4)^5$

C. $(-4)^3$

D. $(-4)^4$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-F1-Q3

Grade 8 • Expressions & Equations • 8.EE.A.1 • Set 19 • FORWARD 1 • Q3/5

586

QUESTION ID

S19-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. $(-4)^4$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-F1-Q3

Grade 8 • Expressions & Equations • 8.EE.A.1 • Set 19 • FORWARD 1 • Q3/5

True or false?

$(-5)^2 = 25$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-F1-Q4

Grade 8 • Expressions & Equations • 8.EE.A.1 • Set 19 • FORWARD 1 • Q4/5

587

QUESTION ID

S19-F1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-F1-Q4

Grade 8 • Expressions & Equations • 8.EE.A.1 • Set 19 • FORWARD 1 • Q4/5

True or false?

$-2^4 = 16$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-F1-Q5

Grade 8 • Expressions & Equations • 8.EE.A.1 • Set 19 • FORWARD 1 • Q5/5

588

QUESTION ID

S19-F1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE $-2^4 = -(2^4) = -16$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-F1-Q5

Grade 8 • Expressions & Equations • 8.EE.A.1 • Set 19 • FORWARD 1 • Q5/5

SET 19 • 7.NS.A.2.a

FORWARD 2

Multiply polynomials • Grade 9 (High School) • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1

Multiply.

$-3x(2x - 5)$

Product: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-F2-Q1

Grade 9 (High School) • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 19 • FORWARD 2 • Q1/5

590

QUESTION ID

S19-F2-Q1

TYPE

Short answer

CORRECT ANSWER

$-6x^2 + 15x$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-F2-Q1

Grade 9 (High School) • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 19 • FORWARD 2 • Q1/5

Multiply.

$(x - 4)(x + 6)$

Product: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-F2-Q2

Grade 9 (High School) • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 19 • FORWARD 2 • Q2/5

591

QUESTION ID

S19-F2-Q2

TYPE

Short answer

CORRECT ANSWER

$x^2 + 2x - 24$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-F2-Q2

Grade 9 (High School) • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 19 • FORWARD 2 • Q2/5

Multiply.

$(-2x)(-7x)$

A. $14x$

B. $14x^2$

C. $-14x^2$

D. $-9x$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-F2-Q3

Grade 9 (High School) • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 19 • FORWARD 2 • Q3/5

592

QUESTION ID

S19-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. $14x^2$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-F2-Q3

Grade 9 (High School) • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 19 • FORWARD 2 • Q3/5

True or false?

$(x - 3)(x - 3) = x^2 - 6x + 9$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-F2-Q4

Grade 9 (High School) • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 19 • FORWARD 2 • Q4/5

593

QUESTION ID

S19-F2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-F2-Q4

Grade 9 (High School) • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 19 • FORWARD 2 • Q4/5

True or false?

$-4(x - 2) = -4x - 8$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-F2-Q5

Grade 9 (High School) • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 19 • FORWARD 2 • Q5/5

594

QUESTION ID

S19-F2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE $-4(x - 2) = -4x + 8$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-F2-Q5

Grade 9 (High School) • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 19 • FORWARD 2 • Q5/5

SET 20 • 7.NS.A.2.a • THE NUMBER SYSTEM

Interpret products of rational numbers in real-world situations

MAIN	Grade 7 main question	**7.NS.A.2.a**
BACKWARD 1	Multiply fractions and mixed numbers in word problems	**5.NF.B.6**
BACKWARD 2	Explain the meaning of signed numbers and 0	**6.NS.C.5**
FORWARD 1	Solve real-world equations with negative rates	**8.EE.C.7.b**
FORWARD 2	Create a one-variable equation for a situation with a negative rate	**HSA-CED.A.1**

SET 20 • 7.NS.A.2.a

MAIN

Grade 7 main question • Grade 7 • The Number System • 7.NS.A.2.a

A stock's price drops $1.75 each day for 6 days.

What is the total change in price?

Change: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-M-Q1

Grade 7 • The Number System • 7.NS.A.2.a • Set 20 • MAIN • Q1/5

597

QUESTION ID

S20-M-Q1

TYPE

Short answer

CORRECT ANSWER

−$10.50

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-M-Q1

Grade 7 • The Number System • 7.NS.A.2.a • Set 20 • MAIN • Q1/5

A diver descends 2.5 feet each second.

What is the change in her position after 8 seconds?

Change: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-M-Q2

Grade 7 • The Number System • 7.NS.A.2.a • Set 20 • MAIN • Q2/5

598

QUESTION ID

S20-M-Q2

TYPE

Short answer

CORRECT ANSWER

−20 feet

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-M-Q2

Grade 7 • The Number System • 7.NS.A.2.a • Set 20 • MAIN • Q2/5

A tank loses 3 gallons each minute.

Which product gives the change in the amount of water after 12 minutes?

A. $12 \times 3 = 36$

B. $12 \times (-3) = -36$

C. $12 + (-3) = 9$

D. $-12 \times (-3) = 36$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-M-Q3

Grade 7 • The Number System • 7.NS.A.2.a • Set 20 • MAIN • Q3/5

599

QUESTION ID

S20-M-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. $12 \times (-3) = -36$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-M-Q3

Grade 7 • The Number System • 7.NS.A.2.a • Set 20 • MAIN • Q3/5

True or false?

The temperature falls 4°F each hour for 5 hours. The total change is −20°F.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-M-Q4

Grade 7 • The Number System • 7.NS.A.2.a • Set 20 • MAIN • Q4/5

600

QUESTION ID

S20-M-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-M-Q4

Grade 7 • The Number System • 7.NS.A.2.a • Set 20 • MAIN • Q4/5

True or false?

An account balance changes by −$15 each week. The change over 3 weeks is $45.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-M-Q5

Grade 7 • The Number System • 7.NS.A.2.a • Set 20 • MAIN • Q5/5

601

QUESTION ID

S20-M-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE The change is −$45.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-M-Q5

Grade 7 • The Number System • 7.NS.A.2.a • Set 20 • MAIN • Q5/5

SET 20 • 7.NS.A.2.a

BACKWARD 1

Multiply fractions and mixed numbers in word problems • Grade 5 • Number & Operations–Fractions • 5.NF.B.6

A recipe uses $\frac{3}{4}$ cup of oats for each batch.

How many cups of oats are needed for 6 batches?

Cups: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-B1-Q1

Grade 5 • Number & Operations–Fractions • 5.NF.B.6 • Set 20 • BACKWARD 1 • Q1/5

603

QUESTION ID

S20-B1-Q1

TYPE

Short answer

CORRECT ANSWER

4 1/2

GRADING NOTE 9/2 is also correct.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-B1-Q1

Grade 5 • Number & Operations–Fractions • 5.NF.B.6 • Set 20 • BACKWARD 1 • Q1/5

A path is $2\frac{1}{2}$ miles long. Sam walks $\frac{2}{5}$ of it.

How far does he walk?

Distance: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-B1-Q2

Grade 5 • Number & Operations–Fractions • 5.NF.B.6 • Set 20 • BACKWARD 1 • Q2/5

604

QUESTION ID

S20-B1-Q2

TYPE

Short answer

CORRECT ANSWER

1 mile

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-B1-Q2

Grade 5 • Number & Operations–Fractions • 5.NF.B.6 • Set 20 • BACKWARD 1 • Q2/5

A board is $\frac{7}{8}$ yard long. Ty cuts a piece that is $\frac{1}{2}$ as long as the board.

How long is the piece?

A. $1\frac{3}{8}$ yards

B. $\frac{7}{4}$ yards

C. $\frac{8}{14}$ yard

D. $\frac{7}{16}$ yard

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-B1-Q3

Grade 5 • Number & Operations–Fractions • 5.NF.B.6 • Set 20 • BACKWARD 1 • Q3/5

605

QUESTION ID

S20-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. 7/16 yard

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-B1-Q3

Grade 5 • Number & Operations–Fractions • 5.NF.B.6 • Set 20 • BACKWARD 1 • Q3/5

True or false?

$\frac{2}{3}$ of a 12-foot rope is 8 feet.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-B1-Q4

Grade 5 • Number & Operations–Fractions • 5.NF.B.6 • Set 20 • BACKWARD 1 • Q4/5

606

QUESTION ID

S20-B1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-B1-Q4

Grade 5 • Number & Operations–Fractions • 5.NF.B.6 • Set 20 • BACKWARD 1 • Q4/5

True or false?

4 bags each hold $\frac{2}{3}$ pound of nuts. Together they hold 3 pounds.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-B1-Q5

Grade 5 • Number & Operations–Fractions • 5.NF.B.6 • Set 20 • BACKWARD 1 • Q5/5

607

QUESTION ID

S20-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE They hold 8/3 = 2 2/3 pounds.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-B1-Q5

Grade 5 • Number & Operations–Fractions • 5.NF.B.6 • Set 20 • BACKWARD 1 • Q5/5

SET 20 • 7.NS.A.2.a

BACKWARD 2

Explain the meaning of signed numbers and 0 • Grade 6 • The Number System • 6.NS.C.5

In a game, a score of 0 means no points won or lost.

What does a score of −12 mean?

Meaning: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-B2-Q1

Grade 6 • The Number System • 6.NS.C.5 • Set 20 • BACKWARD 2 • Q1/5

609

QUESTION ID

S20-B2-Q1

TYPE

Short answer

CORRECT ANSWER

12 points were lost (12 points below 0).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-B2-Q1

Grade 6 • The Number System • 6.NS.C.5 • Set 20 • BACKWARD 2 • Q1/5

Sea level is 0 meters.

Which elevation is 30 meters below sea level?

A. −30 m

B. 30 m

C. 0 m

D. −3 m

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-B2-Q2

Grade 6 • The Number System • 6.NS.C.5 • Set 20 • BACKWARD 2 • Q2/5

610

QUESTION ID

S20-B2-Q2

TYPE

Multiple choice

CORRECT ANSWER

A. −30 m

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-B2-Q2

Grade 6 • The Number System • 6.NS.C.5 • Set 20 • BACKWARD 2 • Q2/5

What does a change of −$8 in a bank account mean?

Meaning: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-B2-Q3

Grade 6 • The Number System • 6.NS.C.5 • Set 20 • BACKWARD 2 • Q3/5

611

QUESTION ID

S20-B2-Q3

TYPE

Short answer

CORRECT ANSWER

$8 was taken out of the account.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-B2-Q3

Grade 6 • The Number System • 6.NS.C.5 • Set 20 • BACKWARD 2 • Q3/5

True or false?

A temperature change of −5° means it got colder.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-B2-Q4

Grade 6 • The Number System • 6.NS.C.5 • Set 20 • BACKWARD 2 • Q4/5

612

QUESTION ID

S20-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-B2-Q4

Grade 6 • The Number System • 6.NS.C.5 • Set 20 • BACKWARD 2 • Q4/5

True or false?

On a horizontal number line, positive numbers are to the left of 0.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-B2-Q5

Grade 6 • The Number System • 6.NS.C.5 • Set 20 • BACKWARD 2 • Q5/5

QUESTION ID

S20-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-B2-Q5

Grade 6 • The Number System • 6.NS.C.5 • Set 20 • BACKWARD 2 • Q5/5

SET 20 • 7.NS.A.2.a

FORWARD 1

Solve real-world equations with negative rates • Grade 8 • Expressions & Equations • 8.EE.C.7.b

A balloon at 300 m descends 12.5 m each minute. Solve $300 - 12.5t = 175$ to find when it is at 175 m.

t = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-F1-Q1

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 20 • FORWARD 1 • Q1/5

615

QUESTION ID

S20-F1-Q1

TYPE

Short answer

CORRECT ANSWER

10 minutes

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-F1-Q1

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 20 • FORWARD 1 • Q1/5

A diver at −8 feet descends 1.5 feet each second. Solve $-8 - 1.5t = -26$ to find when she is at −26 feet.

t = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-F1-Q2

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 20 • FORWARD 1 • Q2/5

616

QUESTION ID

S20-F1-Q2

TYPE

Short answer

CORRECT ANSWER

12 seconds

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-F1-Q2

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 20 • FORWARD 1 • Q2/5

Solve for x.

$-4x + 10 = -14$

A. $x = -6$

B. $x = -1$

C. $x = 6$

D. $x = 1.5$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-F1-Q3

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 20 • FORWARD 1 • Q3/5

617

QUESTION ID

S20-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. x = 6

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-F1-Q3

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 20 • FORWARD 1 • Q3/5

True or false?

t = 5 is the solution of 20 − 3t = 5.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-F1-Q4

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 20 • FORWARD 1 • Q4/5

618

QUESTION ID

S20-F1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-F1-Q4

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 20 • FORWARD 1 • Q4/5

True or false?

$x = -3$ is the solution of $-2x - 7 = 13$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-F1-Q5

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 20 • FORWARD 1 • Q5/5

619

QUESTION ID

S20-F1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE The solution is $x = -10$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-F1-Q5

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 20 • FORWARD 1 • Q5/5

SET 20 • 7.NS.A.2.a

FORWARD 2

Create a one-variable equation for a situation with a negative rate • Grade 9 (High School) • Algebra: Creating Equations • HSA-CED.A.1

A phone battery is at 90% and drops 6% each hour.

Write and solve an equation to find when it reaches 30%.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-F2-Q1

Grade 9 (High School) • Algebra: Creating Equations • HSA-CED.A.1 • Set 20 • FORWARD 2 • Q1/5

621

QUESTION ID

S20-F2-Q1

TYPE

Short answer

CORRECT ANSWER

90 − 6h = 30; h = 10 hours

GRADING NOTE Any correct equation earns credit.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-F2-Q1

Grade 9 (High School) • Algebra: Creating Equations • HSA-CED.A.1 • Set 20 • FORWARD 2 • Q1/5

A submarine at −50 m dives 20 m each minute.

Write and solve an equation to find when it reaches −170 m.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-F2-Q2

Grade 9 (High School) • Algebra: Creating Equations • HSA-CED.A.1 • Set 20 • FORWARD 2 • Q2/5

622

QUESTION ID

S20-F2-Q2

TYPE

Short answer

CORRECT ANSWER

−50 − 20t = −170;
t = 6 minutes

GRADING NOTE Any correct equation earns credit.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-F2-Q2

Grade 9 (High School) • Algebra: Creating Equations • HSA-CED.A.1 • Set 20 • FORWARD 2 • Q2/5

A store's profit was −$400 and goes up $125 each month.

Which equation finds the month m when the profit is $600?

A. −400 + 125m = 600

B. 125 − 400m = 600

C. −400 − 125m = 600

D. 400 + 125m = 600

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-F2-Q3

Grade 9 (High School) • Algebra: Creating Equations • HSA-CED.A.1 • Set 20 • FORWARD 2 • Q3/5

623

QUESTION ID

S20-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

A.

−400 + 125m = 600

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-F2-Q3

Grade 9 (High School) • Algebra: Creating Equations • HSA-CED.A.1 • Set 20 • FORWARD 2 • Q3/5

True or false?

A 72°F room cools 4°F each hour. The equation $72 - 4t = 40$ finds when the room is 40°F.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-F2-Q4

Grade 9 (High School) • Algebra: Creating Equations • HSA-CED.A.1 • Set 20 • FORWARD 2 • Q4/5

624

QUESTION ID

S20-F2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-F2-Q4

Grade 9 (High School) • Algebra: Creating Equations • HSA-CED.A.1 • Set 20 • FORWARD 2 • Q4/5

True or false?

The solution of $72 - 4t = 40$ is $t = 28$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-F2-Q5

Grade 9 (High School) • Algebra: Creating Equations • HSA-CED.A.1 • Set 20 • FORWARD 2 • Q5/5

625

QUESTION ID

S20-F2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE $t = 8$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-F2-Q5

Grade 9 (High School) • Algebra: Creating Equations • HSA-CED.A.1 • Set 20 • FORWARD 2 • Q5/5

SET 21 • 7.NS.A.2.b • THE NUMBER SYSTEM

Divide integers and explain the sign of a quotient

MAIN	Grade 7 main question	**7.NS.A.2.b**
BACKWARD 1	Divide multi-digit whole numbers	**6.NS.B.2**
BACKWARD 2	Divide fractions	**6.NS.A.1**
FORWARD 1	Classify numbers as rational or irrational	**8.NS.A.1**
FORWARD 2	Decide whether a product is rational or irrational	**HSN-RN.B.3**

SET 21 • 7.NS.A.2.b

MAIN

Grade 7 main question • Grade 7 • The Number System • 7.NS.A.2.b

Divide.

$-48 \div 6$

Quotient: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-M-Q1

Grade 7 • The Number System • 7.NS.A.2.b • Set 21 • MAIN • Q1/5

628

QUESTION ID

S21-M-Q1

TYPE

Short answer

CORRECT ANSWER

−8

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-M-Q1

Grade 7 • The Number System • 7.NS.A.2.b • Set 21 • MAIN • Q1/5

Divide.

$-63 \div (-9)$

Quotient: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-M-Q2

Grade 7 • The Number System • 7.NS.A.2.b • Set 21 • MAIN • Q2/5

629

QUESTION ID

S21-M-Q2

TYPE

Short answer

CORRECT ANSWER

7

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-M-Q2

Grade 7 • The Number System • 7.NS.A.2.b • Set 21 • MAIN • Q2/5

Which fraction is equal to $-\frac{3}{4}$?

A. $\frac{-3}{4}$

B. $\frac{4}{-3}$

C. $\frac{-3}{-4}$

D. $\frac{3}{4}$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-M-Q3

Grade 7 • The Number System • 7.NS.A.2.b • Set 21 • MAIN • Q3/5

630

QUESTION ID

S21-M-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. (−3)/4

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-M-Q3

Grade 7 • The Number System • 7.NS.A.2.b • Set 21 • MAIN • Q3/5

True or false?

$$\frac{-10}{2} = \frac{10}{-2} = -5$$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-M-Q4

Grade 7 • The Number System • 7.NS.A.2.b • Set 21 • MAIN • Q4/5

631

QUESTION ID

S21-M-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-M-Q4

Grade 7 • The Number System • 7.NS.A.2.b • Set 21 • MAIN • Q4/5

True or false?

$-7 \div 0 = 0$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-M-Q5

Grade 7 • The Number System • 7.NS.A.2.b • Set 21 • MAIN • Q5/5

632

QUESTION ID

S21-M-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Division by 0 is undefined.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-M-Q5

Grade 7 • The Number System • 7.NS.A.2.b • Set 21 • MAIN • Q5/5

SET 21 • 7.NS.A.2.b

BACKWARD 1

Divide multi-digit whole numbers • Grade 6 • The Number System • 6.NS.B.2

Divide.

1,872 ÷ 24

Quotient: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-B1-Q1

Grade 6 • The Number System • 6.NS.B.2 • Set 21 • BACKWARD 1 • Q1/5

634

QUESTION ID

S21-B1-Q1

TYPE

Short answer

CORRECT ANSWER

78

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-B1-Q1

Grade 6 • The Number System • 6.NS.B.2 • Set 21 • BACKWARD 1 • Q1/5

Divide.

3,915 ÷ 45

Quotient: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-B1-Q2

Grade 6 • The Number System • 6.NS.B.2 • Set 21 • BACKWARD 1 • Q2/5

635

QUESTION ID

S21-B1-Q2

TYPE

Short answer

CORRECT ANSWER

87

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-B1-Q2

Grade 6 • The Number System • 6.NS.B.2 • Set 21 • BACKWARD 1 • Q2/5

Divide.

2,436 ÷ 12

A. 203

B. 2,030

C. 23

D. 230

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-B1-Q3

Grade 6 • The Number System • 6.NS.B.2 • Set 21 • BACKWARD 1 • Q3/5

QUESTION ID

S21-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. 203

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-B1-Q3

Grade 6 • The Number System • 6.NS.B.2 • Set 21 • BACKWARD 1 • Q3/5

True or false?

5,040 ÷ 35 = 144

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-B1-Q4

Grade 6 • The Number System • 6.NS.B.2 • Set 21 • BACKWARD 1 • Q4/5

637

QUESTION ID

S21-B1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-B1-Q4

Grade 6 • The Number System • 6.NS.B.2 • Set 21 • BACKWARD 1 • Q4/5

True or false?

1,950 ÷ 15 = 13

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-B1-Q5

Grade 6 • The Number System • 6.NS.B.2 • Set 21 • BACKWARD 1 • Q5/5

638

QUESTION ID

S21-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-B1-Q5

Grade 6 • The Number System • 6.NS.B.2 • Set 21 • BACKWARD 1 • Q5/5

SET 21 • 7.NS.A.2.b

BACKWARD 2

Divide fractions • Grade 6 • The Number System • 6.NS.A.1

Divide.

$\frac{4}{5} \div \frac{2}{15}$

Quotient: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-B2-Q1

Grade 6 • The Number System • 6.NS.A.1 • Set 21 • BACKWARD 2 • Q1/5

640

QUESTION ID

S21-B2-Q1

TYPE

Short answer

CORRECT ANSWER

6

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-B2-Q1

Grade 6 • The Number System • 6.NS.A.1 • Set 21 • BACKWARD 2 • Q1/5

Divide.

$3 \div \frac{3}{8}$

Quotient: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-B2-Q2

Grade 6 • The Number System • 6.NS.A.1 • Set 21 • BACKWARD 2 • Q2/5

641

QUESTION ID

S21-B2-Q2

TYPE

Short answer

CORRECT ANSWER

8

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-B2-Q2

Grade 6 • The Number System • 6.NS.A.1 • Set 21 • BACKWARD 2 • Q2/5

Divide.

$\frac{7}{10} \div \frac{1}{5}$

A. $\frac{2}{7}$

B. $\frac{7}{50}$

C. $1\frac{2}{5}$

D. $3\frac{1}{2}$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-B2-Q3

Grade 6 • The Number System • 6.NS.A.1 • Set 21 • BACKWARD 2 • Q3/5

642

QUESTION ID

S21-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. 3 1/2

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-B2-Q3

Grade 6 • The Number System • 6.NS.A.1 • Set 21 • BACKWARD 2 • Q3/5

True or false?

$\frac{3}{8} \div \frac{3}{4} = \frac{1}{2}$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-B2-Q4

Grade 6 • The Number System • 6.NS.A.1 • Set 21 • BACKWARD 2 • Q4/5

643

QUESTION ID

S21-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-B2-Q4

Grade 6 • The Number System • 6.NS.A.1 • Set 21 • BACKWARD 2 • Q4/5

True or false?

$\frac{5}{6} \div \frac{5}{12} = \frac{1}{2}$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-B2-Q5

Grade 6 • The Number System • 6.NS.A.1 • Set 21 • BACKWARD 2 • Q5/5

644

QUESTION ID

S21-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-B2-Q5

Grade 6 • The Number System • 6.NS.A.1 • Set 21 • BACKWARD 2 • Q5/5

SET 21 • 7.NS.A.2.b

FORWARD 1

Classify numbers as rational or irrational • Grade 8 • The Number System • 8.NS.A.1

Is $\frac{-5}{8}$ rational? Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-F1-Q1

Grade 8 • The Number System • 8.NS.A.1 • Set 21 • FORWARD 1 • Q1/5

646

QUESTION ID

S21-F1-Q1

TYPE

Short answer

CORRECT ANSWER

Yes. It is a quotient of the integers −5 and 8.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-F1-Q1

Grade 8 • The Number System • 8.NS.A.1 • Set 21 • FORWARD 1 • Q1/5

Which number is irrational?

A. $\sqrt{7}$

B. 0.25

C. $\frac{-22}{7}$

D. $-\sqrt{16}$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-F1-Q2

Grade 8 • The Number System • 8.NS.A.1 • Set 21 • FORWARD 1 • Q2/5

647

QUESTION ID

S21-F1-Q2

TYPE

Multiple choice

CORRECT ANSWER

A. √7

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-F1-Q2

Grade 8 • The Number System • 8.NS.A.1 • Set 21 • FORWARD 1 • Q2/5

Write -2.4 as a quotient of two integers.

Quotient: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-F1-Q3

Grade 8 • The Number System • 8.NS.A.1 • Set 21 • FORWARD 1 • Q3/5

QUESTION ID

S21-F1-Q3

TYPE

Short answer

CORRECT ANSWER

$-12/5$

GRADING NOTE Any equivalent quotient, such as $-24/10$, is correct.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-F1-Q3

Grade 8 • The Number System • 8.NS.A.1 • Set 21 • FORWARD 1 • Q3/5

True or false?

Every integer is a rational number.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-F1-Q4

Grade 8 • The Number System • 8.NS.A.1 • Set 21 • FORWARD 1 • Q4/5

649

QUESTION ID

S21-F1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-F1-Q4

Grade 8 • The Number System • 8.NS.A.1 • Set 21 • FORWARD 1 • Q4/5

True or false?

√25 is irrational.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-F1-Q5

Grade 8 • The Number System • 8.NS.A.1 • Set 21 • FORWARD 1 • Q5/5

650

QUESTION ID

S21-F1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-F1-Q5

Grade 8 • The Number System • 8.NS.A.1 • Set 21 • FORWARD 1 • Q5/5

SET 21 • 7.NS.A.2.b

FORWARD 2

Decide whether a product is rational or irrational • Grade 9 (High School) • Number & Quantity: The Real Number System • HSN-RN.B.3

Is $-3 \times \sqrt{5}$ rational or irrational?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-F2-Q1

Grade 9 (High School) • Number & Quantity: The Real Number System • HSN-RN.B.3 • Set 21 • FORWARD 2 • Q1/5

652

QUESTION ID

S21-F2-Q1

TYPE

Short answer

CORRECT ANSWER

Irrational

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-F2-Q1

Grade 9 (High School) • Number & Quantity: The Real Number System • HSN-RN.B.3 • Set 21 • FORWARD 2 • Q1/5

Is $\frac{-2}{3} \times \frac{9}{4}$ rational or irrational? Give its value.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-F2-Q2

Grade 9 (High School) • Number & Quantity: The Real Number System • HSN-RN.B.3 • Set 21 • FORWARD 2 • Q2/5

653

QUESTION ID

S21-F2-Q2

TYPE

Short answer

CORRECT ANSWER

Rational; −3/2

GRADING NOTE −1 1/2 or −1.5 is also correct for the value.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-F2-Q2

Grade 9 (High School) • Number & Quantity: The Real Number System • HSN-RN.B.3 • Set 21 • FORWARD 2 • Q2/5

Which product is rational?

A. $-1 \times \pi$

B. $5 \times \sqrt{2}$

C. $2 \times \sqrt{3}$

D. $\sqrt{2} \times \sqrt{8}$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-F2-Q3

Grade 9 (High School) • Number & Quantity: The Real Number System • HSN-RN.B.3 • Set 21 • FORWARD 2 • Q3/5

654

QUESTION ID

S21-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. $\sqrt{2} \times \sqrt{8}$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-F2-Q3

Grade 9 (High School) • Number & Quantity: The Real Number System • HSN-RN.B.3 • Set 21 • FORWARD 2 • Q3/5

True or false?

The product of a nonzero rational number and an irrational number is irrational.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-F2-Q4

Grade 9 (High School) • Number & Quantity: The Real Number System • HSN-RN.B.3 • Set 21 • FORWARD 2 • Q4/5

655

QUESTION ID

S21-F2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-F2-Q4

Grade 9 (High School) • Number & Quantity: The Real Number System • HSN-RN.B.3 • Set 21 • FORWARD 2 • Q4/5

True or false?

The quotient of two rational numbers (with a nonzero divisor) can be irrational.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-F2-Q5

Grade 9 (High School) • Number & Quantity: The Real Number System • HSN-RN.B.3 • Set 21 • FORWARD 2 • Q5/5

656

QUESTION ID

S21-F2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-F2-Q5

Grade 9 (High School) • Number & Quantity: The Real Number System • HSN-RN.B.3 • Set 21 • FORWARD 2 • Q5/5

SET 22 • 7.NS.A.2.b • THE NUMBER SYSTEM

Interpret quotients of rational numbers in real-world situations

MAIN	Grade 7 main question	**7.NS.A.2.b**
BACKWARD 1	Divide fractions in word problems	**6.NS.A.1**
BACKWARD 2	Divide unit fractions and whole numbers in word problems	**5.NF.B.7.c**
FORWARD 1	Find a rate of change from a description	**8.F.B.4**
FORWARD 2	Calculate and interpret a negative average rate of change	**HSF-IF.B.6**

SET 22 • 7.NS.A.2.b

MAIN

Grade 7 main question • Grade 7 • The Number System • 7.NS.A.2.b

A stock lost $18 in value over 6 days at a steady rate.

What was the change in value per day?

Change per day: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S22-M-Q1

Grade 7 • The Number System • 7.NS.A.2.b • Set 22 • MAIN • Q1/5

659

QUESTION ID

S22-M-Q1

TYPE

Short answer

CORRECT ANSWER

−$3

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S22-M-Q1

Grade 7 • The Number System • 7.NS.A.2.b • Set 22 • MAIN • Q1/5

A diver's position changed by −45 feet in 9 seconds at a steady rate.

What was the change in position per second?

Change per second: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-M-Q2

Grade 7 • The Number System • 7.NS.A.2.b • Set 22 • MAIN • Q2/5

660

QUESTION ID

S22-M-Q2

TYPE

Short answer

CORRECT ANSWER

−5 feet

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-M-Q2

Grade 7 • The Number System • 7.NS.A.2.b • Set 22 • MAIN • Q2/5

The temperature changed by −14°F over 4 hours at a steady rate.

What was the change per hour?

A. −10°F

B. 3.5°F

C. −3.5°F

D. −56°F

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S22-M-Q3

Grade 7 • The Number System • 7.NS.A.2.b • Set 22 • MAIN • Q3/5

661

QUESTION ID

S22-M-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. −3.5°F

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S22-M-Q3

Grade 7 • The Number System • 7.NS.A.2.b • Set 22 • MAIN • Q3/5

True or false?

A balance changes by −$120 over 8 weeks at a steady rate. The change is −$15 per week.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S22-M-Q4

Grade 7 • The Number System • 7.NS.A.2.b • Set 22 • MAIN • Q4/5

662

QUESTION ID

S22-M-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S22-M-Q4

Grade 7 • The Number System • 7.NS.A.2.b • Set 22 • MAIN • Q4/5

True or false?

A hiker descends 600 feet in 4 hours at a steady rate. Her change in elevation is 150 feet per hour.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-M-Q5

Grade 7 • The Number System • 7.NS.A.2.b • Set 22 • MAIN • Q5/5

663

QUESTION ID

S22-M-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE The change is −150 feet per hour.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-M-Q5

Grade 7 • The Number System • 7.NS.A.2.b • Set 22 • MAIN • Q5/5

SET 22 • 7.NS.A.2.b

BACKWARD 1

Divide fractions in word problems • Grade 6 • The Number System • 6.NS.A.1

How many $\frac{3}{4}$-cup servings are in 6 cups of rice?

Servings: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S22-B1-Q1

Grade 6 • The Number System • 6.NS.A.1 • Set 22 • BACKWARD 1 • Q1/5

665

QUESTION ID

S22-B1-Q1

TYPE

Short answer

CORRECT ANSWER

8

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S22-B1-Q1

Grade 6 • The Number System • 6.NS.A.1 • Set 22 • BACKWARD 1 • Q1/5

A $\frac{2}{3}$-mile trail is split into 4 equal parts.

How long is each part?

Length: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-B1-Q2

Grade 6 • The Number System • 6.NS.A.1 • Set 22 • BACKWARD 1 • Q2/5

666

QUESTION ID

S22-B1-Q2

TYPE

Short answer

CORRECT ANSWER

1/6 mile

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-B1-Q2

Grade 6 • The Number System • 6.NS.A.1 • Set 22 • BACKWARD 1 • Q2/5

A strip $\frac{5}{6}$ meter long is cut into pieces $\frac{1}{12}$ meter long.

How many pieces are there?

A. $\frac{5}{72}$

B. 12

C. 6

D. 10

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-B1-Q3

Grade 6 • The Number System • 6.NS.A.1 • Set 22 • BACKWARD 1 • Q3/5

667

QUESTION ID

S22-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. 10

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-B1-Q3

Grade 6 • The Number System • 6.NS.A.1 • Set 22 • BACKWARD 1 • Q3/5

True or false?

There are 6 half-pound portions in 3 pounds.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S22-B1-Q4

Grade 6 • The Number System • 6.NS.A.1 • Set 22 • BACKWARD 1 • Q4/5

668

QUESTION ID

S22-B1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S22-B1-Q4

Grade 6 • The Number System • 6.NS.A.1 • Set 22 • BACKWARD 1 • Q4/5

True or false?

3 people share $\frac{3}{4}$ pound of nuts equally. Each gets $\frac{1}{2}$ pound.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-B1-Q5

Grade 6 • The Number System • 6.NS.A.1 • Set 22 • BACKWARD 1 • Q5/5

669

QUESTION ID

S22-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Each gets 1/4 pound.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-B1-Q5

Grade 6 • The Number System • 6.NS.A.1 • Set 22 • BACKWARD 1 • Q5/5

SET 22 • 7.NS.A.2.b

BACKWARD 2

Divide unit fractions and whole numbers in word problems • Grade 5 • Number & Operations–Fractions • 5.NF.B.7.c

How many $\frac{1}{4}$-pound burgers can be made from 5 pounds of meat?

Burgers: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-B2-Q1

Grade 5 • Number & Operations–Fractions • 5.NF.B.7.c • Set 22 • BACKWARD 2 • Q1/5

671

QUESTION ID

S22-B2-Q1

TYPE

Short answer

CORRECT ANSWER

20

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-B2-Q1

Grade 5 • Number & Operations–Fractions • 5.NF.B.7.c • Set 22 • BACKWARD 2 • Q1/5

3 friends share $\frac{1}{2}$ pound of grapes equally.

How much does each friend get?

Amount: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-B2-Q2

Grade 5 • Number & Operations–Fractions • 5.NF.B.7.c • Set 22 • BACKWARD 2 • Q2/5

672

QUESTION ID

S22-B2-Q2

TYPE

Short answer

CORRECT ANSWER

1/6 pound

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-B2-Q2

Grade 5 • Number & Operations–Fractions • 5.NF.B.7.c • Set 22 • BACKWARD 2 • Q2/5

2 people share $\frac{1}{3}$ of a pizza equally.

How much of the whole pizza does each person get?

A. $\frac{2}{3}$ of the pizza

B. $\frac{1}{6}$ of the pizza

C. 6 pizzas

D. $\frac{3}{2}$ of the pizza

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-B2-Q3

Grade 5 • Number & Operations–Fractions • 5.NF.B.7.c • Set 22 • BACKWARD 2 • Q3/5

673

QUESTION ID

S22-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. 1/6 of the pizza

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-B2-Q3

Grade 5 • Number & Operations–Fractions • 5.NF.B.7.c • Set 22 • BACKWARD 2 • Q3/5

True or false?

$4 \div \frac{1}{5} = 20$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-B2-Q4

Grade 5 • Number & Operations–Fractions • 5.NF.B.7.c • Set 22 • BACKWARD 2 • Q4/5

674

QUESTION ID

S22-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-B2-Q4

Grade 5 • Number & Operations–Fractions • 5.NF.B.7.c • Set 22 • BACKWARD 2 • Q4/5

True or false?

$\frac{1}{8} \div 2 = \frac{1}{4}$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-B2-Q5

Grade 5 • Number & Operations–Fractions • 5.NF.B.7.c • Set 22 • BACKWARD 2 • Q5/5

675

QUESTION ID

S22-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-B2-Q5

Grade 5 • Number & Operations–Fractions • 5.NF.B.7.c • Set 22 • BACKWARD 2 • Q5/5

SET 22 • 7.NS.A.2.b

FORWARD 1

Find a rate of change from a description • Grade 8 • Functions • 8.F.B.4

A tank held 240 gallons at 2:00 and 180 gallons at 2:15. It drained at a constant rate.

What is the rate of change in gallons per minute?

Rate of change: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-F1-Q1

Grade 8 • Functions • 8.F.B.4 • Set 22 • FORWARD 1 • Q1/5

677

QUESTION ID

S22-F1-Q1

TYPE

Short answer

CORRECT ANSWER

−4 gallons per minute

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-F1-Q1

Grade 8 • Functions • 8.F.B.4 • Set 22 • FORWARD 1 • Q1/5

A plane descends from 30,000 feet to 21,000 feet in 15 minutes at a constant rate.

What is the rate of change in feet per minute?

Rate of change: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-F1-Q2

Grade 8 • Functions • 8.F.B.4 • Set 22 • FORWARD 1 • Q2/5

678

QUESTION ID

S22-F1-Q2

TYPE

Short answer

CORRECT ANSWER

−600 feet per minute

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-F1-Q2

Grade 8 • Functions • 8.F.B.4 • Set 22 • FORWARD 1 • Q2/5

A line passes through (0, 50) and (10, 20).

What is its rate of change?

A. $-\frac{1}{3}$

B. 3

C. -30

D. -3

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-F1-Q3

Grade 8 • Functions • 8.F.B.4 • Set 22 • FORWARD 1 • Q3/5

679

QUESTION ID

S22-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. −3

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-F1-Q3

Grade 8 • Functions • 8.F.B.4 • Set 22 • FORWARD 1 • Q3/5

True or false?

A balance that goes from $80 to $20 in 12 weeks at a constant rate changes by −$5 per week.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S22-F1-Q4

Grade 8 • Functions • 8.F.B.4 • Set 22 • FORWARD 1 • Q4/5

680

QUESTION ID

S22-F1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S22-F1-Q4

Grade 8 • Functions • 8.F.B.4 • Set 22 • FORWARD 1 • Q4/5

True or false?

A temperature that goes from 10° to −6° in 4 hours at a constant rate changes by −1° per hour.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-F1-Q5

Grade 8 • Functions • 8.F.B.4 • Set 22 • FORWARD 1 • Q5/5

681

QUESTION ID

S22-F1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE It changes by −4° per hour.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-F1-Q5

Grade 8 • Functions • 8.F.B.4 • Set 22 • FORWARD 1 • Q5/5

SET 22 • 7.NS.A.2.b

FORWARD 2

Calculate and interpret a negative average rate of change • Grade 9 (High School) • Functions: Interpreting Functions • HSF-IF.B.6

The table shows temperatures.

What is the average rate of change from 0 hours to 6 hours?

Time (h)	0	3	6
Temperature (°F)	12	3	−6

Average rate: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-F2-Q1

Grade 9 (High School) • Functions: Interpreting Functions • HSF-IF.B.6 • Set 22 • FORWARD 2 • Q1/5

683

QUESTION ID

S22-F2-Q1

TYPE

Short answer

CORRECT ANSWER

−3°F per hour

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-F2-Q1

Grade 9 (High School) • Functions: Interpreting Functions • HSF-IF.B.6 • Set 22 • FORWARD 2 • Q1/5

$f(x) = -x^2 + 10$

What is the average rate of change from $x = 1$ to $x = 3$?

Average rate: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-F2-Q2

Grade 9 (High School) • Functions: Interpreting Functions • HSF-IF.B.6 • Set 22 • FORWARD 2 • Q2/5

684

QUESTION ID

S22-F2-Q2

TYPE

Short answer

CORRECT ANSWER

−4

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-F2-Q2

Grade 9 (High School) • Functions: Interpreting Functions • HSF-IF.B.6 • Set 22 • FORWARD 2 • Q2/5

A company's profit was $5,000 in year 2 and $2,000 in year 5.

What is the average rate of change?

A. −$600 per year

B. $1,000 per year

C. −$3,000 per year

D. −$1,000 per year

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-F2-Q3

Grade 9 (High School) • Functions: Interpreting Functions • HSF-IF.B.6 • Set 22 • FORWARD 2 • Q3/5

685

QUESTION ID

S22-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. −$1,000 per year

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-F2-Q3

Grade 9 (High School) • Functions: Interpreting Functions • HSF-IF.B.6 • Set 22 • FORWARD 2 • Q3/5

True or false?

A negative average rate of change means the quantity decreased over the interval.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-F2-Q4

Grade 9 (High School) • Functions: Interpreting Functions • HSF-IF.B.6 • Set 22 • FORWARD 2 • Q4/5

686

QUESTION ID

S22-F2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-F2-Q4

Grade 9 (High School) • Functions: Interpreting Functions • HSF-IF.B.6 • Set 22 • FORWARD 2 • Q4/5

True or false?

For f(x) = 5 − 2x, the average rate of change from x = 0 to x = 4 is 2.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-F2-Q5

Grade 9 (High School) • Functions: Interpreting Functions • HSF-IF.B.6 • Set 22 • FORWARD 2 • Q5/5

687

QUESTION ID

S22-F2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE It is −2.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-F2-Q5

Grade 9 (High School) • Functions: Interpreting Functions • HSF-IF.B.6 • Set 22 • FORWARD 2 • Q5/5

SET 23 • 7.NS.A.2.c • THE NUMBER SYSTEM

Use properties of operations to multiply and divide rational numbers

MAIN	Grade 7 main question	**7.NS.A.2.c**
BACKWARD 1	Apply the distributive property	**6.EE.A.3**
BACKWARD 2	Use the commutative and associative properties of multiplication	**3.OA.B.5**
FORWARD 1	Solve equations by multiplying by a reciprocal	**8.EE.C.7.b**
FORWARD 2	Rewrite expressions with rational exponents	**HSN-RN.A.2**

SET 23 • 7.NS.A.2.c

MAIN

Grade 7 main question • Grade 7 • The Number System • 7.NS.A.2.c

Use properties to multiply. Show how you grouped or ordered the factors.

$-4 \times 7.3 \times 25$

Product: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-M-Q1

Grade 7 • The Number System • 7.NS.A.2.c • Set 23 • MAIN • Q1/5

690

QUESTION ID

S23-M-Q1

TYPE

Answer with required work

CORRECT ANSWER

−730

GRADING NOTE REQUIRED WORK: Any correct use of the commutative and associative properties. Example: $(-4 \times 25) \times 7.3 = -100 \times 7.3$. Other valid groupings are also correct.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-M-Q1

Grade 7 • The Number System • 7.NS.A.2.c • Set 23 • MAIN • Q1/5

Multiply.

$\frac{3}{5} \times (-12) \times \frac{5}{3}$

Product: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-M-Q2

Grade 7 • The Number System • 7.NS.A.2.c • Set 23 • MAIN • Q2/5

691

QUESTION ID

S23-M-Q2

TYPE

Short answer

CORRECT ANSWER

−12

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-M-Q2

Grade 7 • The Number System • 7.NS.A.2.c • Set 23 • MAIN • Q2/5

Which property is shown?

$-2 \times (5 \times 3.1) = (-2 \times 5) \times 3.1$

A. Multiplicative identity property

B. Associative property of multiplication

C. Distributive property

D. Commutative property of multiplication

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-M-Q3

Grade 7 • The Number System • 7.NS.A.2.c • Set 23 • MAIN • Q3/5

692

QUESTION ID

S23-M-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. Associative property of multiplication

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-M-Q3

Grade 7 • The Number System • 7.NS.A.2.c • Set 23 • MAIN • Q3/5

True or false?

$-6 \times (10 + \frac{1}{2}) = -60 + (-3)$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-M-Q4

Grade 7 • The Number System • 7.NS.A.2.c • Set 23 • MAIN • Q4/5

693

QUESTION ID

S23-M-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-M-Q4

Grade 7 • The Number System • 7.NS.A.2.c • Set 23 • MAIN • Q4/5

True or false?

$-8 \times \frac{1}{8} = 1$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-M-Q5

Grade 7 • The Number System • 7.NS.A.2.c • Set 23 • MAIN • Q5/5

694

QUESTION ID

S23-M-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE The product is −1.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-M-Q5

Grade 7 • The Number System • 7.NS.A.2.c • Set 23 • MAIN • Q5/5

SET 23 • 7.NS.A.2.c

BACKWARD 1

Apply the distributive property • Grade 6 • Expressions & Equations • 6.EE.A.3

ANSWER KEY Reduced copy of the question page, with its answer

Rewrite using the distributive property.

7(2y + 3)

Expression: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-B1-Q1

Grade 6 • Expressions & Equations • 6.EE.A.3 • Set 23 • BACKWARD 1 • Q1/5

696

QUESTION ID

S23-B1-Q1

TYPE

Short answer

CORRECT ANSWER

14y + 21

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-B1-Q1

Grade 6 • Expressions & Equations • 6.EE.A.3 • Set 23 • BACKWARD 1 • Q1/5

Use the distributive property to find 8 × 47 as 8 × (40 + 7).

Product: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-B1-Q2

Grade 6 • Expressions & Equations • 6.EE.A.3 • Set 23 • BACKWARD 1 • Q2/5

697

QUESTION ID

S23-B1-Q2

TYPE

Short answer

CORRECT ANSWER

376

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-B1-Q2

Grade 6 • Expressions & Equations • 6.EE.A.3 • Set 23 • BACKWARD 1 • Q2/5

Which expression is equivalent to 5(x + 9)?

A. 5x + 14

B. x + 45

C. 5x + 45

D. 5x + 9

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-B1-Q3

Grade 6 • Expressions & Equations • 6.EE.A.3 • Set 23 • BACKWARD 1 • Q3/5

698

QUESTION ID

S23-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. 5x + 45

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-B1-Q3

Grade 6 • Expressions & Equations • 6.EE.A.3 • Set 23 • BACKWARD 1 • Q3/5

True or false?

$4(3 + 2.5) = 12 + 10$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-B1-Q4

Grade 6 • Expressions & Equations • 6.EE.A.3 • Set 23 • BACKWARD 1 • Q4/5

699

QUESTION ID

S23-B1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-B1-Q4

Grade 6 • Expressions & Equations • 6.EE.A.3 • Set 23 • BACKWARD 1 • Q4/5

True or false?

$3(a + 4) = 3a + 7$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-B1-Q5

Grade 6 • Expressions & Equations • 6.EE.A.3 • Set 23 • BACKWARD 1 • Q5/5

700

QUESTION ID

S23-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-B1-Q5

Grade 6 • Expressions & Equations • 6.EE.A.3 • Set 23 • BACKWARD 1 • Q5/5

SET 23 • 7.NS.A.2.c

BACKWARD 2

Use the commutative and associative properties of multiplication • Grade 3 • Operations & Algebraic Thinking • 3.OA.B.5

Fill in the blank.

$6 \times 4 = 4 \times ___$

Blank: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-B2-Q1

Grade 3 • Operations & Algebraic Thinking • 3.OA.B.5 • Set 23 • BACKWARD 2 • Q1/5

QUESTION ID

S23-B2-Q1

TYPE

Short answer

CORRECT ANSWER

6

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-B2-Q1

Grade 3 • Operations & Algebraic Thinking • 3.OA.B.5 • Set 23 • BACKWARD 2 • Q1/5

Find 2 × 7 × 5 by first multiplying 2 × 5.

Product: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-B2-Q2

Grade 3 • Operations & Algebraic Thinking • 3.OA.B.5 • Set 23 • BACKWARD 2 • Q2/5

703

QUESTION ID

S23-B2-Q2

TYPE

Short answer

CORRECT ANSWER

70

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-B2-Q2

Grade 3 • Operations & Algebraic Thinking • 3.OA.B.5 • Set 23 • BACKWARD 2 • Q2/5

Which expression is equal to (3 × 2) × 4?

A. 3 × 2 + 4

B. 3 + (2 × 4)

C. 3 × (2 × 4)

D. (3 + 2) × 4

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-B2-Q3

Grade 3 • Operations & Algebraic Thinking • 3.OA.B.5 • Set 23 • BACKWARD 2 • Q3/5

704

QUESTION ID

S23-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. 3 × (2 × 4)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-B2-Q3

Grade 3 • Operations & Algebraic Thinking • 3.OA.B.5 • Set 23 • BACKWARD 2 • Q3/5

True or false?

$8 \times 3 = 3 \times 8$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-B2-Q4

Grade 3 • Operations & Algebraic Thinking • 3.OA.B.5 • Set 23 • BACKWARD 2 • Q4/5

705

QUESTION ID

S23-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-B2-Q4

Grade 3 • Operations & Algebraic Thinking • 3.OA.B.5 • Set 23 • BACKWARD 2 • Q4/5

True or false?

$(2 \times 3) \times 5 = 2 \times 3 + 5$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-B2-Q5

Grade 3 • Operations & Algebraic Thinking • 3.OA.B.5 • Set 23 • BACKWARD 2 • Q5/5

706

QUESTION ID

S23-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-B2-Q5

Grade 3 • Operations & Algebraic Thinking • 3.OA.B.5 • Set 23 • BACKWARD 2 • Q5/5

SET 23 • 7.NS.A.2.c

FORWARD 1

Solve equations by multiplying by a reciprocal • Grade 8 • Expressions & Equations • 8.EE.C.7.b

Solve for x.

$-\frac{3}{4}x = 18$

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-F1-Q1

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 23 • FORWARD 1 • Q1/5

708

QUESTION ID

S23-F1-Q1

TYPE

Short answer

CORRECT ANSWER

−24

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-F1-Q1

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 23 • FORWARD 1 • Q1/5

Solve for x.

$-0.6x = -4.2$

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-F1-Q2

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 23 • FORWARD 1 • Q2/5

709

QUESTION ID

S23-F1-Q2

TYPE

Short answer

CORRECT ANSWER

7

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-F1-Q2

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 23 • FORWARD 1 • Q2/5

Solve for x.

$\frac{2}{5}(x - 10) = -6$

A. x = −5

B. x = 25

C. x = −25

D. x = 5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-F1-Q3

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 23 • FORWARD 1 • Q3/5

710

QUESTION ID

S23-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. x = −5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-F1-Q3

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 23 • FORWARD 1 • Q3/5

True or false?

$x = -10$ is the solution of $-\frac{1}{2}x = 5$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-F1-Q4

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 23 • FORWARD 1 • Q4/5

711

QUESTION ID

S23-F1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-F1-Q4

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 23 • FORWARD 1 • Q4/5

True or false?

$x = 3$ is the solution of $-1.5x = 4.5$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-F1-Q5

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 23 • FORWARD 1 • Q5/5

712

QUESTION ID

S23-F1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE The solution is $x = -3$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-F1-Q5

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 23 • FORWARD 1 • Q5/5

SET 23 • 7.NS.A.2.c

FORWARD 2

Rewrite expressions with rational exponents • Grade 9 (High School) • Number & Quantity: The Real Number System • HSN-RN.A.2

Find the value.

27^(1/3)

Value: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-F2-Q1

Grade 9 (High School) • Number & Quantity: The Real Number System • HSN-RN.A.2 • Set 23 • FORWARD 2 • Q1/5

714

QUESTION ID

S23-F2-Q1

TYPE

Short answer

CORRECT ANSWER

3

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-F2-Q1

Grade 9 (High School) • Number & Quantity: The Real Number System • HSN-RN.A.2 • Set 23 • FORWARD 2 • Q1/5

Write √x using a rational exponent.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-F2-Q2

Grade 9 (High School) • Number & Quantity: The Real Number System • HSN-RN.A.2 • Set 23 • FORWARD 2 • Q2/5

715

QUESTION ID

S23-F2-Q2

TYPE

Short answer

CORRECT ANSWER

x^(1/2)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-F2-Q2

Grade 9 (High School) • Number & Quantity: The Real Number System • HSN-RN.A.2 • Set 23 • FORWARD 2 • Q2/5

Which number is equal to 16^(1/2)?

A. 4

B. 32

C. 2

D. 8

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-F2-Q3

Grade 9 (High School) • Number & Quantity: The Real Number System • HSN-RN.A.2 • Set 23 • FORWARD 2 • Q3/5

716

QUESTION ID

S23-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. 4

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-F2-Q3

Grade 9 (High School) • Number & Quantity: The Real Number System • HSN-RN.A.2 • Set 23 • FORWARD 2 • Q3/5

True or false?

$(x^3)^{(1/3)} = x$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-F2-Q4

Grade 9 (High School) • Number & Quantity: The Real Number System • HSN-RN.A.2 • Set 23 • FORWARD 2 • Q4/5

717

QUESTION ID

S23-F2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-F2-Q4

Grade 9 (High School) • Number & Quantity: The Real Number System • HSN-RN.A.2 • Set 23 • FORWARD 2 • Q4/5

True or false?

25^(1/2) = 12.5

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-F2-Q5

Grade 9 (High School) • Number & Quantity: The Real Number System • HSN-RN.A.2 • Set 23 • FORWARD 2 • Q5/5

718

QUESTION ID

S23-F2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE 25^(1/2) = 5.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-F2-Q5

Grade 9 (High School) • Number & Quantity: The Real Number System • HSN-RN.A.2 • Set 23 • FORWARD 2 • Q5/5

SET 24 • 7.NS.A.2.d • THE NUMBER SYSTEM

Use long division to write a rational number as a terminating or repeating decimal

MAIN	Grade 7 main question	**7.NS.A.2.d**
BACKWARD 1	Write fractions with denominators of 10 or 100 as decimals	**4.NF.C.6**
BACKWARD 2	Divide multi-digit whole numbers	**6.NS.B.2**
FORWARD 1	Write a repeating decimal as a fraction	**8.NS.A.1**
FORWARD 2	Report a decimal result to a sensible level of accuracy	**HSN-Q.A.3**

SET 24 • 7.NS.A.2.d

MAIN

Grade 7 main question • Grade 7 • The Number System • 7.NS.A.2.d

Use long division to write $\frac{3}{8}$ as a decimal. Show your work.

Decimal: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-M-Q1

Grade 7 • The Number System • 7.NS.A.2.d • Set 24 • MAIN • Q1/5

721

QUESTION ID

S24-M-Q1

TYPE

Answer with required work

CORRECT ANSWER

0.375

GRADING NOTE REQUIRED WORK: Long division of the numerator by the denominator, shown step by step, until the remainder is 0 or the digits start to repeat. A correct decimal without the division earns partial credit.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-M-Q1

Grade 7 • The Number System • 7.NS.A.2.d • Set 24 • MAIN • Q1/5

Use long division to write $\frac{5}{11}$ as a decimal. Show your work.

Decimal: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-M-Q2

Grade 7 • The Number System • 7.NS.A.2.d • Set 24 • MAIN • Q2/5

722

QUESTION ID

S24-M-Q2

TYPE

Answer with required work

CORRECT ANSWER

0.4545... (the digits 45 repeat)

GRADING NOTE REQUIRED WORK: Long division of the numerator by the denominator, shown step by step, until the remainder is 0 or the digits start to repeat. A correct decimal without the division earns partial credit.

A bar over 45 is also correct.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-M-Q2

Grade 7 • The Number System • 7.NS.A.2.d • Set 24 • MAIN • Q2/5

Which decimal is equal to $-\frac{7}{12}$?

A. −0.58333... (the 3 repeats)

B. −0.712

C. −1.714...

D. −0.5833 (the decimal ends)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S24-M-Q3

Grade 7 • The Number System • 7.NS.A.2.d • Set 24 • MAIN • Q3/5

723

QUESTION ID

S24-M-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. −0.58333... (the 3 repeats)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S24-M-Q3

Grade 7 • The Number System • 7.NS.A.2.d • Set 24 • MAIN • Q3/5

True or false?

$\frac{2}{9} = 0.222...$, a repeating decimal.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S24-M-Q4

Grade 7 • The Number System • 7.NS.A.2.d • Set 24 • MAIN • Q4/5

724

QUESTION ID

S24-M-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S24-M-Q4

Grade 7 • The Number System • 7.NS.A.2.d • Set 24 • MAIN • Q4/5

True or false?

$\frac{7}{20}$ is a repeating decimal.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-M-Q5

Grade 7 • The Number System • 7.NS.A.2.d • Set 24 • MAIN • Q5/5

725

QUESTION ID

S24-M-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE 7/20 = 0.35, a terminating decimal.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-M-Q5

Grade 7 • The Number System • 7.NS.A.2.d • Set 24 • MAIN • Q5/5

SET 24 • 7.NS.A.2.d

BACKWARD 1

Write fractions with denominators of 10 or 100 as decimals • Grade 4 • Number & Operations–Fractions • 4.NF.C.6

Write $\frac{7}{10}$ as a decimal.

Decimal: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-B1-Q1

Grade 4 • Number & Operations–Fractions • 4.NF.C.6 • Set 24 • BACKWARD 1 • Q1/5

727

QUESTION ID

S24-B1-Q1

TYPE

Short answer

CORRECT ANSWER

0.7

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-B1-Q1

Grade 4 • Number & Operations–Fractions • 4.NF.C.6 • Set 24 • BACKWARD 1 • Q1/5

Write $\frac{43}{100}$ as a decimal.

Decimal: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-B1-Q2

Grade 4 • Number & Operations–Fractions • 4.NF.C.6 • Set 24 • BACKWARD 1 • Q2/5

728

QUESTION ID

S24-B1-Q2

TYPE

Short answer

CORRECT ANSWER

0.43

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-B1-Q2

Grade 4 • Number & Operations–Fractions • 4.NF.C.6 • Set 24 • BACKWARD 1 • Q2/5

Which decimal is equal to $\frac{6}{100}$?

A. 0.06

B. 0.006

C. 0.6

D. 6.0

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-B1-Q3

Grade 4 • Number & Operations–Fractions • 4.NF.C.6 • Set 24 • BACKWARD 1 • Q3/5

729

QUESTION ID

S24-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. 0.06

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-B1-Q3

Grade 4 • Number & Operations–Fractions • 4.NF.C.6 • Set 24 • BACKWARD 1 • Q3/5

True or false?

$\frac{9}{10} = 0.9$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S24-B1-Q4

Grade 4 • Number & Operations–Fractions • 4.NF.C.6 • Set 24 • BACKWARD 1 • Q4/5

730

QUESTION ID

S24-B1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S24-B1-Q4

Grade 4 • Number & Operations–Fractions • 4.NF.C.6 • Set 24 • BACKWARD 1 • Q4/5

True or false?

$\frac{25}{100} = 2.5$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-B1-Q5

Grade 4 • Number & Operations–Fractions • 4.NF.C.6 • Set 24 • BACKWARD 1 • Q5/5

731

QUESTION ID

S24-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-B1-Q5

Grade 4 • Number & Operations–Fractions • 4.NF.C.6 • Set 24 • BACKWARD 1 • Q5/5

SET 24 • 7.NS.A.2.d

BACKWARD 2

Divide multi-digit whole numbers • Grade 6 • The Number System • 6.NS.B.2

ANSWER KEY Reduced copy of the question page, with its answer

Divide.

7,425 ÷ 25

Quotient: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-B2-Q1

Grade 6 • The Number System • 6.NS.B.2 • Set 24 • BACKWARD 2 • Q1/5

733

QUESTION ID

S24-B2-Q1

TYPE

Short answer

CORRECT ANSWER

297

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-B2-Q1

Grade 6 • The Number System • 6.NS.B.2 • Set 24 • BACKWARD 2 • Q1/5

Divide.

4,368 ÷ 52

Quotient: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S24-B2-Q2

Grade 6 • The Number System • 6.NS.B.2 • Set 24 • BACKWARD 2 • Q2/5

734

QUESTION ID

S24-B2-Q2

TYPE

Short answer

CORRECT ANSWER

84

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S24-B2-Q2

Grade 6 • The Number System • 6.NS.B.2 • Set 24 • BACKWARD 2 • Q2/5

Divide.

9,072 ÷ 36

A. 2,520

B. 252

C. 25.2

D. 242

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S24-B2-Q3

Grade 6 • The Number System • 6.NS.B.2 • Set 24 • BACKWARD 2 • Q3/5

735

QUESTION ID

S24-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. 252

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S24-B2-Q3

Grade 6 • The Number System • 6.NS.B.2 • Set 24 • BACKWARD 2 • Q3/5

True or false?

6,912 ÷ 48 = 144

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S24-B2-Q4

Grade 6 • The Number System • 6.NS.B.2 • Set 24 • BACKWARD 2 • Q4/5

736

QUESTION ID

S24-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S24-B2-Q4

Grade 6 • The Number System • 6.NS.B.2 • Set 24 • BACKWARD 2 • Q4/5

True or false?

2,925 ÷ 75 = 49

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S24-B2-Q5

Grade 6 • The Number System • 6.NS.B.2 • Set 24 • BACKWARD 2 • Q5/5

737

QUESTION ID

S24-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S24-B2-Q5

Grade 6 • The Number System • 6.NS.B.2 • Set 24 • BACKWARD 2 • Q5/5

SET 24 • 7.NS.A.2.d

FORWARD 1

Write a repeating decimal as a fraction • Grade 8 • The Number System • 8.NS.A.1

Write 0.777... (the 7 repeats) as a fraction.

Fraction: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S24-F1-Q1

Grade 8 • The Number System • 8.NS.A.1 • Set 24 • FORWARD 1 • Q1/5

739

QUESTION ID

S24-F1-Q1

TYPE

Short answer

CORRECT ANSWER

7/9

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S24-F1-Q1

Grade 8 • The Number System • 8.NS.A.1 • Set 24 • FORWARD 1 • Q1/5

Write 0.1818... (the digits 18 repeat) as a fraction.

Fraction: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-F1-Q2

Grade 8 • The Number System • 8.NS.A.1 • Set 24 • FORWARD 1 • Q2/5

740

QUESTION ID

S24-F1-Q2

TYPE

Short answer

CORRECT ANSWER

2/11

GRADING NOTE 18/99 is also correct.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-F1-Q2

Grade 8 • The Number System • 8.NS.A.1 • Set 24 • FORWARD 1 • Q2/5

Which fraction is equal to 0.333... (the 3 repeats)?

A. $\frac{1}{3}$

B. $\frac{3}{100}$

C. $\frac{3}{10}$

D. $\frac{33}{100}$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S24-F1-Q3

Grade 8 • The Number System • 8.NS.A.1 • Set 24 • FORWARD 1 • Q3/5

741

QUESTION ID

S24-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. 1/3

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S24-F1-Q3

Grade 8 • The Number System • 8.NS.A.1 • Set 24 • FORWARD 1 • Q3/5

True or false?

0.999... (the 9 repeats) is equal to 1.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S24-F1-Q4

Grade 8 • The Number System • 8.NS.A.1 • Set 24 • FORWARD 1 • Q4/5

742

QUESTION ID

S24-F1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S24-F1-Q4

Grade 8 • The Number System • 8.NS.A.1 • Set 24 • FORWARD 1 • Q4/5

True or false?

0.4545... (the digits 45 repeat) is equal to $\frac{45}{100}$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-F1-Q5

Grade 8 • The Number System • 8.NS.A.1 • Set 24 • FORWARD 1 • Q5/5

743

QUESTION ID

S24-F1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE It is equal to 45/99 = 5/11.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-F1-Q5

Grade 8 • The Number System • 8.NS.A.1 • Set 24 • FORWARD 1 • Q5/5

SET 24 • 7.NS.A.2.d

FORWARD 2

Report a decimal result to a sensible level of accuracy • Grade 9 (High School) • Number & Quantity: Quantities • HSN-Q.A.3

A 10-foot board is cut into 3 equal pieces. The tape measure reads to the nearest hundredth of a foot.

Report the length of each piece to that accuracy.

Length: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-F2-Q1

Grade 9 (High School) • Number & Quantity: Quantities • HSN-Q.A.3 • Set 24 • FORWARD 2 • Q1/5

745

QUESTION ID

S24-F2-Q1

TYPE

Short answer

CORRECT ANSWER

3.33 feet

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-F2-Q1

Grade 9 (High School) • Number & Quantity: Quantities • HSN-Q.A.3 • Set 24 • FORWARD 2 • Q1/5

$\frac{2}{7}$ = 0.285714... Round it to the nearest thousandth.

Rounded: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-F2-Q2

Grade 9 (High School) • Number & Quantity: Quantities • HSN-Q.A.3 • Set 24 • FORWARD 2 • Q2/5

746

QUESTION ID

S24-F2-Q2

TYPE

Short answer

CORRECT ANSWER

0.286

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-F2-Q2

Grade 9 (High School) • Number & Quantity: Quantities • HSN-Q.A.3 • Set 24 • FORWARD 2 • Q2/5

A scale reads to the nearest 0.1 kg. 5 kg of flour is split into 3 equal bags.

Which report of each bag's mass fits the scale?

A. 1.666666 kg

B. 1.7 kg

C. 2 kg

D. 1.67 kg

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-F2-Q3

Grade 9 (High School) • Number & Quantity: Quantities • HSN-Q.A.3 • Set 24 • FORWARD 2 • Q3/5

747

QUESTION ID

S24-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. 1.7 kg

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-F2-Q3

Grade 9 (High School) • Number & Quantity: Quantities • HSN-Q.A.3 • Set 24 • FORWARD 2 • Q3/5

True or false?

Reporting $\frac{1}{3}$ hour as 0.33 hour is accurate to the nearest hundredth of an hour.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-F2-Q4

Grade 9 (High School) • Number & Quantity: Quantities • HSN-Q.A.3 • Set 24 • FORWARD 2 • Q4/5

748

QUESTION ID

S24-F2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-F2-Q4

Grade 9 (High School) • Number & Quantity: Quantities • HSN-Q.A.3 • Set 24 • FORWARD 2 • Q4/5

True or false?

A length reported as 4.0 cm is less precise than a length reported as 4 cm.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-F2-Q5

Grade 9 (High School) • Number & Quantity: Quantities • HSN-Q.A.3 • Set 24 • FORWARD 2 • Q5/5

749

QUESTION ID

S24-F2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-F2-Q5

Grade 9 (High School) • Number & Quantity: Quantities • HSN-Q.A.3 • Set 24 • FORWARD 2 • Q5/5

SET 25 • 7.NS.A.3 • THE NUMBER SYSTEM

Solve real-world problems with the four operations on rational numbers

MAIN	Grade 7 main question	**7.NS.A.3**
BACKWARD 1	Multiply and divide decimals in word problems	**6.NS.B.3**
BACKWARD 2	Solve one-step equations	**6.EE.B.7**
FORWARD 1	Solve real-world problems with scientific notation	**8.EE.A.4**
FORWARD 2	Use units to solve multistep problems	**HSN-Q.A.1**

SET 25 • 7.NS.A.3

MAIN

Grade 7 main question • Grade 7 • The Number System • 7.NS.A.3

A freezer is at 4°C. It cools 1.5°C every hour.

What is the temperature after 6 hours?

Temperature: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S25-M-Q1

Grade 7 • The Number System • 7.NS.A.3 • Set 25 • MAIN • Q1/5

752

QUESTION ID

S25-M-Q1

TYPE

Short answer

CORRECT ANSWER

−5°C

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S25-M-Q1

Grade 7 • The Number System • 7.NS.A.3 • Set 25 • MAIN • Q1/5

Maya owes $42.60. She pays it off in 6 equal payments.

How much is each payment?

Payment: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S25-M-Q2

Grade 7 • The Number System • 7.NS.A.3 • Set 25 • MAIN • Q2/5

753

QUESTION ID

S25-M-Q2

TYPE

Short answer

CORRECT ANSWER

$7.10

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S25-M-Q2

Grade 7 • The Number System • 7.NS.A.3 • Set 25 • MAIN • Q2/5

A diver is at −18 m. She rises $\frac{1}{3}$ of the distance to the surface.

What is her new position?

A. −6 m

B. 12 m

C. −12 m

D. −24 m

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S25-M-Q3

Grade 7 • The Number System • 7.NS.A.3 • Set 25 • MAIN • Q3/5

754

QUESTION ID

S25-M-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. −12 m

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S25-M-Q3

Grade 7 • The Number System • 7.NS.A.3 • Set 25 • MAIN • Q3/5

True or false?

A hiker is at −86 m. She climbs 120 m. She ends 34 m above sea level.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-M-Q4

Grade 7 • The Number System • 7.NS.A.3 • Set 25 • MAIN • Q4/5

755

QUESTION ID

S25-M-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-M-Q4

Grade 7 • The Number System • 7.NS.A.3 • Set 25 • MAIN • Q4/5

True or false?

Three stock prices change by −$2.50, +$4, and −$3.25. The total change is +$1.75.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S25-M-Q5

Grade 7 • The Number System • 7.NS.A.3 • Set 25 • MAIN • Q5/5

756

QUESTION ID

S25-M-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE The total change is −$1.75.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S25-M-Q5

Grade 7 • The Number System • 7.NS.A.3 • Set 25 • MAIN • Q5/5

SET 25 • 7.NS.A.3

BACKWARD 1

Multiply and divide decimals in word problems • Grade 6 • The Number System • 6.NS.B.3

Gas costs $3.85 per gallon.

How much do 12 gallons cost?

Cost: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-B1-Q1

Grade 6 • The Number System • 6.NS.B.3 • Set 25 • BACKWARD 1 • Q1/5

758

QUESTION ID

S25-B1-Q1

TYPE

Short answer

CORRECT ANSWER

$46.20

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-B1-Q1

Grade 6 • The Number System • 6.NS.B.3 • Set 25 • BACKWARD 1 • Q1/5

A 7.5-meter rope is cut into 3 equal pieces.

How long is each piece?

Length: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-B1-Q2

Grade 6 • The Number System • 6.NS.B.3 • Set 25 • BACKWARD 1 • Q2/5

759

QUESTION ID

S25-B1-Q2

TYPE

Short answer

CORRECT ANSWER

2.5 m

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-B1-Q2

Grade 6 • The Number System • 6.NS.B.3 • Set 25 • BACKWARD 1 • Q2/5

A 4.8-liter jug fills cups that hold 0.3 liter each.

How many cups can it fill?

A. 14.4

B. 16

C. 160

D. 1.6

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S25-B1-Q3

Grade 6 • The Number System • 6.NS.B.3 • Set 25 • BACKWARD 1 • Q3/5

760

QUESTION ID

S25-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. 16

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S25-B1-Q3

Grade 6 • The Number System • 6.NS.B.3 • Set 25 • BACKWARD 1 • Q3/5

True or false?

A 2.5-pound bag of nuts costs $3.20 per pound. The bag costs $8.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-B1-Q4

Grade 6 • The Number System • 6.NS.B.3 • Set 25 • BACKWARD 1 • Q4/5

761

QUESTION ID

S25-B1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-B1-Q4

Grade 6 • The Number System • 6.NS.B.3 • Set 25 • BACKWARD 1 • Q4/5

True or false?

$9.60 shared equally by 4 people is $2.60 each.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-B1-Q5

Grade 6 • The Number System • 6.NS.B.3 • Set 25 • BACKWARD 1 • Q5/5

762

QUESTION ID

S25-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Each gets $2.40.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-B1-Q5

Grade 6 • The Number System • 6.NS.B.3 • Set 25 • BACKWARD 1 • Q5/5

SET 25 • 7.NS.A.3

BACKWARD 2

Solve one-step equations • Grade 6 • Expressions & Equations • 6.EE.B.7

Solve for x.

x + 4.5 = 12

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-B2-Q1

Grade 6 • Expressions & Equations • 6.EE.B.7 • Set 25 • BACKWARD 2 • Q1/5

764

QUESTION ID

S25-B2-Q1

TYPE

Short answer

CORRECT ANSWER

7.5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-B2-Q1

Grade 6 • Expressions & Equations • 6.EE.B.7 • Set 25 • BACKWARD 2 • Q1/5

Solve for y.

6y = 39

y = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-B2-Q2

Grade 6 • Expressions & Equations • 6.EE.B.7 • Set 25 • BACKWARD 2 • Q2/5

765

QUESTION ID

S25-B2-Q2

TYPE

Short answer

CORRECT ANSWER

6.5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-B2-Q2

Grade 6 • Expressions & Equations • 6.EE.B.7 • Set 25 • BACKWARD 2 • Q2/5

Solve for n.

$\frac{2}{3}n = 10$

A. $n = 30$

B. $n = \frac{20}{3}$

C. $n = 15$

D. $n = 9\frac{1}{3}$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S25-B2-Q3

Grade 6 • Expressions & Equations • 6.EE.B.7 • Set 25 • BACKWARD 2 • Q3/5

766

QUESTION ID

S25-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. n = 15

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S25-B2-Q3

Grade 6 • Expressions & Equations • 6.EE.B.7 • Set 25 • BACKWARD 2 • Q3/5

True or false?

m = 8 is the solution of m − 2.75 = 5.25.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S25-B2-Q4

Grade 6 • Expressions & Equations • 6.EE.B.7 • Set 25 • BACKWARD 2 • Q4/5

767

QUESTION ID

S25-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S25-B2-Q4

Grade 6 • Expressions & Equations • 6.EE.B.7 • Set 25 • BACKWARD 2 • Q4/5

True or false?

k = 3 is the solution of 4k = 1.2.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-B2-Q5

Grade 6 • Expressions & Equations • 6.EE.B.7 • Set 25 • BACKWARD 2 • Q5/5

768

QUESTION ID

S25-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE The solution is k = 0.3.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-B2-Q5

Grade 6 • Expressions & Equations • 6.EE.B.7 • Set 25 • BACKWARD 2 • Q5/5

SET 25 • 7.NS.A.3

FORWARD 1

Solve real-world problems with scientific notation • Grade 8 • Expressions & Equations • 8.EE.A.4

ANSWER KEY Reduced copy of the question page, with its answer

A hard drive holds 2×10^{12} bytes. A file is 4×10^{8} bytes.

How many files of this size fit on the drive?

Files: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-F1-Q1

Grade 8 • Expressions & Equations • 8.EE.A.4 • Set 25 • FORWARD 1 • Q1/5

770

QUESTION ID

S25-F1-Q1

TYPE

Short answer

CORRECT ANSWER

5,000

GRADING NOTE 5×10^{3} is also correct.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-F1-Q1

Grade 8 • Expressions & Equations • 8.EE.A.4 • Set 25 • FORWARD 1 • Q1/5

Earth is about 1.5×10^8 km from the Sun. Mars is about 2.3×10^8 km from the Sun.

About how much farther from the Sun is Mars? Write the answer in scientific notation.

Distance: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-F1-Q2

Grade 8 • Expressions & Equations • 8.EE.A.4 • Set 25 • FORWARD 1 • Q2/5

771

QUESTION ID

S25-F1-Q2

TYPE

Short answer

CORRECT ANSWER

8×10^7 km

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-F1-Q2

Grade 8 • Expressions & Equations • 8.EE.A.4 • Set 25 • FORWARD 1 • Q2/5

A cell is 2×10^{-5} m wide.

How wide are 3,000 of these cells placed side by side?

A. 6×10^{-8} m

B. 5×10^{-2} m

C. 6×10^{-2} m

D. 6×10^{2} m

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-F1-Q3

Grade 8 • Expressions & Equations • 8.EE.A.4 • Set 25 • FORWARD 1 • Q3/5

772

QUESTION ID

S25-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. 6×10^{-2} m

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-F1-Q3

Grade 8 • Expressions & Equations • 8.EE.A.4 • Set 25 • FORWARD 1 • Q3/5

True or false?

$(6 \times 10^4) \times 5 = 3 \times 10^5$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S25-F1-Q4

Grade 8 • Expressions & Equations • 8.EE.A.4 • Set 25 • FORWARD 1 • Q4/5

773

QUESTION ID

S25-F1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S25-F1-Q4

Grade 8 • Expressions & Equations • 8.EE.A.4 • Set 25 • FORWARD 1 • Q4/5

True or false?

$(9 \times 10^6) \div (3 \times 10^2) = 3 \times 10^3$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-F1-Q5

Grade 8 • Expressions & Equations • 8.EE.A.4 • Set 25 • FORWARD 1 • Q5/5

774

QUESTION ID

S25-F1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE The quotient is 3×10^4.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-F1-Q5

Grade 8 • Expressions & Equations • 8.EE.A.4 • Set 25 • FORWARD 1 • Q5/5

SET 25 • 7.NS.A.3

FORWARD 2

Use units to solve multistep problems • Grade 9 (High School) • Number & Quantity: Quantities • HSN-Q.A.1

A car uses 0.04 gallon of gas per mile. Gas costs $3.50 per gallon.

How much does the gas cost for a 250-mile trip?

Cost: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-F2-Q1

Grade 9 (High School) • Number & Quantity: Quantities • HSN-Q.A.1 • Set 25 • FORWARD 2 • Q1/5

776

QUESTION ID

S25-F2-Q1

TYPE

Short answer

CORRECT ANSWER

$35

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-F2-Q1

Grade 9 (High School) • Number & Quantity: Quantities • HSN-Q.A.1 • Set 25 • FORWARD 2 • Q1/5

A faucet leaks 0.2 liter per hour.

How many liters leak in 1 week?

Liters: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-F2-Q2

Grade 9 (High School) • Number & Quantity: Quantities • HSN-Q.A.1 • Set 25 • FORWARD 2 • Q2/5

777

QUESTION ID

S25-F2-Q2

TYPE

Short answer

CORRECT ANSWER

33.6

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-F2-Q2

Grade 9 (High School) • Number & Quantity: Quantities • HSN-Q.A.1 • Set 25 • FORWARD 2 • Q2/5

A worker earns $18.50 per hour. She works 7.5 hours per day for 4 days.

How much does she earn?

A. $555

B. $74

C. $540

D. $138.75

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-F2-Q3

Grade 9 (High School) • Number & Quantity: Quantities • HSN-Q.A.1 • Set 25 • FORWARD 2 • Q3/5

778

QUESTION ID

S25-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. $555

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-F2-Q3

Grade 9 (High School) • Number & Quantity: Quantities • HSN-Q.A.1 • Set 25 • FORWARD 2 • Q3/5

True or false?

A speed of 12 meters per second is 43.2 kilometers per hour.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-F2-Q4

Grade 9 (High School) • Number & Quantity: Quantities • HSN-Q.A.1 • Set 25 • FORWARD 2 • Q4/5

779

QUESTION ID

S25-F2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-F2-Q4

Grade 9 (High School) • Number & Quantity: Quantities • HSN-Q.A.1 • Set 25 • FORWARD 2 • Q4/5

True or false?

Walking 3 miles per hour for 45 minutes covers 2.5 miles.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-F2-Q5

Grade 9 (High School) • Number & Quantity: Quantities • HSN-Q.A.1 • Set 25 • FORWARD 2 • Q5/5

780

QUESTION ID

S25-F2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE It covers 2.25 miles.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-F2-Q5

Grade 9 (High School) • Number & Quantity: Quantities • HSN-Q.A.1 • Set 25 • FORWARD 2 • Q5/5

SET 26 • 7.EE.A.1 • EXPRESSIONS & EQUATIONS

Add and subtract linear expressions with rational coefficients

MAIN	Grade 7 main question	**7.EE.A.1**
BACKWARD 1	Combine like terms	**6.EE.A.3**
BACKWARD 2	Identify equivalent expressions	**6.EE.A.4**
FORWARD 1	Solve equations by expanding and collecting like terms	**8.EE.C.7.b**
FORWARD 2	Interpret the terms, factors, and coefficients of an expression	**HSA-SSE.A.1.a**

SET 26 • 7.EE.A.1

MAIN

Grade 7 main question • Grade 7 • Expressions & Equations • 7.EE.A.1

Simplify.

$(3x + 5) + (2x - 8)$

Expression: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S26-M-Q1

Grade 7 • Expressions & Equations • 7.EE.A.1 • Set 26 • MAIN • Q1/5

783

QUESTION ID

S26-M-Q1

TYPE

Short answer

CORRECT ANSWER

5x − 3

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S26-M-Q1

Grade 7 • Expressions & Equations • 7.EE.A.1 • Set 26 • MAIN • Q1/5

Simplify.

$(4.5y - 2) - (1.5y + 6)$

Expression: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S26-M-Q2

Grade 7 • Expressions & Equations • 7.EE.A.1 • Set 26 • MAIN • Q2/5

784

QUESTION ID

S26-M-Q2

TYPE

Short answer

CORRECT ANSWER

3y − 8

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S26-M-Q2

Grade 7 • Expressions & Equations • 7.EE.A.1 • Set 26 • MAIN • Q2/5

Simplify.

$\frac{1}{2}a + 3 - \frac{3}{4}a + 1$

A. $-\frac{1}{4}a + 4$

B. $-\frac{1}{4}a + 2$

C. $\frac{1}{4}a + 4$

D. $\frac{5}{4}a + 4$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S26-M-Q3

Grade 7 • Expressions & Equations • 7.EE.A.1 • Set 26 • MAIN • Q3/5

785

QUESTION ID

S26-M-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. −(1/4)a + 4

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S26-M-Q3

Grade 7 • Expressions & Equations • 7.EE.A.1 • Set 26 • MAIN • Q3/5

True or false?

$(6m - 1) - (2m - 4) = 4m + 3$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S26-M-Q4

Grade 7 • Expressions & Equations • 7.EE.A.1 • Set 26 • MAIN • Q4/5

786

QUESTION ID

S26-M-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S26-M-Q4

Grade 7 • Expressions & Equations • 7.EE.A.1 • Set 26 • MAIN • Q4/5

True or false?

$-3k + 7 + 5k - 2 = 8k + 5$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S26-M-Q5

Grade 7 • Expressions & Equations • 7.EE.A.1 • Set 26 • MAIN • Q5/5

787

QUESTION ID

S26-M-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE The expression simplifies to 2k + 5.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S26-M-Q5

Grade 7 • Expressions & Equations • 7.EE.A.1 • Set 26 • MAIN • Q5/5

SET 26 • 7.EE.A.1

BACKWARD 1

Combine like terms • Grade 6 • Expressions & Equations • 6.EE.A.3

Combine like terms.

5n + 3n + 4

Expression: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S26-B1-Q1

Grade 6 • Expressions & Equations • 6.EE.A.3 • Set 26 • BACKWARD 1 • Q1/5

789

QUESTION ID

S26-B1-Q1

TYPE

Short answer

CORRECT ANSWER

8n + 4

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S26-B1-Q1

Grade 6 • Expressions & Equations • 6.EE.A.3 • Set 26 • BACKWARD 1 • Q1/5

Combine like terms.

2y + 7 + y

Expression: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-B1-Q2

Grade 6 • Expressions & Equations • 6.EE.A.3 • Set 26 • BACKWARD 1 • Q2/5

790

QUESTION ID

S26-B1-Q2

TYPE

Short answer

CORRECT ANSWER

3y + 7

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-B1-Q2

Grade 6 • Expressions & Equations • 6.EE.A.3 • Set 26 • BACKWARD 1 • Q2/5

Which expression is equivalent to x + x + x + 6?

A. 9x

B. 3x + 2

C. 3x + 6

D. $x^3 + 6$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-B1-Q3

Grade 6 • Expressions & Equations • 6.EE.A.3 • Set 26 • BACKWARD 1 • Q3/5

791

QUESTION ID

S26-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. 3x + 6

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-B1-Q3

Grade 6 • Expressions & Equations • 6.EE.A.3 • Set 26 • BACKWARD 1 • Q3/5

True or false?

4b + 2b = 6b

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S26-B1-Q4

Grade 6 • Expressions & Equations • 6.EE.A.3 • Set 26 • BACKWARD 1 • Q4/5

792

QUESTION ID

S26-B1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S26-B1-Q4

Grade 6 • Expressions & Equations • 6.EE.A.3 • Set 26 • BACKWARD 1 • Q4/5

True or false?

$3a + 2 = 5a$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S26-B1-Q5

Grade 6 • Expressions & Equations • 6.EE.A.3 • Set 26 • BACKWARD 1 • Q5/5

793

QUESTION ID

S26-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S26-B1-Q5

Grade 6 • Expressions & Equations • 6.EE.A.3 • Set 26 • BACKWARD 1 • Q5/5

SET 26 • 7.EE.A.1

BACKWARD 2

Identify equivalent expressions • Grade 6 • Expressions & Equations • 6.EE.A.4

Are 2(x + 3) and 2x + 6 equivalent? Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-B2-Q1

Grade 6 • Expressions & Equations • 6.EE.A.4 • Set 26 • BACKWARD 2 • Q1/5

795

QUESTION ID

S26-B2-Q1

TYPE

Short answer

CORRECT ANSWER

Yes. Using the distributive property, 2(x + 3) = 2x + 6 for every value of x.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-B2-Q1

Grade 6 • Expressions & Equations • 6.EE.A.4 • Set 26 • BACKWARD 2 • Q1/5

Which expression is equivalent to y + y + y?

A. 3y

B. y + 3

C. y^3

D. 3 + y

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-B2-Q2

Grade 6 • Expressions & Equations • 6.EE.A.4 • Set 26 • BACKWARD 2 • Q2/5

796

QUESTION ID

S26-B2-Q2

TYPE

Multiple choice

CORRECT ANSWER

A. 3y

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-B2-Q2

Grade 6 • Expressions & Equations • 6.EE.A.4 • Set 26 • BACKWARD 2 • Q2/5

Write an expression with no parentheses that is equivalent to 4(n + 1).

Expression: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S26-B2-Q3

Grade 6 • Expressions & Equations • 6.EE.A.4 • Set 26 • BACKWARD 2 • Q3/5

797

QUESTION ID

S26-B2-Q3

TYPE

Short answer

CORRECT ANSWER

4n + 4

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S26-B2-Q3

Grade 6 • Expressions & Equations • 6.EE.A.4 • Set 26 • BACKWARD 2 • Q3/5

True or false?

5m − 2m and 3m are equivalent.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S26-B2-Q4

Grade 6 • Expressions & Equations • 6.EE.A.4 • Set 26 • BACKWARD 2 • Q4/5

798

QUESTION ID

S26-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S26-B2-Q4

Grade 6 • Expressions & Equations • 6.EE.A.4 • Set 26 • BACKWARD 2 • Q4/5

True or false?

a + 4 and 4a are equivalent.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S26-B2-Q5

Grade 6 • Expressions & Equations • 6.EE.A.4 • Set 26 • BACKWARD 2 • Q5/5

799

QUESTION ID

S26-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S26-B2-Q5

Grade 6 • Expressions & Equations • 6.EE.A.4 • Set 26 • BACKWARD 2 • Q5/5

SET 26 • 7.EE.A.1

FORWARD 1

Solve equations by expanding and collecting like terms • Grade 8 • Expressions & Equations • 8.EE.C.7.b